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18 June 1985
SUPERSEDING
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(See 6.9)

MILITARY SPECIFICATION

CABLE AND CORD, ELECTRICAL, FOR SHIPBOARD USE, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electrical cable and cord for Navy shipboard applications.

1.2 Classification. Cables and cords covered by this specification shall be classified as watertight and non-watertight constructions, armored and unarmored, and further classified for flexing and nonflexing service for power, lighting, control, communications, instrumentation and electronic applications, as specified in 1.2.1 through 1.2.4 (see 6.2.1).

1.2.1 Watertight, nonflexing service. Cable and cord types for watertight, nonflexing service are as follows:

Power and lighting:

	Unarmored	Armored		
Type	DRW	DRWA	Two conductors	3000 volts (V)
	—	DPS	Two conductors	600 V
	DSGU	DSGA	Two conductors	1000 V
	—	FPS	Four conductors	600 V
	FSGU	FSGA	Four conductors	1000 V
	MDU	MDY	Nineteen conductors	600 V
	SRW	SRWA	Single conductor	3000 V
	SSGU	SSGA	Single conductor	1000 V
	—	TPS	Three conductors	600 V
	TRW	TRWA	Three conductors	3000 V
	TSGU	TSGA	Three conductors	1000 V

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523 Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6145

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Unarmored	Armored		
5KVTSGU	5KVTSGA	Three conductors	5000 V
6SGU	6SGA	Six conductors	1000 V
---	7PS	Seven conductors	600 V
7SGU	7SGA	Seven conductors	1000 V
Control:			
Type MSCU	MSCA	Seven through 91 conductors	1000 V
MSCS	---	Double overall shield	
Electronic, communication and instrumentation:			
Type ECM	ECMA	56 single conductors plus 8 shielded pairs	600 V
MSPW	---	59 conductors (8 shielded pairs, 3 shielded triads, 34 shielded singles)	
PBTMU	PBTM	Five, 15 and 30 pairs	
TCJU	TCJA	One pair	
---	TCJX	Three, 7 and 12 pairs	
---	TCKX	One, 3, 7 and 12 pairs	
TCTU	TCTA	One pair	
---	TCTX	Three, 7 and 12 pairs	
TSP	TSPA	11 and 31 pairs	300 V
TTSU	TTSA	1-1/2 through 60 pairs	300 V
1SMWU	1SMWA	70 shielded singles	
1SWU	1SWA	Two, 14, 20 and 30 shielded singles	
2SWAU	2SWA	Three through 61 shielded pairs	
2SWU	2SWUA	One through 61 shielded pairs	
2WAU	2WA	40 pairs with overall shield	600 V
3SWU	3SWA	Three through 44 shielded triads	
3SWUS	---	Three through 44 shielded triads with double overall shield	
2SWL-7	2SWLA-7	Seven shielded pairs	
2UW-42	2UWA-42	42 pairs with overall shield	
2UWS-42		42 pairs with double overall shield	

1.2.2 Watertight, flexing service. Cable and cord types for watertight, flexing service are as follows:

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Power:

	Unarmored	Armored		
Type	DSWS	---	Two conductors shielded	
	MCSF	---	Four conductors (2-#6 and 2-#1 AWG)	600 V

Control:

Type	MWF	---	Seven through 37 conductors	600 V

Electronic, communication and instrumentation:

Type	DSS	---	Two conductors with overall shield	600 V
	FSS	---	Four conductors with overall shield	600 V
	S2S	---	Two conductor (concentric)	
	TSS	---	Three conductors with overall shield	600 V
	1SWF	---	Two shielded singles	
	2SWF	---	Three, 4 and 7 shielded pairs	
	7SS	---	Seven conductors with overall shield	600 V
	TPUM-6	---	Six pairs	

1.2.3 Non-watertight, nonflexing service. Cable and cord types for non-watertight, nonflexing service are as follows:

Power and lighting:

Type	DNW	---	DNWA	Two conductors	1000 V
	FNW	---	FNWA	Four conductors	1000 V
	TNW	---	TNWA	Three conductors	1000 V
	2SJ	---	2SJA	Two conductors with overall shield	600 V
	3SJ	---	3SJA	Three conductors with overall shield	600 V
	4SJ	---	4SJA	Four conductors with overall shield	600 V

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Control:

	Unarmored	Armored		
Type	MNW	MNWA	Seven through 44 conductors	1000 V
	MS	MSA	37 conductors with overall shield	300 V
	MU	MA	14 conductors	300 V
	MUS	---	14 conductors with double overall shield	300 V
	4NW8	4NWA8	Four conductors	
	8NW6	8NWA6	Eight conductors	

Electronic, communication and instrumentation:

Type	MRI	---	Twisted pair and twisted triad	
	MSP	---	59 conductors (8 shielded pairs, 3 shielded triads, 34 shielded singles)	
	---	PI	Three, 7, and 12 shielded pairs	
	TPNW	TPNWA	1-1/2 through 40 pairs	300 V
	1SAU	1SA	44 shielded singles	
	1SMU	1SMA	Five shielded singles	
	1SU	1SUA	36 and 60 shielded singles	
	1S50MU	1S50MA	16, 20, 40 and 70 shielded singles	
	1S50MUS	---	16, 20, 40 and 70 shielded singles	
	1S75MU	1S75MA	Eight shielded singles	
	2AU	2A	40 pairs with overall shield	600 V
	2AUS	---	40 pairs with overall shield	
	2SJ	2SJA	Two conductors with overall shield	600 V
	2SU	2SA	Three through 61 shielded pairs	
	2SUS	---	Three through 61 shielded pairs	
	2U	2UA	10 through 60 pairs with overall shield	300 V
	3SJ	3SJA	Three conductors with overall shield	600 V
	3SU	3SA	Three through 44 shielded triads	
	3SUS	---	Three through 44 shielded triads	
	3U	3UA	Three, 7 and 12 triads	
	4SJ	4SJA	Four conductors with overall shield	600 V
	2CS	---	Six, 18, 42, 60, and 77 pairs with double overall shield	300 V

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Unarmored	Armored	(Not for inboard use)
1PR-A20E	——	One twisted pair
1PR-16	——	One twisted pair
7PR-16	——	Seven twisted pairs
2SPR-16	——	Two shielded twisted pairs
3PR-16	——	Three twisted pairs
1Q-16	——	Twisted quad
1TR-16	——	Twisted triad
7SPR-16S	——	Seven shielded twisted pairs with overall shield
1SPR-16	——	One shielded twisted pair

1.2.4 Non-watertight flexing service. Cable and cord types for non-watertight flexing service are as follows:

Power and lighting:

Type	CVSF	——	Four conductors (3-#3 and 1-#5 AWG)	600 V
	DHOF	——	Two conductors	600 V
	DLT	——	Four conductors	300 V
	FHOF	——	Four conductors	600 V
	JAS	——	Four conductors (2-#250 and 2-#6 AWG)	600 V
	SHOF	——	Single conductors sizes 3 through 800 AWG	600 V
	SSF	——	Single conductor	600 V
	THOF	——	Three conductors, sizes 3 through 600 AWG	600 V
	TRF	——	Single conductor	600 V
	TRXF	——	Single conductor	125 V

Control:

Type	MHOF		Seven through 61 conductors	600 V

Electronic, communication and instrumentation:

Type	DCOP	——	Two conductors	300 V
	MCOS	——	Multi conductors-shielded	600 V
	MMOP	——	Five conductors	300 V
	TCOP	——	Three conductors	300 V
	TTOP	——	Three through 15 pairs	300 V
	TTRS	TTRSA	Two through 16 shielded pairs	300 V
	3SP	——	Seven shielded triads	600 V
	5SS	——	Five conductors (4 singles and one shielded single) with overall shield	600 V

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-P-320	- Pigment, Aluminum: Powder and Paste for Paint.
PPP-V-205	- Veneer, Paper Overlaid, Container Grade.

MILITARY

MIL-C-572	- Cords, Yarns and Monofilaments Organic Synthetic Fiber.
MIL-Y-1140	- Yarn, Cord, Sleeving, Cloth, and Tape-Glass.
MIL-C-12000	- Cable, Cord, and Wire, Electric, Packaging of.
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-S-24235	- Stuffing Tubes, Metal, and Packing Assemblies for Electrical Cables, General Specification for.
MIL-S-24235/1	- Stuffing Tube, Bulkhead, Pressureproof.
MIL-S-24235/2	- Stuffing Tubes, Metal, and Packing Assemblies for Electric Cables, Packing Assemblies for Pressureproof Bulkhead.
MIL-W-81381	- Wire, Electric, Polyimide-Insulated, Copper or Copper Alloy.
MIL-W-83420	- Wire Rope, Flexible, For Aircraft Control.

(See supplement 1 for list of applicable specification sheets.)

STANDARDS

FEDERAL

FED-STD-228	- Cable and Wire, Insulated; Methods of Testing.
FED-STD-595	- Colors.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.

2.1.2 Government drawing. The following Government drawing forms a part of this specification to the extent specified herein.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-5001027 - Electric Plant Installation Standard Methods.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 3 - Soft or Annealed Copper Wire. (DoD adopted)
B 8 - Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft. (DoD adopted)
B 33 - Tinned Soft or Annealed Copper Wire for Electrical Purposes.
B 172 - Rope-Lay-Stranded Copper Conductors Having Bunch-Stranded Members, for Electrical Conductors. (DoD adopted)
B 173 - Rope-Lay-Stranded Copper Conductors Having Concentric-Stranded Members, for Electrical Conductors. (DoD adopted)
B 174 - Bunch-Stranded Copper Conductors for Electrical Conductors. (DoD adopted)
B 189 - Lead-Coated and Lead Alloy-Coated Soft Copper Wire for Electrical Purposes. (DoD adopted)
B 193 - Resistivity of Electrical Conductor Materials. (DoD adopted)
B 228 - Concentric-Lay-Stranded Copper-Clad Steel Conductors. (DoD adopted)
B 258 - Standard Nominal Diameters and Cross-Sectional Areas of AWG Sizes of Solid Round Wires Used as Electrical Conductors.
B 286 - Copper Conductors for Use in Hookup Wire for Electronic Equipment. (DoD adopted)
B 298 - Silver-Coated Soft or Annealed Copper Wire. (DoD adopted)
B 344 - Drawn or Rolled Nickel-Chromium and Nickel-Chromium-Iron Alloys for Electrical Heating Elements.
B 355 - Nickel-Coated Soft or Annealed Copper Wire. (DoD adopted)
D 297 - Rubber Products-Chemical Analysis, Methods for. (DoD adopted)
D 470 - Crosslinked Insulations and Jackets for Wire and Cable, Methods for Testing.
D 1458 - Fully Cured Silicone Rubber-Coated Glass Fabric and Tapes for Electrical Insulation, Methods for Testing.
D 2240 - Rubber Property-Durometer Hardness, Test Methods for. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B46.1 - Surface Texture, Surface Roughness, Waviness and Lay. (DoD adopted)
MC96.1 - Temperature Measurement Thermocouples.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)
383 - Class 1E Electric Cables, Field Splices, and Connections for Nuclear Power Generating Stations, Type Test of.

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

UNDERWRITERS' LABORATORIES, INC. (UL)

(Application for copies of publications should be addressed to Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the general requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Cables and cords furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.5).

3.2.1 Requalification. Changes in materials and construction which have been qualified shall require the written approval of the Naval Sea Systems Command (NAVSEA). Incorporation of any changes which have not been so approved shall require requalification of the cable or cord or groups of cables or cords in question.

3.3. Materials. Materials used in the construction of those cables or cords requiring qualification shall have the written approval of NAVSEA. In the case of cables or cords not requiring qualification, NAVSEA approval will not be required, and materials in these cables shall be in accordance with the requirements shown in the applicable specification sheets.

3.3.1 Conductors.

3.3.1.1 Copper conductors. Copper conductors shall be composed of soft or annealed copper strands in accordance with ASTM B 3.

3.3.1.2 High strength conductors. Unless otherwise specified on the specification sheet, high strength conductors shall be composed of 30 percent conductivity, high strength, copper-covered steel strands in accordance with ASTM B 228.

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3.3.1.3 Tin coating. Tin coating shall be in accordance with ASTM B 33.

3.3.1.4 Lead or lead alloy coating. Lead or lead alloy coating shall be in accordance with ASTM B 189.

3.3.1.5 Nickel coating. Nickel coating shall be in accordance with class 2 of ASTM B 355.

3.3.1.6 Silver coating. Silver coating shall be in accordance with ASTM B 298.

3.3.1.7 Thermocouple wire. Conductors for thermocouple cables shall be in accordance with ANSI MC96.1.

3.3.2 Insulation. Insulations shall be as specified on the applicable specification sheet except the silicone rubber coated glass fabric tape insulation shall, before application to a conductor, have a dielectric strength of not less than 475 V per mil for tapes which are less than 8.8 mils thick, and 575 V per mil for tapes which are 8.9 mils or more in thickness. These dielectric strength values shall be met after the tapes have been conditioned for 24 hours at 23 degrees Celsius (°C) and 96 percent relative humidity and tested in accordance with the short time dielectric breakdown test in accordance with ASTM D 1458 while maintaining temperature and humidity conditions. Extruded silicone rubber insulation shall have a specific gravity of not more than 1.55. Other physical and electrical properties shall be as specified by the specification sheets.

3.3.3 Jackets. The material used for jackets over insulated conductors and groups of insulated conductors, and for the cable or cord jacket, shall have the physical and electrical properties as required by the applicable specification sheet.

3.3.3.1 Thermoplastic jackets. Thermoplastic jackets shall be polyvinyl chloride and are classified as standard thermoplastic or special thermoplastic. The special thermoplastic jacket shall be used where necessary to provide extra protection. The standard thermoplastic jacket shall be used for lightweight cables or cords which are classified for either flexing or nonflexing service, and the jacket shall be compounded to meet the requirements of the specification and have the physical and electrical properties as required by the specification sheet. Unless otherwise specified on the specification sheet, thermoplastic cable or cord jackets shall be gray in accordance with color no. 26270 of FED-STD-595.

3.3.4 Shields. The materials and constructions for shields of insulated conductors, groups of insulated conductors and overall cable or cord shall be as specified in the specification sheet. When AWG sizes are specified for wire shields, they shall be in accordance with ASTM B 258..

3.3.5 Fillers. Unless otherwise specified in the applicable specification sheet, fillers for cables or cords which are not required to be watertight may be fibrous. Fillers for cables or cords which are required to be watertight shall be nonfibrous or a combination of nonfibrous and fibrous as specified in the specification sheet.

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3.3.5.1 Fibrous fillers. [illegible] fillers shall consist of either cotton, jute, synthetic fibers or glass [illegible] material shall be treated for flame or moisture resistance, or a combination of both, to meet the requirements for the particular type of cable or cord.

3.3.5.2 Asbestos fillers. The use of asbestos or asbestos form containing materials shall be prohibited throughout the construction and manufacturing of cables.

3.3.5.3 Nonfibrous fillers. Nonfibrous fillers shall consist of elastomeric material which is readily removable from insulation of conductors and insulating tapes over shields without the aid of solvents, cleansers or tools. The acceptability of the material shall be in accordance with 4.7.7 or 4.7.8 and 4.7.9, as applicable.

3.3.6 Tapes. Tapes shall be of the material as specified (see 3.1) or shall be a type approved by NAVSEA for the specific cable or cord construction.

3.3.7 Braids (identification). Colored braids used for conductor or group identification shall be in accordance with MIL-C-572.

3.3.8 Braids (glass). Glass braids for use on silicone insulated conductors shall be composed of the appropriate size of either staple or continuous fiber in accordance with MIL-Y-1140.

3.3.9 Tie cords. Tie cords shall be of cotton or synthetic fiber having a maximum diameter of 0.065 inch and a minimum breaking strength of 30 pounds.

3.3.10 Separators. Unless otherwise specified (see 3.1) separators, when required, shall consist of a winding of glass fibers, synthetic fibers or cotton. They may also be of a tape of cotton, synthetic fiber, paper or polyester. Separators shall be opaque.

3.3.11 Reinforcement. Reinforcing binders or reinforcement for a single or double layer jacket shall be a size 3 nylon seine twine having a minimum breaking strength of 15 pounds or of a material approved by NAVSEA.

3.3.12 Binders. Binders shall be of the material specified (see 3.1) or of a material approved by NAVSEA.

3.3.13 Armor wires. The armor wires shall consist of Alclad 5056 aluminum alloy having a diameter of 0.0126 ± 0.0005 inch, a minimum tensile strength of 50,000 pounds per square inch (lb/in^2) and a minimum elongation (before application to the cable) of 2 percent in 10-inch length.

3.3.14 Armor paint. Armor paint shall consist of aluminum pigment (paste form) in accordance with TT-P-320, in a vehicle of the synthetic resinous (phenol-formaldehyde or glyceride) type designed for good penetration, high bending power, resistance to water and oil, and with flexibility at temperatures up to 100°C.

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3.3.15 Prohibited materials. Materials or parts containing mercury shall not be used (see 4.4.7).

3.3.16 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Construction.

3.4.1 Conductor stranding. The size and quantity of individual conductor strands and the total circular-mil area of the conductor shall be in accordance with ASTM B 8 or B 286 as specified in the specification sheet and tables I, II or III and ASTM B 172, 173 or 174 as specified in the specification sheet and tables IV, V, VI and VII. Navy standard sized conductors shall be in accordance with table VIII.

TABLE I. Conductor data, concentric-lay stranded.

Conductor size ASTM B 286 designation	Number of strands (minimum)	Strand diameter nominal (inch)	Conductor diameter nominal (inch)	Conductor cross-sectional area, nominal (circular mils)	Weight per 1000 feet approximate (pounds)
12-37	37	0.0126	0.088	5874	20.20
12-19	19	.0179	.091	6088	20.20
14-19	19	.0142	.072	3831	12.65
16-19	19	.0113	.057	2426	7.97
18-19	19	.0100	.051	1900	5.02
18-7	7	.0159	.049	1770	5.02
20-19	19	.0080	.041	1216	3.16
20-7	7	.0126	.038	1111	3.16
22-19	19	.0063	.032	754	1.98
22-7	7	.0100	.031	700	1.98
24-19	19	.0050	.026	475	1.24
24-7	7	.0080	.025	448	1.24
26-19	19	.0040	.021	304	.780
26-7	7	.0063	.020	278	.780
28-19	19	.0031	.016	183	.490
28-7	7	.0050	.016	175	.490
30-7	7	.0040	.013	112	.309

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TABLE II. Construction requirements of concentric-lay stranded copper conductors, ASTM B 8.

Area of cross-section (circular mils)	Size, American (or B & S) wire gauge	Class B		Class C		Class D	
		Number of strands	Diameter of strands (mils)	Number of strands	Diameter of strands (mils)	Number of strands	Diameter of strands (mils)
2,000,000	---	127	125.5	---	---	---	---
1,600,000	---	127	112.2	---	---	---	---
1,300,000	---	---	---	127	101.2	---	---
1,000,000	---	---	---	---	---	127	88.7
800,000	---	---	---	---	---	127	79.4
650,000	---	---	---	---	---	127	71.5
550,000	---	---	---	---	---	127	65.8
350,000	---	---	---	---	---	91	62.0
300,000	---	---	---	---	---	91	57.4
250,000	---	---	---	61	64.0	---	---
211,600	0000	---	---	---	---	61	58.9
167,800	000	---	---	---	---	61	52.4
133,100	00	---	---	---	---	61	46.7
105,600	0	---	---	---	---	61	41.6
83,690	1	---	---	37	47.6	---	---
66,360	2	---	---	---	---	37	42.4
52,620	3	---	---	19	52.6	---	---
41,740	4	---	---	19	46.9	---	---
33,090	5	---	---	19	41.7	---	---
26,240	6	7	61.2	---	---	---	---
20,820	7	7	54.5	---	---	---	---
13,090	9	7	43.2	---	---	---	---
10,380	10	7	38.5	---	---	---	---
6,530	12	7	30.5	---	---	---	---
4,110	14	7	24.2	---	---	---	---
2,580	16	7	19.2	19	11.7	---	---
1,620	18	7	15.2	---	---	---	---
1,020	20	7	12.1	---	---	---	---

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TABLE III. Diameters, areas, and weights of concentric-lay-stranded copper conductors, ASTM B 8.

Size of conductor, circular mil or AWG numbers	Nominal conductor diameter, class B, (inches)	Area (square inches)	Approximate weight, pounds	
			per 1000 feet	per mile
2,000,000 cmil	1.632	1.571	6,175	32,600
1,600,000 cmil	1.459	1.257	4,940	26,080
1,300,000 cmil	1.315	1.021	4,014	21,190
1,000,000 cmil	1.152	0.7854	3,088	16,300
800,000 cmil	1.031	0.6283	2,470	13,040
650,000 cmil	0.929	0.5105	2,007	10,600
550,000 cmil	0.855	0.4320	1,698	8,966
350,000 cmil	0.681	0.2749	1,081	5,706
300,000 cmil	0.630	0.2356	926.3	4,891
250,000 cmil	0.575	0.1963	771.9	4,076
No. 0000	0.528	0.1662	653.3	3,450
No. 000	0.470	0.1318	518.1	2,736
No. 00	0.419	0.1045	410.9	2,169
No. 0	0.373	0.08289	325.8	1,720
No. 1	0.332	0.06573	258.4	1,364
No. 2	0.292	0.05213	204.9	1,082
No. 3	0.260	0.04134	162.5	858.1
No. 4	0.232	0.03278	128.9	680.5
No. 5	0.206	0.02600	102.2	539.6
No. 6	0.184	0.02062	81.05	428.0
No. 7	0.164	0.01635	64.28	339.4
No. 9	0.130	0.01028	40.42	213.4
No. 10	0.116	0.008155	32.06	169.3
No. 12	0.0915	0.005129	20.16	106.5
No. 14	0.0726	0.003225	12.68	66.95
No. 16	0.0576	0.002028	7.974	42.10
No. 18	0.0456	0.001276	5.015	26.48
No. 20	0.0363	0.0008023	3.154	16.65

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TABLE IV. Construction requirements of rope-lay stranded copper conductors having bunch-stranded members 1/ ASTM B 172.

Area of cross-section, (circular mils)	Size (AWG)	Class I		Class K		Class M	
		Nominal number of strands 0.0201 inch in diameter (no. 24 AWG)	Strand construction	Nominal number of strands 0.0100 inch in diameter (no. 30 AWG)	Strand construction	Nominal number of strands 0.0063 inch in diameter (no. 34 AWG)	Strand construction
650,000	---	1,596	19 by 7 by 12	6,517	19 by 7 by 49	16,226	61 by 7 by 38
600,000	---	1,470	7 by 7 by 30	5,985	19 by 7 by 45	14,945	61 by 7 by 35
500,000	---	1,225	7 by 7 by 25	5,054	19 by 7 by 38	12,691	37 by 7 by 49
300,000	---	735	7 by 7 by 15	2,989	7 by 7 by 61	7,581	19 by 7 by 57
250,000	---	637	7 by 7 by 13	2,499	7 by 7 by 51	6,384	19 by 7 by 48
211,600	0000	532	19 by 28	2,107	7 by 7 by 43	5,320	19 by 7 by 40
167,800	000	418	19 by 22	1,666	7 by 7 by 34	4,256	19 by 7 by 32
133,100	00	342	19 by 18	1,323	7 by 7 by 27	3,325	19 by 7 by 25
105,600	0	266	19 by 14	1,064	19 by 56	2,646	7 by 7 by 54
83,690	1	210	7 by 30	836	19 by 44	2,107	7 by 7 by 43
66,360	2	161	7 by 23	665	19 by 35	1,666	7 by 7 by 34
52,620	3	133	7 by 19	532	19 by 28	1,323	7 by 7 by 27
41,740	4	105	7 by 15	420	7 by 60	1,064	19 by 56
33,090	5	84	7 by 12	336	7 by 48	836	19 by 44
26,240	6	63	7 by 9	266	7 by 38	665	19 by 35
20,820	7	---	---	210	7 by 30	532	19 by 28
13,090	9	---	---	133	7 by 19	336	7 by 48
6,530	12	---	---	---	---	168	7 by 24

1/ The designations of strand construction in accordance with table I indicate by means of the last number (at the right hand) the number of individual wires contained in the smallest bunch-stranded member of the conductor. The preceding number (to the left of the word "by") indicates the number of these stranded members which are cabled together to form a rope-stranded conductor or a rope-stranded member of a larger conductor. A third number preceding these indicates the number of these rope-stranded members which are cabled together to form a larger rope-stranded conductor. For example, 19 by 7 by 25 indicates that 25 wires are bunch-stranded into a stranded member; 7 of these members are stranded into a rope-lay-stranded member; and then 19 of these 7 by 25 members are formed into a larger rope-stranded conductor having a total of 3325 wires.

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TABLE V. Construction requirements of rope-lay-stranded copper conductors having concentric-stranded members, ASTM B 173.

Area of cross section (circular mils)	Size (AWG)	Class C			Class H		
		Number of strands	Diameter of strands (mils)	Number of strands in each member	Number of strands	Diameter of strands (mils)	Number of strands in each member
650,000	---	427	39.0	7	---	---	---
600,000	---	427	37.5	7	---	---	---
500,000	---	259	43.9	7	---	---	---
300,000	---	259	34.0	7	---	---	---
167,800	000	---	---	---	259	25.5	7
133,100	00	---	---	---	259	22.7	7
105,600	0	---	---	---	259	20.2	7
26,240	6	49	23.1	7	---	---	---
20,820	7	49	20.6	7	---	---	---

TABLE VI. Approximate diameters and weights of completed conductors, ASTM B 173.

Area of cross section (circular mils)	Size (AWG)	Class C			Class H		
		Number of strands	Outside diameter (inches)	Weight (lb/1000 feet)	Number of strands	Outside diameter (inches)	Weight lb/1000 feet
650,000	---	427	1.053	2,070	---	---	---
600,000	---	427	1.013	1,910	---	---	---
500,000	---	259	0.922	1,585	---	---	---
300,000	---	259	.714	945	---	---	---
167,800	000	---	---	---	259	0.536	533
133,100	00	---	---	---	259	.477	422
105,600	0	---	---	---	259	.424	334
26,240	6	49	0.208	82	---	---	---
20,820	7	49	.185	65	---	---	---

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TABLE VII. Construction requirements of bunch-stranded conductors ASTM B 174.

Area of cross section (circular mils)	Size (AWG)	Class K Minimum number of strands 0.0100 inch in diameter (no. 30 AWG)	Class M Minimum number of strands 0.0063 inch in diameter (no. 34 AWG)	Class O Minimum number of strands 0.0050 inch in diameter (no. 36 AWG)
6,530	12	65	---	---
4,110	14	41	104	---
2,580	16	26	65	104
1,620	18	16	41	65
1,020	20	1/ 10	26	41
640	22	7	19	---
404	24	---	---	19

1/ As an alternative to the construction shown for hookup wire, size 20 AWG may be supplied with 8 strands of 0.0100 inch diameter around 1 strand of 0.0142 inch diameter.

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TABLE VIII. Navy standard sizes.

Conductor size Navy standard	Number of strands (minimum)	Strand diameter nominal (inch)	Conductor diameter nominal (inches)	Conductor cross-sectional area (circular mils)		Maximum conductor resistance (dc) per 1000 feet at 25°C		Weight per 1000 feet approximate (pounds)
				Nominal 1/	Minimum	Bare (ohms)	Coated (ohms)	
Concentric-lay-stranded:								
400(127)	127	0.057	0.742	413,600	405,400	0.0268	0.0273	1,300
Bunch-lay-stranded:								
9(90)	90	.010	.120	9,045	8,864	1.22	1.28	28
14(140)	140	.010	.145	14,070	13,790	.786	.823	43
Rope-lay-stranded:								
23(228)	228	.010	.190	22,910	22,460	.499	.523	73
42(209)	209	.014	.260	42,100	41,280	.272	.284	130
60(304)	304	.014	.310	61,260	60,040	.187	.196	190
83(418)	418	.014	.380	84,230	82,560	.136	.142	270
133(684)	684	.014	.480	137,800	135,100	.0830	.0867	440
150(760)	760	.014	.510	153,100	150,100	.0747	.0780	490
200(998)	998	.014	.580	199,100	195,100	.0575	.0600	630
250(1254)	1254	.014	.680	252,700	247,700	.0453	.0472	800
400(2052)	2052	.014	.850	413,500	405,300	.0277	.0289	1,300
800(4033)	4033	.014	1.150	812,700	796,500	.0141	.0148	2,600

1/ Valves are for information only.

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3.4.1.1 Concentric-lay-stranded. The length and direction of lay and the type and number of joints in concentric-lay-stranded conductors shall be in accordance with ASTM B 8 or B 286, as applicable.

3.4.1.2 Bunch-stranded. The length and direction of lay and the type and number of joints in bunch-stranded conductors shall be in accordance with ASTM B 174.

3.4.1.3 Rope-lay-stranded. The length and direction of lay and the type and number of joints in rope-lay-stranded conductors shall be in accordance with ASTM B 172 or B 173, as applicable.

3.4.2 Separators. Separators employed directly over conductors shall be applied to give not less than 100 percent coverage to the conductors.

3.4.3 Insulation. Where the thickness of conductor insulation is specified on the applicable specification sheet as a nominal value, the minimum thickness measured at any cross section shall be not less than 90 percent of the value specified as nominal on the specification sheet.

3.4.3.1 Extruded insulation. Extruded insulation shall be applied concentrically to the dimensions required by the specification sheet.

3.4.3.2 Taped insulation. Tapes used as insulation shall be applied in such a manner that they lie smoothly and free from wrinkles. The tape width shall be proper for the diameter over which it is applied so as to minimize splits, creases and edge tears when the completed cable is subjected to bending. Registrations shall be held to a minimum.

3.4.3.2.1 Silicone rubber-glass tapes. Silicone rubber-glass tapes shall be held from unwinding from conductors when the conductors are exposed and subjected to normal handling during splicing and terminating. The tapes may be held from unwinding by the use of self-bonding tapes, by the use of a silicone rubber adhesive, or by using self-bonding or heat sealable polyester tapes over the insulating tapes. Voids within the insulating wall shall be filled with a compound so as to exclude air and moisture and prevent the passage of water through the insulation. Any adhesive or sealing compound shall have no deleterious effect on the other cable parts during tests or during the normal service life of the cable.

3.4.4 Glass braids. Glass braids for use on extruded silicone rubber insulation shall have a thickness of 0.006 ± 0.005 inch. The braid angle shall be such that the completed cable shall meet all inspection and test requirements.

3.4.5 Braid covering (over glass braid). When a specification sheet requires a covering over the glass braid, the covering shall consist of a layer or layers of a black material that provides high resistance to moisture, abrasion and hydraulic fluids and also provides a smooth surface for printed identification marking.

3.4.6 Fillers. Fillers shall be used to provide firmness and roundness of completed cables or cords and to provide watertightness when required by the specification sheet.

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3.4.7 Binder. A binder shall be employed over the assembled core of cables or cords to provide a firm core and to serve as a barrier between the core and cable or cord jacket.

3.4.8 Shields. The material and construction of the shields shall be as specified in the specification sheet.

3.4.8.1 Braided wire shield. Braided wire shield shall have the angle of application and the percent coverage as required in the specification sheet. The percent coverage and the angle of application shall be determined by the following formula:

$$\text{Percent coverage } (k) = 100\ (2F - F^2)$$

Where $F = NPd/\sin a$

$\tan a = 2 \times 3.14159\ DP/C$

a = acute angle of braid with axis of cable or cord
d = diameter (inch) of individual braid wires
D = diameter (inch) of cable under braid
N = number of wires per carrier
C = number of carriers
P = picks per inch of cable or cord length.

3.4.8.2 Double shield. Double shields shall have the angle of application and the percent coverage as required in the specification sheet. The percent coverage and the angle of application shall be determined by the formula specified in 3.4.8.1. The material and percent of separator tape coverage between double shields shall be as required by the applicable specification sheet.

3.4.9 Identification codes and methods. Individual conductors and groups of conductors shall be separately identified. The applicable identification code and the method by which the code is applied shall be as specified in the specification sheet.

3.4.9.1 Identification codes.

3.4.9.1.1 Standard identification code. Standard identification code shall be in accordance with table IX.

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TABLE IX. Standard identification code.

Color, conductor or group number	Background or base color	First tracer color	Second tracer color
1	Black	---	---
2	White	---	---
3	Red	---	---
4	Green	---	---
5	Orange	---	---
6	Blue	---	---
7	White	Black	---
8	Red	Black	---
9	Green	Black	---
10	Orange	Black	---
11	Blue	Black	---
12	Black	White	---
13	Red	White	---
14	Green	White	---
15	Blue	White	---
16	Black	Red	---
17	White	Red	---
18	Orange	Red	---
19	Blue	Red	---
20	Red	Green	---
21	Orange	Green	---
22	Black	White	Red
23	White	Black	Red
24	Red	Black	White
25	Green	Black	White
26	Orange	Black	White
27	Blue	Black	White
28	Black	Red	Green
29	White	Red	Green
30	Red	Black	Green
31	Green	Black	Orange
32	Orange	Black	Green
33	Blue	White	Orange
34	Black	White	Orange
35	White	Red	Orange
36	Orange	White	Blue
37	White	Red	Blue
38	Brown	---	---
39	Brown	Black	---
40	Brown	White	---

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TABLE IX. Standard identification code. - Continued

Color, conductor or group number	Background or base color	First tracer color	Second tracer color
41	Brown	Red	---
42	Brown	Green	---
43	Brown	Orange	---
44	Brown	Blue	---
45	White	Black	Blue
46	Red	White	Blue
47	Green	Orange	Red
48	Orange	Red	Blue
49	Blue	Red	Orange
50	Black	Orange	Red
51	White	Black	Orange
52	Red	Orange	Black
53	Green	Red	Blue
54	Orange	Black	Blue
55	Blue	Black	Orange
56	Black	Orange	Green
57	White	Orange	Green
58	Red	Orange	Green
59	Green	Black	Blue
60	Orange	Green	Blue
61	Blue	Green	Orange
62	Black	Red	Blue
63	White	Orange	Blue
64	Red	Black	Blue
65	Green	Orange	Blue
66	Orange	White	Red
67	Blue	White	Red
68	Black	Green	Blue
69	White	Green	Blue
70	Red	Green	Blue
71	Green	White	Red
72	Orange	Red	Black
73	Blue	Red	Black
74	Black	Orange	Blue
75	Red	Orange	Blue
76	Green	Red	Black
77	Orange	White	Green
78	Blue	White	Green
79	Red	White	Orange
80	Green	White	Orange

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TABLE IX. Standard identification code. - Continued

Color, conductor or group number	Background or base color	First tracer color	Second tracer color
81	Blue	Black	Green
82	Orange	White	—
83	Green	Red	—
84	Black	Green	—
85	White	Green	—
86	Blue	Green	—
87	Black	Orange	—
88	White	Orange	—
89	Red	Orange	—
90	Green	Orange	—
91	Blue	Orange	—
92	Black	Blue	—
93	White	Blue	—
94	Red	Blue	—
95	Green	Blue	—
96	Orange	Blue	—
97	Yellow	—	—
98	Yellow	Black	—
99	Yellow	White	—
100	Yellow	Red	—
101	Yellow	Green	—
102	Yellow	Orange	—
103	Yellow	Blue	—
104	Black	Yellow	—
105	White	Yellow	—
106	Red	Yellow	—
107	Green	Yellow	—
108	Orange	Yellow	—
109	Blue	Yellow	—
110	Black	Yellow	Red
111	White	Yellow	Red
112	Green	Yellow	Red
113	Orange	Yellow	Red
114	Blue	Yellow	Red
115	Black	Yellow	White
116	Red	Yellow	White
117	Green	Yellow	White
118	Orange	Yellow	White
119	Blue	Yellow	White
120	Black	Yellow	Green

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TABLE IX. Standard identification code. - Continued

Color, conductor or group number	Background or base color	First tracer color	Second tracer color
121	White	Yellow	Green
122	Red	Yellow	Green
123	Orange	Yellow	Green
124	Blue	Yellow	Green
125	Black	Yellow	Blue
126	White	Yellow	Blue
127	Red	Yellow	Blue

3.4.9.1.2 Telephone identification code. Conductor identification code for telephone cables shall be as follows:

Color or conductor no.	Color	Color or conductor no.	Color
1	Black	7	Brown
2	White	8	Gray
3	Red	9	Yellow
4	Green	10	Purple
5	Orange	11	Tan
6	Blue	12	Pink

The pairing of conductors for forming pairs shall be as follows:

No. 1 paired with nos. 2 thru 12 for first 11 pairs
No. 2 paired with nos. 3 thru 12 for next 10 pairs
No. 3 paired with nos. 4 thru 12 for next 9 pairs
No. 4 paired with nos. 5 thru 12 for next 8 pairs
No. 5 paired with nos. 6 thru 12 for next 7 pairs
No. 6 paired with nos. 7 thru 12 for next 6 pairs
No. 7 paired with nos. 8 thru 12 for next 5 pairs
No. 8 paired with nos. 9 thru 12 for next 4 pairs
No. 9 paired with nos. 10 thru 12 for next 3 pairs
No. 10 paired with nos. 11 thru 12 for next 2 pairs
No. 11 paired with no. 12

3.4.9.1.3 Special identification code. Special identification code shall be in accordance with the following:

Color or conductor no.	Color	Color or conductor no.	Color
1	Black	7	Brown
2	White	8	Gray
3	Red	9	Yellow
4	Green	10	Purple
5	Orange	11	Tan
6	Blue	12	Pink

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3.4.9.1.4 Letter identification code. Letter identification code shall consist of the letters A, B, C and D printed in block type and with black, white, red and green ink, respectively.

3.4.9.1.5 Twisted pair identification code. This code shall consist of numbers in sequence running from 1 through the number corresponding to the total quantity of twisted pairs in the cable. Both conductors in each pair shall be numbered the same, denoting the sequence number of the pair. Distinction between the two conductors is provided by different colored insulation. Conductors of a cable with a single pair need not be numbered.

3.4.9.1.6 Twisted triad identification code. This code shall consist of numbers in sequence running from 1 through the number corresponding to the total quantity of twisted triads in the cable. Three conductors shall be numbered the same, denoting the sequence number of the triad; distinction between the three conductors is provided by different colored insulation. Conductors of a cable with a single triad need not be numbered.

3.4.9.2 Identification methods.

3.4.9.2.1 Method 1. Identification method 1 shall be surface printing of both number and color designations. The legend shall be printed in contrasting color: preferably white ink on black or dark background or black ink on white or light background. The printing can be on the conductor insulation provided that the jacket is transparent or on the jacket if the jacket is not transparent. The legend shall be repeated at intervals not exceeding 3 inches and alternate legends shall be inverted. For example: 10-ORANGE-BLACK 10-ORANGE-BLACK. The character type shall be block or italic and shall have a height in accordance with the diameter over which it is applied as follows:

Diameter range (inch)	Height of character approximate (inch)
0.045 to 0.070	0.025
.070 to .095	.031
.095 to .115	.047
.115 to .200	.062
.190 to .250	.078
.235 to .325	.094
.330 and larger	.125

3.4.9.2.2 Method 2. Identification method 2 shall be the use of opaque white polyester tapes which have been printed with both the number and color designation prior to application. The legend shall be printed with black ink and shall be repeated at intervals not exceeding 3 inches and alternate legends shall be inverted. The character type shall be block and shall be approximately 3/32-inch high.

3.4.9.2.3 Method 3. Identification method 3 shall be the use of solid base colors or solid base colors with tracers as required. The base color may be either the color of the insulation or the color of a coating applied to the insulation. The tracers shall be approximately 1/32 inch wide ink stripes of the required color applied helically with 1-1/2 ± 1/4-inch lay. If two tracers are required, the second tracer shall be half the width of the first tracer.

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3.4.9.2.4 Method 4. Identification method 4 shall be the use of colored braids. Tracers shall consist of the required colors applied by three adjacent carriers. Where two tracers are required, they shall be applied with reverse lay.

3.4.9.2.5 Method 5. Identification method 5 shall be the use of the printed letter on the outermost insulating tape or the printed letter on a polyester binder tape over the insulating tapes. The letters shall be approximately 3/16-inch high and shall have been printed at intervals not exceeding 3 inches prior to the application of the tape to the conductor. If the insulating tapes are white, no printing is required on the B (white) conductor.

3.4.9.2.6 Method 6. Identification method 6 shall consist of numerals printed in ink on the conductor insulation. For conductors having a jacket directly over the insulation, the numerals may be printed in ink on the jacket, at the manufacturer's option. White ink shall be used for a red or black background; black ink shall be used for a white background. Numerals shall be upright on the longitudinal axis of the conductor (see figure 1). Numeral width shall be proportional to conductor outside diameter (od) as follows:

Diameter range (inch)	Height of character approximate (inch)
0.045 to 0.095	0.025
0.096 to 0.120	.047
0.121 to 0.175	.063
0.176 to 0.330	.094
0.331 and larger	.125

Numeral height shall be 2-1/2 to 3 times numeral width. Each numeric legend shall be underlined. Two-digit legends shall have the bottom numeral underlined. Legends shall be alternately inverted and shall be repeated at intervals not greater than 1-1/2 inch.

3.4.10 Manufacturer's identification tape. Unless otherwise indicated on the specification sheet, cables and cords shall contain a continuous, thin, moisture-resistant marker tape, not less than 1/10-inch wide. The marker tape shall be placed directly under the cable or cord binder tape or jacket. The tape shall be printed to show the following information at intervals not greater than 1 foot: Name and location of manufacture; year of manufacture; specification number (MIL-C-915); and progressive serial number. The serial number is not necessarily a footage marker. A serial number shall not be repeated by a manufacturer in any 1 year for any one type and size of cable or cord.

3.4.10.1 Marker threads. The use of marker threads in lieu of the manufacturer's identification tape, as specified in 3.4.10, in small cables or cords, shall have written approval of NAVSEA. If approved, marker threads shall be in accordance with those assigned by Underwriter's Laboratory (UL) Incorporated to indicate the manufacturer of the cable or cord. When UL marker threads are used, yearly marker threads shall also be used to indicate the year of manufacture, as follows:

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Year	Color codes: Background or base color	Color codes: Color of tracer
1981	Blue	Black
1982	Orange	Slate
1983	Orange	Green
1984	Orange	Brown
1985	Orange	Black
1986	Orange	Pink
1987	White	Yellow
1988	White	Slate
1989	White	Green
1990	White	Brown
1991	White	Black

3.4.11 Cable or surface marking. Unless otherwise specified (see 3.1), where overall cable or cord jacket surface marking is specified by the specification sheet, ink marking shall be used. The legend shall be printed in contrasting color: preferably white ink on a dark background or black ink on a light background. When indented or embossed marking is specified by the specification sheet, the depth shall be not greater than 0.020 inch. The legend shall consist of the manufacturer's name, the cable or cord type and size designation, when applicable, the type of jacket material and the year of manufacture. The legend shall be repeated at intervals not exceeding 1 foot and the year of manufacture need not be in line with the balance of the legend.

For example: "Manufacturer's name - MWP-24-NEOP - 1986"

The character type may be either block or italic and shall have a minimum height in accordance with the diameter over which it is applied as follows:

Diameter range (inch)	Height of character approximate (inch)
0.125 to 0.200	0.047
.200 to .285	.063
.285 to .350	.078
.350 to .500	.094
.500 and larger	.125

Jacket material names shall be abbreviated as follows:

Polychloroprene	-	NEOP
Standard thermoplastic	-	STD PVC
Special thermoplastic	-	SPL PVC
Chlorosulfonated polyethylene	-	CSPE

3.4.12 Watertightness. Where watertight cable construction is specified (see 3.1), voids within the cable or cord construction shall be filled to prevent the passage of water longitudinally through the cable or cord.

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3.4.13 Jacket reinforcement. Where a reinforced polychloroprene jacket is specified (see 3.1), the jacket shall be applied in two layers with a size 3 nylon cord applied as a braid or two reverse serves between the layers. The minimum thickness of the outer layer of jacket shall be 50 percent of the total. The reinforcement shall be applied with approximately four picks or crossovers per inch.

3.4.14 Braided armor. The armor shall be applied in the form of a braid. The weave of the braid shall be of either the "one over-one under" or the "two over-two under" type. The weave shall be such that the wires of a carrier are laid closely together, flat and parallel. The braid shall be applied with maximum tension possible so as to prevent loosening and creeping but not to cause broken ends. The braid shall remain snug to the cable jacket and not spring away when cut. Splices in the braid strands shall be infrequent, staggered and inconspicuous so as not to increase the nominal diameter of the cable or result in rough spots. The selection of the number of wires per carrier and the number of carriers per braider shall be such as to produce a basket weave to give a minimum of 88 percent coverage and a braid angle within the limits as specified in table X. The number of wires per carrier shall be not more than as specified in table XI.

TABLE X. Armor braid angle.

Diameter over jacket (inches)	Braid angle	
	Minimum (degrees)	Maximum (degrees)
0.100 - 0.600	30	60
.601 - 1.000	35	60
1.001 - 1.500	40	70
1.501 - 2.000	45	70
2.001 - 3.000	50	80

TABLE XI. Wires per carrier.

Diameter under braid (inches)	Maximum number of ends per carrier	
	One over-one under	Two over-two under
0.100 - 0.400	8	5
.401 - .800	12	10
.801 - 3.000	16	10

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3.4.14.1 Coverage. The coverage shall be as determined by the following formula:

$$\text{Percent coverage } (K) = 100\,(2F - F^2)$$

Where $F = NPd/\text{Sin } a$

$\text{Tan } a = 2 \times 3.14159 \times DP/C$

a = acute angle of braid with axis of cable
d = diameter (inch) of individual braid wires
D = diameter (inch) of cable under braid
N = number of wires in carrier
C = number of carriers
P = picks per inch of cable length

3.4.15 Armor paint. Aluminum paint shall be applied to the armor as soon as practicable after completion of manufacturing operations so as to prevent accumulation of dirt, moisture and foreign matter on the braid or in the interstices. The paint shall be properly dried and shall not be tacky. The painting method (immersion method preferred) shall be such that all interstices to the outer cable surface are filled. The paint finish shall provide a relatively smooth overall coating that will facilitate making watertight glands with stuffing tubes and packing. The coating shall be flexible and adhere firmly so that it will not crack, peel or flake off.

3.4.16 Dimensional tolerances. Where minimum or maximum dimensions, or both, are specified, no minus or plus tolerances, respectively, shall be permitted. Where a dimension is specified as nominal, the average dimension shall be not less than the specified nominal. Where no minimum overall cable or cord diameter is specified, the minimum permissible diameter shall be not less than 92-1/2 percent of the specified maximum overall cable or cord diameter. Where a specification sheet gives the maximum overall diameter for unarmored cables and the cable is acquired with armor, the maximum overall diameter shall be increased by 0.050 inch.

3.4.16.1 Extruded and taped insulation wall thickness. For conductor insulation wall thickness specified as nominal, the average thickness shall be not less than the specified nominal. The minimum thickness, measured at any cross section, shall be not less than 90 percent of the specified nominal.

3.4.16.2 Cable or cord jacket thickness. The average thickness of a cable or cord jacket measured at any cross section shall be not less than the specified nominal. In case of multiconductor cables or cords, the jacket thickness shall be determined from the measurements made at the high point of each conductor taken on a line through the center of the cable or cord and through the center of each conductor in the outer layer. The minimum thickness at any cross section shall be not less than 80 percent of the specified nominal.

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3.4.17 Centering and circularity.

3.4.17.1 Insulation. The insulation on the individual conductors shall be uniform in diameter throughout the conductor length. At any cross section, the maximum wall thicknesses shall not exceed the minimum by more than 25 percent for specified thickness greater than 0.025 inch, nor by more than 40 percent for specified thicknesses of 0.025 inch and less.

3.4.17.2 Cable or cord jacket. The cable or cord jacket shall be applied concentrically to the cable or cord core in a manner to maintain circularity in the completed cable or cord. The maximum wall thickness of the jacket at any cross section shall not exceed the minimum by more than 66 percent.

3.4.18 Surface condition (cable or cord jacket). The surface of the cable or cord jacket of unarmored cables or cords shall be dry and free of any coating, film or treatment which would tend to interfere with the bonding to it of encapsulating or molding materials normally used in splicing and terminating.

3.5 Electrical properties. Electrical properties of the completed cable or cord shall be as specified by the specification sheet.

3.6 Physical properties. Physical properties of the completed cable or cord and cable or cord components shall be as specified by the specification sheet.

3.7 Repair of insulation or cable or cord jacket. Repair of cable or cord jacket will not be permitted on unarmored cables which are required to pass the hydrostatic (open end) test. Repair of cable or cord jacket will be permitted on cables or cords which are not required to pass the hydrostatic (open end) test, provided the materials and techniques used are such that the finished cable or cord complies with all the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with the data ordering documents (see 6.2.2).

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4.2 Classification of inspection. The examination and testing of all cable and cord covered by this specification shall be classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Conductor strand inspection (see 4.5).

4.3 Qualification inspection. Qualification inspection shall consist of the examination and tests specified in tables XII and XIV and in accordance with the applicable specification sheet.

TABLE XII. Qualification inspection.

Examination or test	Requirement paragraph	Test method paragraph
Accelerated service	Specification sheet	4.7.2
Cable aging and compatibility (95°C)	Specification sheet	4.7.7
Cable aging and compatibility (125°C)	Specification sheet	4.7.8
Cold working (minus 20°C)	Specification sheet	4.7.12
Cold working (minus 54°C)	Specification sheet	4.7.13
Durometer hardness (jacket)	Specification sheet	4.7.16
Flame propagation	Specification sheet	4.7.17
Gas flame (3 hour)	Specification sheet	4.7.19
Hydrostatic (open end)	Specification sheet	4.7.21
Pressure cycling	Specification sheet	4.7.25
Specific gravity	Specification sheet	4.7.26
Stress endurance	Specification sheet	4.7.27
Insulation resistance	Specification sheet	4.8.5

4.3.1 Qualification specimens. A manufacturer seeking qualified products listing shall manufacture and be responsible for testing at least one specimen of cable or cord of each type, size or group for which qualification is sought. The cable or cord type and size required for group qualification shall be in accordance with table XIII. When a manufacturer desires qualification of an individual type or size, the selection of test specimens shall be subject to review by NAVSEA. A sufficient length of any test specimen shall be manufactured at one time. The groups are based on similar characteristics and requirements. Qualification will apply for both armored and unarmored cables.

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TABLE XIII. Qualification samples and groups.

Group number	Qualification test: Additional cable types and sizes for flame propagation test only	Qualification test: All tests specified in the applicable specification sheet (type and size)	Types comprising the group 1/
1	None	DLT	DLT
2	None	DRW	SRW, DRW, TRW
3	None	DSS-3	All sizes of DSS, TSS, FSS, 5SS and 7SS
4	None	DSWS	DSWS
5	None	ECM	ECM, MSP and MSPW
6	None	JAS	JAS
7	None	MCOS-6	All sizes of MCOS
8	None	MCSF	MCSF
9	MDU-6	MDU-23	All sizes of MDU
10	MDY-6	MDY-23	All sizes of MDY
11	None	MHOF-30	All sizes of MHOF
12	None	MMOP	All sizes of DCOP, TCOP and MMOP
13	MNW-44	MNW-10	All sizes of MNW
14	MSCU-44 and TSCU-14	MSCU-10	All those sizes of DSCU, TSCU, FSCU, MSCU, 7SCU, TCJU 2/ and TCTU which have conductor sizes 23 and smaller
15	None	MWP-24	All sizes of MWP
16	PBTMU-15	PBTMU-15	All sizes of PBTMU and TSP
17	TPS-9	PI-12	All sizes of PI, DPS, TPS, FPS and 7PS
18	None	S2S	S2S
19	TCTX-7	TCTX-7	All sizes of TCJX, TCKX and TCTX
20	None	THOF-9	Sizes 42 and smaller of types DHOF, THOF and FHOF, except type-and-size THOF-14
21	None	THOF-150	Sizes 60 and larger of types DHOF, THOF and FHOF; type-and-size THOF-14; type GVSF
22	TNW-14, and TNW-50	TNW-150	All sizes of DNW, TNW, FNW, 4NWB, and 8NW6
23	None	TPNW-20	All sizes of TPNW
24	TSCU-50	TSCU-150	All those sizes of DSCU, DSCU, TSCU, FSCU and 6SCU 3/ which have conductor sizes 23 and larger
25	None	TTOP-10	All sizes of TTOP
26	None	TTBS-8	All sizes of TTBS

1/ See footnotes at end of table.

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TABLE XIII. Qualification samples and groups. - Continued

Group number	Qualification test		Types comprising the group[1/]
	Additional cable types and sizes for flame propagation test only	All tests specified in the applicable specification sheet (type and size)	
27	TTSU-10	TTSU-10	All sizes of TTSU
28	MU	MS	All sizes of MS, MU, 2SJ, 3SJ and 4SJ
29	None	1S50MU-16	All sizes of 1SAU, 1SMU, 1S50MU, 1S75MU and 1SU
30	None	2AU-40	All sizes of 2AU, 2U, 2SU, 3SU, 3U and 2CS
31	None	2SWAU-10	All sizes of 1SMWU, 1SWU, 2SWAU, 2SWU, 2WAU, 3SWU and 2SWL
32	None	2SWF-7	All sizes of 1SWF, 2SWF and 3SF
33	5KVTSGU-100	5KVTSGU-100	All sizes of 5KVTSGU
34	None	2UW-42	2UW
35	None	TPUM	All sizes of TPUM
36	None	1PR-A20E	1PR-A20E
37	None	2SPR-16	1PR-16, 7PR-16, 2SPR-16, 3PR-16, 1Q-16, 1TR-16, 7SPR-16S and 1SPR-16

1/ Includes all sizes and variations of the cable type.
2/ Size 23 with extruded silicone rubber only.
3/ Size 23 with silicone rubber glass tape only.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed on all completed cable or cord in accordance with the procedures specified herein. This inspection shall consist of basic electrical test plus groups A, B and C examination and tests specified in table XIV and in accordance with the specification sheets.

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TABLE XIV. Quality conformance inspection.

Examination or test	Requirement paragraph	Test method
Basic electrical		
Conductor resistance	Specification sheet	4.8.4
Insulation resistance	Specification sheet	4.8.5
Voltage withstand	Specification sheet	4.8.8
Insulation dielectric	Specification sheet	MIL-W-81381
Group A		
Abrasion resistance	Specification sheet	4.7.1
Crack resistance	Specification sheet	4.7.14
Hydrostatic (open end)	Specification sheet	4.7.21
Thermoplastic flow	Specification sheet	4.7.30
Visual and dimensional	Specification sheet	4.7.32
Watertightness	Specification sheet	4.7.33
Capacitance	Specification sheet	4.8.2
Characteristic impedance	Specification sheet	4.8.3
Conductor tensile and elongation	Specification sheet	Method 3212, FED-STD-228
Group B		
Bending endurance	Specification sheet	4.7.4
Breaking strength	Specification sheet	4.7.5
Cold bending cable	Specification sheet	4.7.10
Cold bending cord	Specification sheet	4.7.11
Drip	Specification sheet	4.7.15
Durometer hardness (insulation)	Specification sheet	4.7.16
Gas flame (1 hour)	Specification sheet	4.7.19
Physical tests on insulation and jacket (unaged)	Specification sheet	4.7.24
Tear strength	Specification sheet	4.7.28
Tension	Specification sheet	4.7.29
Twisting endurance	Specification sheet	4.7.31
Attenuation	Specification sheet	4.8.1
Group C		
Armor (abuse)	Specification sheet	4.7.3
Cable aging (260°C)	Specification sheet	4.7.6
Cable filler removability	Specification sheet	4.7.9
Flammability	Specification sheet	4.7.18
Heat distortion	Specification sheet	4.7.20
Permanence of printing		
Conductor insulation	Specification sheet	4.7.22
Jacket	Specification sheet	4.7.23
Physical tests on insulation and jacket (aged)	Specification sheet	4.7.24
Mutual inductance	Specification sheet	4.8.6
Pulse response time	Specification sheet	4.8.7
Steel strength member	Specification sheet	MIL-W-83420

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4.4.1 Inspection lot. An inspection lot shall consist of the total number of units of product of any one type, size, and construction manufactured under the same conditions and offered for delivery at one time.

4.4.2 Basic electrical tests. Basic electrical tests shall consist of voltage withstand, insulation resistance and conductor resistance. These tests shall be performed on each length of completed cable or cord.

4.4.3 Sampling procedure. The required number of samples for groups A, B and C examination and tests shall be selected at random from the inspection lot. Nonconforming starting and finishing ends of completed cable or cord shall be removed by the manufacturer prior to selecting samples.

4.4.3.1 Sampling for group A examination and tests. Samples for group A examination and tests shall be selected from each lot in accordance with table XV.

TABLE XV. Sampling for group A examination and tests.

Units of product in lot	Number of samples	Acceptance number defective units	Rejection number defective units
10 and under	2	0	1
11 thru 30	3	0	1
31 thru 90	7	0	1
91 thru 210	15	1	2
211 and over	25	2	3

4.4.3.2 Sampling for group B tests. Samples for group B tests shall be selected from each lot in accordance with table XVI.

TABLE XVI. Sampling for group B tests.

Units of products in lot	Number of samples
8 and under	1
9 thru 30	2
31 thru 90	3
91 thru 210	4
211 and over	5

4.4.3.3 Sampling for group C tests. Samples for group C tests shall be selected in accordance with table XVII.

TABLE XVII. Sampling for group C tests.

Two months' production (units of product)	Number of samples
8 and under	1
9 thru 30	2
31 thru 90	3
91 thru 210	4
211 and over	5

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4.4.3.4 Tightened sampling and accept-reject criteria. If the number of defective units in a lot exceeds the acceptance number for group A examination and tests (see table XV), or if there are any failures in group B tests, the manufacturer shall have the option of reworking the lot and performing examination and tests on the tightened sampling basis in accordance with table XVIII. If there are any failures on tests performed on the tightened sampling basis, if there are any failures in group C tests, or if the manufacturer chooses not to rework and retest, the lot shall be rejected.

TABLE XVIII. Tightened sampling.

Units of product in lot	Number of samples
10 or under	3
11 thru 30	5
31 thru 90	9
91 thru 210	15

4.4.4 Group A examination and tests. Group A examination and tests as required by the specification sheet shall be performed on samples selected in accordance with 4.4.3.1.

4.4.5 Group B tests. Group B tests as required by the specification sheet shall be performed on specimens taken from samples selected in accordance with 4.4.3.2.

4.4.6 Group C tests. Group C tests as required by the specification sheet shall be performed on specimens taken from samples selected in accordance with 4.4.3.3.

4.4.7 Certified test reports. The contractor shall prepare test reports in accordance with the data ordering document (see 6.2.2). The test report shall contain the following information:

(a) Manufacturer's QPL number and date or serial number and date of NAVSEA letter of approval.
(b) A statement to the effect that the product was constructed from materials listed on the manufacturer's approved details of construction sheet.
(c) A statement that the product meets all of the requirements of this specification.
(d) Results of all quality conformance tests showing actual values obtained.
(e) Year and month cable or cord was manufactured.
(f) Serial numbers of the marker tape taken from each end of each length cable or cord.
(g) Customer's name and contract or order number.
(h) The supplies contain no metallic mercury or mercury compounds and are free from mercury contamination.

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4.5 Conductor strand inspection. Unless otherwise specified (see 3.1), test procedures shall be in accordance with the applicable ASTM standard referenced in 3.3.1. Before individual strands are assembled into a conductor, they shall be tested for coating, when applicable, tensile strength and elongation. Lot sizes and number of specimens per lot shall be as follows:

Diameter of individual strand (inch)	Lot size	Number of test specimens
0.003 to .013	2500 pounds or less	2
	Each additional 2500 pounds or fraction thereof	1
.014 to .040	5000 pounds or less	2
	Each additional 5000 pounds or fraction thereof	1
.041 to .129	10,000 pounds or less	2
	Each additional 10,000 pounds or fraction thereof	1

Records of tests shall be kept, and copies of the applicable records shall become a part of the test report for finished cable or cord.

4.6 Test conditions. Unless otherwise specified (see 3.1), the examination and tests specified in tables XII and XIV shall be made at standard ambient conditions as follows:

(a) Temperature: 23 ± 10°C
(b) Humidity: 50 ± 30 percent

4.7 Test methods (physical).

4.7.1 Abrasion resistance. The ability of glass braid coverings and polypropylene insulation to withstand repeated abrasion shall be determined.

4.7.1.1 Specimens. Each specimen shall consist of a single insulated conductor (including insulation covering, if any) of not less than 22 inches in length. Specimens shall be removed from completed cable or cord as follows:

(a) Cables or cords containing four or fewer conductors - one specimen shall be removed from each insulated conductor.
(b) Cables or cords containing more than four but fewer than 21 conductors - one specimen shall be removed from each of four different insulated conductors (a total of four specimens).

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(c) Cables or cords containing 21 or more conductors - the number of specimens removed shall be equal to the square root (rounded to the nearest whole number) of the total number of conductors. Each specimen shall be removed from a different insulated conductor.

4.7.1.2 Special apparatus. Apparatus shall include the following:

(a) Abrading machine - A cylinder, which incorporates two abrading elements on its surface, which is motor driven to rotate about a horizontal axis at 17 plus 3, minus 2 revolutions per minute (r/min) and over which the specimen is draped, as shown on figures 2 and 3, and as described below:

(1) Each abrading element shall consist of 5/16 inch square (in^2), high-speed tool bit (Cleveland Twist Drill Company number 855 or its equivalent) which has been ground on two adjacent, longitudinal sides to produce a single, sharp 90 degree longitudinal edge, free of visible nicks. A medium grade grinding wheel (Norton Company Number 39C6018VK or its equivalent) shall be used for this grinding. Abrading elements shall be reground as required; in no case shall an abrading element be used in excess of 5 hours without regrinding. Whenever the perpendicular distance between either pair of opposite longitudinal sides of an abrading element becomes less than 0.3085 inch (as by repeated regrinding) then that abrading element shall be discarded and replaced.

(2) The cylinder shall be 8 ± 1/3 inches in diameter and shall be rigidly fabricated from metal.

(3) Each of the two abrading elements shall be maintained in electrical contact with the metal cylinder by securing each in a notch cut into the cylinder surface, parallel to the cylinder axis; the two notches required shall be spaced 180 ± 2 degrees apart around the circumference of the cylinder. These notches shall be cut such that the sharpened 90 degree longitudinal edge of each element shall be facing outward from the cylinder surface, and such that the midpoint of a straight line drawn between the two longitudinal edges adjacent to the sharpened edge is tangential plus or minus 0.003 inch to the cylinder surface.

(4) An automatic counter shall be provided to totalize the number of times that the specimen is scraped by abrading elements during the test.

(5) A 1 ± 1/32 pound weight shall be provided for applying tension to the specimen.

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(b) Short circuit monitor - A voltage source of not less than 12 V (either direct current (dc) or root mean square (rms) alternating current (ac)) which can be applied between the specimen conductor and both abrading elements of the abrading machine as shown on figure 2. (The connection to the abrasion elements may be provided by a wiping contact on the cylinder.) A means (such as an electrical relay) shall be provided whereby an electrical connection between either abrading element and the specimen conductor shall automatically stop rotation of the cylinder on the abrading machine (as by removing motor power).

4.7.1.3 Procedure. Specimens shall be wiped with a clean, dry cloth to remove any lubricant or dirt. Each specimen shall then be tested (either simultaneously or one at a time) as follows. Each specimen shall be hung circumferentially over the cylinder of the abrading machine, such that each specimen shall form an arc of not less than 170 degrees around the cylinder for the remainder of the test. One end of each specimen shall then be secured to a fixed surface. The abrading machine weight shall be attached to the opposite specimen end, such that both this specimen end and the weight shall be freely suspended. The automatic counter shall be set initially to zero, and the short circuit monitor shall be applied between both abrading elements on the abrading machine and the conductor of each specimen. The cylinder motor shall then be turned on, allowing the cylinder to rotate beneath each specimen, from secured end to weighted end.

4.7.1.4 Observation. Specimen failure shall be construed if electrical contact occurs between either abrading element and any specimen conductor (as evidenced by cessation of cylinder rotation) prior to completing the specified number of abrasive scrapes (see 3.1).

4.7.2 Accelerated service. The ability of completed cable and cord to withstand electrical current overload shall be determined.

4.7.2.1 Specimen. The specimen shall consist of 15 foot ± 2 inch length of completed cable or cord.

4.7.2.2 Special apparatus. Apparatus shall include the following and separate voltage and current sources shall be used:

(a) Voltage source - a 3-phase, 60 hertz (Hz) adjustable voltage source, as shown on figure 4, to provide adjustable, single-phase test voltages through separate normal-blow fuses or overload relays to each specimen conductor. This voltage source shall be in accordance with the following:

(1) Specimens rated for 1 kilovolt (kV) - the test voltage shall be 450 ± 10 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of 1 ampere (A).
(2) Specimens rated for 5 kV - the test voltage shall be 4160 ± 100 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of 1/4 A.

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The secondary winding voltage rating of any constituent voltage step-up transformers shall be not less than the required test voltage. Each phase of the voltage source shall supply the specimen with a current which shall be not less than twice the fuse or overload relay rating.

(b) Current source - a 3-phase, 60 Hz adjustable voltage source shall be used with three current transformers, as shown on figure 4, to provide three adjustable, single-phase test currents as specified. The secondary winding of each current transformer shall be isolated from electrical ground. Busbar conducting test current without generating significant amounts of heat shall be used to convey the test currents from the current transformers to the specimen conductors.

(c) Electrical metering - instrumentation shall be provided, as shown on figure 4, such that the test voltage and current applied to each specimen conductor shall be measured with an error of not more than plus or minus 2 percent.

(d) Heating chamber - a heating chamber shall be provided for supporting the requirements as specified in 4.7.2.3. This chamber shall have interior surface finishes and dimensions which discourage the reflection of radiant heat (generated by the test currents) onto the specimen.

(e) Thermocouple devices - four thermocouples and an associated temperature indicating device, shall be provided for the use and temperatures as specified in 4.7.2.3. Temperatures shall be measured with an error of not more than plus or minus 1°C.

4.7.2.3 Procedure. The specimen shall be formed into a single, horizontal loop within the heating chamber (to permit free circulation of convection air currents). Air temperature within the chamber shall be maintained at 30 ± 2°C during the test (except in the immediate vicinity of the specimen) and means shall be provided to prevent forced air drafts from directly striking the specimen. The four thermocouples shall be placed on top of the specimen, near its midpoint, and in firm contact with its jacket. These thermocouples shall be spaced longitudinally along the specimen jacket at 2 ± 1/16-inch intervals. On unarmored specimens, firm jacket contact can be assured by the use of a length of string with a weight on each end as this can be hung over the jacket after forming a single wrapping around the jacket and over the thermocouple bead. On armored specimens, firm jacket contact can be assured by opening a small space between the jacket and an armor carrier (as with an icepick), and by inserting the thermocouple bead within this space, afterwards pulling the armor carrier longitudinally to press the thermocouple bead tightly against the jacket. Under no circumstances shall the jacket be cut, or shall wide tapes be used to hold the thermocouple beads since these may cause premature specimen failure due to elevated jacket temperatures (see 4.7.2.4). Two thermocouples shall be used to measure ambient temperature with the chamber. The test voltages and currents shall be simultaneously applied to the specimen as shown on figure 4. If the specimen is armored, an additional test voltage shall be applied by grounding the armor, as shown on figure 4. This shall effectively apply an additional voltage, which is approximately equal to the test voltage divided by the square

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root of 3 between the armor and the specimen conductors. The test currents and voltages shall remain continuously applied to the specimen for a period of not less than 7 hours, unless specimen failure (see 4.7.2.4) occurs prior to the end of this period, then the test shall be terminated prematurely. The following notes shall apply for specimens with more than three conductors:

(a) Unarmored four-conductor specimens - the green insulated conductor shall be isolated from all test voltages and currents.
(b) Armored four-conductor specimens - the green insulated conductor shall be electrically connected to the armor during the test.
(c) Six-conductor specimens (armored or unarmored) - the radically opposed conductors within the core of the specimen shall be electrically connected, at both specimen ends, to effectively form three conductors during the test.

4.7.2.4 Observations. Any of the following shall constitute specimen failure, if occurring prior to the expiration of the 7 hour minimum test period:

(a) The blowing of any fuse or the tripping of any overload relay.
(b) Any thermocouple indication of a jacket temperature in excess of 135°C.
(c) Jacket sagging on an unarmored specimen.
(d) Jacket exudation through the armor of an armored specimen. (Jacket bulging, without actual flow or cutting, is not considered to be exudation.)

4.7.3 Armor. The ability of cable armor wire to withstand abuse during in-service cable installation shall be determined.

4.7.3.1 Specimens. Each specimen shall consist of a single piece of armor wire, of not less than 6 inches in length, which has been removed from a spool of armor wire prior to cable armoring. Each specimen shall be removed from a different spool. Ten specimens shall be required.

4.7.3.2 Special apparatus. Apparatus shall include the following:

(a) Mandrel - a 1/4 ± 1/64 inch diameter steel cylinder, of length and construction for performing the springiness test as specified in 4.7.3.3.
(b) Scale - a spring scale or other type scale, the measurement error of which shall not exceed plus or minus 5 percent for use in the springiness and toughness tests as specified in 4.7.3.3.
(c) Special vise - a special vise, with two parallel edges of 0.030 ± 0.001-inch radius between which a specimen may be clamped to perform the toughness test as specified in 4.7.3.3.

4.7.3.3 Procedure. The two following tests shall be performed, using not fewer than five specimens. The specimen shall be tested one at a time for each test. The same specimen shall not be used for more than one test.

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(a) Springiness - one specimen shall be selected and secured by one end to the mandrel, such that it is both in contact with the curved surface of the mandrel and perpendicular to the mandrel axis. The opposite end of the specimen shall be attached to the scale, and the specimen shall be wrapped circumferentially around the mandrel, by handling the scale only, for not fewer than four complete turns, while maintaining a specimen tension of not less than 100 nor more than 125 grams. Care shall be taken to pull the scale in such a direction that it accurately indicates the specimen tension. The specimen shall then be released from the scale and allowed to uncoil. Any specimen which is kinked at any place other than at its attachment points at either end shall be considered an invalid specimen, and the test shall be repeated using a different specimen. This test shall be successfully repeated four more times (five times total), using a different specimen each time.

(b) Toughness - one specimen shall be selected and secured by one end between the rounded, horizontal and parallel vise edges, such that not less than 2 inches of the specimen extends vertically above the vise. The scale shall then be attached to the free end of the specimen, and a specimen tension of 100 ± 10 grams established by handling the scale only. The specimen shall be bent 90 plus 15, minus 0 degrees alternately in both directions perpendicular to the rounded vise edges, while maintaining this specified scale tension. Not fewer than ten 90-degree bends shall be made unless the specimen breaks sooner. The bending rate shall not exceed fifty 90-degree bends per minute. This test shall be repeated four more times (five times total), using a different specimen each time.

4.7.3.4 Observation. Either of the following shall constitute specimen failure:

(a) Springiness - the od of the coil formed by any of the specimens around the mandrel, as measured at a middle portion of the specimen, exceeds 3/8 inch.

(b) Toughness - any specimen withstands fewer than ten 90-degree bends without breaking.

4.7.4 Bending endurance. The ability of completed cable and cord to withstand repeated and reversing bending motion while subjected to a specified temperature shall be determined.

4.7.4.1 Specimen. The specimen shall consist of a 30 plus 6, minus 0 inch length of completed cable or cord. Specimen conductor ends shall be electrically interconnected such that a single series electrical circuit is formed by the conductors. The two end conductors of this series circuit shall be connected as specified in 4.7.4.3.

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4.7.4.2 Special apparatus. Apparatus shall include the following:

(a) Flexing machine - a device for suspending the specimen vertically and for automatically bending the upper portion of the specimen alternately to left and to right by 90 plus 5, minus 0 degrees from vertical, over rollers, as shown on figure 5 and as further described below.

(1) The specimen shall be suspended and bent by means of a rigid, motor driven arm, which shall initially be vertical, which shall be horizontally pivoted about its lower end, and which shall incorporate a clamp at its upper end for securing the upper portion of the specimen jacket. The distance between the upper end of the specimen jacket and the nearest edge of this clamp shall be not less than 1 inch. When its motor is turned on, the arm shall move at a rate of not more than 14, nor less than 12 cycles per minute, where a cycle is 180 plus 10, minus 0 degrees of clockwise arm travel plus 180 plus 10, minus 0 degrees of counterclockwise arm travel (360 plus 20, minus 0 degrees total travel). The specimen clamp shall be positioned such that when the arm is vertical (straight up), the longitudinal axis of the specimen shall intersect the pivoting axis, and the lower edge of the clamp shall be 8-1/2 $\pm$ 1/2 inches above a horizontal line drawn across the tops of the upper rollers.

(2) The specimen shall be bent over two solid steel cylindrical rollers (the upper rollers as shown on figure 5) of the diameter specified. Unless otherwise specified in the applicable specification sheet: For cables over 1 inch in diameter, the roller shall have a diameter four times the diameter of the specimen; for cables with a diameter between 0.75 inch and 1 inch, roller size shall be 3.5 times the maximum diameter of the cable; and for cables with a diameter less than 0.75 inch, roller size shall be 3 times the maximum diameter of the cable. Both rollers shall have a smooth surface finish on their curving surfaces, and shall be pivoted about their axes and capable of free rotation at all times. They shall be positioned such that their axes are horizontal and parallel and share a common horizontal centerline. The tops of the rollers shall be below the horizontal centerline of the flexing machine arm pivoting axis by a distance equal to one-half of the maximum specified overall cable or cord diameter plus or minus 10 percent. The rollers shall be located equidistantly on opposite sides of the vertical centerline passing through the flexing machine arm pivoting axis. They shall be spaced apart such that their nearest edges are 1/8 $\pm$ 1/32 inch from the specimen passing between them when the flexing machine arm is vertical.

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(3) Two cylindrical rollers (the lower rollers as shown on figure 5), of not less than 3/8-inch diameter, shall be provided to restrain lateral motion of the specimen. Both rollers shall be pivoted about their axes and capable of free rotation at all times. They shall be positioned such that their axes are horizontal centerline. The axes of these rollers shall be 7 ± 1 inch below the axes of the rollers over which the specimen is bent (the upper rollers as shown on figure 5). The lower rollers shall be located equidistantly on opposite sides of the vertical centerline passing through the flexing machine arm pivoting axis; they shall be spaced apart such that their nearest edges are 1/8 ± 1/32 inch from the specimen passing between them when the flexing machine arm is vertical.

(4) A weight shall be provided for applying tension to the specimen. This weight shall be attached to the specimen by means of a second clamp. The lower end of the specimen jacket shall be a distance of not less than 1 inch below the nearest edge of this clamp. The weight shall be chosen such that the:

Specimen tension produced by the clamp plus the weight in pounds = $10\ (3.14159)d^2 \pm (3.14159)\ d^2/2$

Where: d = specified maximum overall cable or cord diameter, in inches.

The clamp and weight shall be attached such that specimen tension is applied vertically downward along the specimen axis.

(5) Both the specimen clamp on the flexing machine arm and the specimen clamp supporting the tensioning weight shall apply radial compression to the specimen that neither the specimen nor any of its internal components will slip in the clamping area during the test.

(6) An automatic counter shall be provided to total the number of bending cycles performed on the specimen during a test.

(7) The specimen shall not come into contact with any piece of machinery at any time during the test, except for the upper and lower rollers, the two specimen clamps, and the circuit continuity monitor.

(b) Circuit continuity monitor is a means for continuously monitoring the electrical continuity of the series through connected specimen conductors, which shall automatically stop the flexing machine arm (as by removing motor power) when a specimen conductor breaks.

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(c) Test chambers are chambers for maintaining the specified air temperature within plus or minus 2°C for accommodating the flexible machine with specimen attached and for supporting the requirements as specified in 4.7.4.3. The chamber air temperature shall be measured in the immediate vicinity of the upper roller of the flexing machine. The chamber shall incorporate a viewing port for observing the specimen under test.
(d) Voltage withstand test apparatus is as specified in 4.8.8.2.

4.7.4.3 Procedure. The flexing machine shall be installed in the test chamber, and the specimen installed in the flexing machine, as shown on figure 5. The circuit continuity monitor shall be connected between the two end conductors of the single series circuit within the specimen. The test chamber doors shall then be closed, and the air temperature within the chamber shall be brought to the specified value within plus or minus 2°C. After the air temperature within the chamber has been maintained at this value for a period of not less than 1 hour, the automatic counter shall be set initially to zero, and the bending endurance test begun by turning on the motor driven flexing machine arm. The specified number of bending cycles shall be performed. At all times during the test, the air temperature within the chamber shall be maintained as specified within plus or minus 2°C, and the test chamber doors shall remain closed.

4.7.4.4 Observation. Either of the following shall constitute specimen failure, except that specimen damage within 2 inches of the location where either specimen clamp had been fastened shall not constitute specimen failure, but shall require a complete retest using a different specimen from the same length of cable or cord:

(a) Stoppage of the flexing machine arm caused by the circuit continuity monitor (indicating specimen conductor breakage), prior to completion of the specified number of bending cycles.
(b) Rupture of the specimen jacket prior to completion of the specified number of bending cycles.

4.7.4.5 Further procedure. If the specimen shows no failure, it shall be allowed to return to room temperature and shall be subjected to the voltage withstand test as specified in 4.8.8, using the specified voltages. Following this, that portion of the specimen which had been repeatedly bent over the upper rollers of the flexing machine shall be dissected, and each of its constituent components visually inspected for deterioration.

4.7.4.6 Further observation. Either of the following shall constitute specimen failure, except that specimen damage within 2 inches of the location where either specimen clamp had been fastened shall not constitute specimen failure, but shall require a complete retest of bending endurance using a different specimen from the same length of cable or cord:

(a) Specimen fails the voltage withstand test (see 4.8.8.4).
(b) Specimen, upon dissection and inspection, exhibits visible distortion or cracking of any specimen component, including strand breakage on any conductor, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable or cord in service.

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4.7.5 Breaking strength. This test shall be to determine the axial tension necessary to break completed cable and cord.

4.7.5.1 Specimen. The specimen shall consist of a piece of completed cable or cord which shall have sufficient length for use in the test as specified in 4.7.5.3. Additional specimens may be required in the event of invalid test results (see 4.7.5.3).

4.7.5.2 Special apparatus. Apparatus shall include a motor driven tensile machine, for applying increasing axial tension to the specimen, which shall automatically indicate within plus or minus 1 percent the maximum axial tension experienced by the specimen prior to rupture. This tensile machine shall incorporate two parallel and opposing, rigid and smooth cylindrical mandrels, between which the specimen can be secured, as specified in 4.7.5.3. The two mandrels shall increase their separation at the uniform rate of 1 ± 1/8 inch per minute during the test by means of the tensile machine motor.

4.7.5.3 Procedure. The ends of the specimen shall be attached to opposing tensile machine mandrels, such that each specimen end is tightly wrapped circumferentially around its mandrel for not fewer than two complete revolutions and then firmly secured, at its extremity, to that mandrel. This is to assure that neither the specimen nor any of its internal components shall slip during the test. Care shall be taken during specimen mounting to assure that the specimen shall be subjected only to axial forces, and not to lateral or torsional forces during the test. The specimen shall be mounted such that it has a free span of not less than 6 inches between its nearest points of contact on the opposing mandrels at all times during the test. The test shall be initiated by starting the tensile machine motor, thereby increasing the mandrel separation and applying increasing tension to the specimen. If specimen rupture occurs in a location other than in the free span between mandrels, then the test result shall be considered invalid, and the test shall be repeated using a different specimen.

4.7.5.4 Observation. Specimen failure shall be construed if the maximum recorded specimen tension is less than that specified (see 3.1).

4.7.6 Cable aging (260°C). This test shall be to determine the ability of armored cable with silicone-rubber insulation and jacketing to withstand accelerated heat aging.

4.7.6.1 Specimen. The specimen shall consist of a 48 ± 1 inch length of completed cable or cord.

4.7.6.2 Special apparatus. Apparatus shall include the following:

(a) A forced-fresh-air circulating oven shall be used for supporting the requirements as specified in 4.7.6.3. The oven air temperature shall be measured in the immediate vicinity of the specimen.
(b) Tensile strength and elongation test apparatus shall be as specified in 4.7.24.2.

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4.7.6.3 Procedure. The specimen shall be bent at the approximate middle of its length to form a symmetrical U shape, such that its inside diameter (id) shall be six times the specified maximum overall cable diameter (see 3.1) plus or minus 1/4 inch. The specimen shall be secured in this position using heat-resistant twine, and shall be freely suspended vertically such that the specimen ends face downwards within the oven. The air temperature within the oven shall be raised in a uniform manner from room temperature to a temperature of 260 ± 10°C, over a period of not more than 3 hours nor less than 2 hours. The oven air temperature shall be maintained at 260 ± 10°C for a period of not less than 24 hours, and shall then be returned to room temperature. Provision shall be made to catch, for later weighing, any material which may fall away from the specimen ends during this test. Following the test, the air within the oven shall be allowed to return to room temperature. The specimen shall then be removed from the oven.

4.7.6.4 Observation. Any of the following shall constitute specimen failure:

(a) The falling away of any material in excess of 1 gram from the specimen ends during the test.
(b) Jacket exudation through the armor of an armored cable. (Jacket bulging, without actual flow or cutting, is not considered to be exudation.)
(c) Jacket shrinkback in excess of 1/8 inch from either end of the specimen.

4.7.6.5 Further procedure. If the specimen shows no failure, its jacket shall be subjected to the tensile strength and elongation tests as specified in 4.7.24.4.

4.7.6.6 Further observation. Specimen failure shall be construed if either the jacket tensile strength or jacket elongation is less than that specified (see 3.1).

4.7.7 Cable aging and compatibility (95°C). This test shall detect any significant degradation due to component incompatibility or prolonged overheating of completed cable and cord containing thermoplastic insulation.

4.7.7.1 Specimens. Two specimens of completed cable or cord shall be required: specimen number 1 shall have a length of 30 ± 3 feet, and specimen number 2 shall have a length of not less than the sum of 12 inches plus 20 times the maximum specified overall cable or cord diameter (see 3.1). Both specimens shall be removed from the same length of cable or cord.

4.7.7.2 Special apparatus. Apparatus shall include the following:

(a) A heating chamber shall support the requirements as specified in 4.7.7.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.
(b) The mandrel shall be rigid, smooth cylinder or partial cylinder with a continuous, curved surface of not less than 180 degrees, of suitable length and construction for the specimen bending

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as specified in 4.7.7.3(b). The mandrel diameter shall be not more than 13 nor less than 11 times the specified maximum overall cable or cord diameter (see 3.1).

(c) Watertightness test apparatus shall be as specified in 4.7.33.2.
(d) Voltage withstand test apparatus shall be as specified in 4.8.8.2.
(e) Insulation resistance test apparatus shall be as specified in 4.8.5.3.

4.7.7.3 Procedure. Both specimens shall be simultaneously subjected to heat aging within the heating chamber after preliminary procedures (a) and (b) have been completed. Heat aging shall be exposed to a temperature of 95 ± 3°C for a continuous period of not less than 400 hours. The chamber air temperature shall be recorded at intervals of not more than 15 minutes during the first hour after attaining the heat aging temperature, at intervals of not more than 1 hour for the following 5 hours, and not less than twice daily thereafter for the duration of heat aging. The interval between consecutive temperature recordings shall not exceed 17 hours at any time during heat aging. The following additional details shall apply for each specimen:

(a) Prior to heat aging, specimen number 1 shall be subjected to the insulation resistance and voltage withstand tests as specified in 4.8.5 and 4.8.8, respectively. The voltage withstand test shall use the voltages specified (see 3.1). The specimen shall then be formed into the largest practicable horizontal coil within the heating chamber, and shall be subjected to heat aging. Following heat aging, the specimen shall be allowed to cool to room temperature, and shall again be subjected to the insulation resistance and voltage withstand test as specified in 4.8.5 and 4.8.8, respectively. The specimen shall then be cut into two samples: one sample (see 4.7.33.1) shall be subjected to the watertightness test as specified in 4.7.33; the other sample (see 4.7.9.1) shall be subjected to the cable filler removability test as specified in 4.7.9.
(b) Specimen number 2 shall be placed within the heating chamber, such that it is both straight and horizontal, and shall then be subjected to heat aging. Following heat aging, the specimen shall be allowed to cool to room temperature. The mandrel shall be secured to a fixed surface, and one end of the specimen shall be secured such that it is both in contact with the curved surface of the mandrel, and also perpendicular to the mandrel center of curvature axis. The opposite end of the specimen shall then be bent circumferentially around the curved mandrel surface through an arc of not less than 170 degrees; bending shall be accomplished at a rate of approximately 20 degrees per second. The specimen shall next be secured to maintain its bent shape (as with a piece of twine, stretched taut and tied between each end of the specimen) and shall be removed from the mandrel. Not less than two-thirds of the bent portion of the specimen shall then be dissected, and each of its constituent components shall be visually inspected for deterioration.

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4.7.7.4 Observation. Any of the following shall constitute specimen failure:

(a) The falling away of any material from either specimen during heat aging.
(b) Jacket sagging on any unarmored specimen.
(c) Jacket exudation through the armor of any armored specimen. (Jacket bulging, without actual flow or cutting, is not considered to be exudation.)
(d) Specimen number 1 fails either of the insulation resistance tests (see 4.8.5.4).
(e) Specimen number 1 fails either of the voltage withstand tests (see 4.8.8.4).
(f) The indicated sample from specimen number 1 fails the watertightness test (see 4.7.33.4).
(g) The indicated sample from specimen number 1 fails the cable filler removability test (see 4.7.9.3).
(h) Specimen number 2, upon dissection and inspection, exhibits visible deterioration of such a nature or extent as to impair the performance of the cable or cord in service. Deterioration sufficient to constitute specimen failure shall include, but shall not be limited to: distortion or cracking of any constituent component, hardening of filler material, or any discoloration indicating material incompatibility.

4.7.8 Cable aging and compatibility (125°C). This test shall detect any significant degradation due to component incompatibility or prolonged overheating of completed cable and cord containing thermosetting or silicone-rubber and glass-tape insulation. This test permits either of two methods (the current overload method or the hot air oven method) to be used for heat aging.

4.7.8.1 Specimens. Specimens for cable aging and compatibility tests shall be as follows:

(a) For the current overload method, the specimen shall consist of a 40 ± 4 foot length of a completed cable or cord.
(b) For the hot air method, two specimens of completed cable or cord shall be required: specimen number 1 shall have a length of 30 ± 3 feet; specimen number 2 shall have a length of not less than the sum of 12 inches plus 20 times the maximum specified overall cable or cord diameter (see 3.1). Both specimens shall be removed from the same length of cable or cord.

4.7.8.2 Special apparatus. Apparatus shall include the following:

(a) The oven (required for either heat aging method) shall support the requirements as specified in 4.7.8.3 as applicable. The oven air temperature shall be measured in the immediate vicinity of the specimen.

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(b) The mandrel (required for either heat aging method) shall be rigid, smooth cylinder or partial cylinder, with a continuous, curved surface of not less than 180 degrees, of suitable length and construction for the specimen bending as specified in 4.7.8.5. The mandrel diameter shall be not more than 13 nor less than 11 times the specified maximum overall cable or cord diameter (see 3.1).

(c) Voltage withstand test apparatus (required for either heat aging method) shall be as specified in 4.8.8.2.

(d) Insulation resistance test apparatus (required for either heat aging method) shall be as specified in 4.8.5.3.

(e) Thermocouple device (required for the current overload method if submethod 4.7.8.3(a)(1) is used) and an associated temperature indicating device shall be provided for the use and temperatures specified in 4.7.8.3(a). Temperatures shall be measured with an error of not more than plus or minus 1°C.

(f) Current source (required for the current overload method only) shall be an adjustable source of either dc or 60 Hz single-phase ac, for maintaining specimen conductors at an elevated temperature as specified in 4.7.8.3(a). A means shall be provided for measuring the current produced by this source with an error of not more than plus or minus 5 percent.

(g) Resistance meter (required for the current overload method if submethod 4.7.8.3(a)(2) is used) shall be provided for measuring the resistance of a single specimen conductor at room temperature and at elevated temperatures, as specified in 4.7.8.3(a). The meter shall exhibit a measurement error of not more than plus or minus 1 percent.

4.7.8.3 Procedure. Select and perform either of the two heat aging methods, the current overload method or the hot air oven method, specified in (a) and (b) herein, respectively. The oven air temperature (in both methods) and the conductor temperature and current (in the current overload method only) shall be recorded at intervals of not more than 15 minutes during the first hour after attaining the heat aging temperature, at intervals of not more than 1 hour for the following 5 hours, and not less than twice daily thereafter for the duration of heat aging. The interval between consecutive temperature recordings shall not exceed 17 hours at any time during heat aging.

(a) Prior to heat aging, the current overload method specimen shall be subjected to the insulation resistance and voltage withstand tests in accordance with 4.8.5 and 4.8.8, respectively. The voltage withstand test shall use the specified voltages (see 3.1). Following these, specimen conductor ends shall be electrically interconnected such that a single, series electrical circuit is formed of the conductors. Provision shall then be made to measure specimen conductor temperature, using either of the following submethods:

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(1) The smallest practicable "V"-shaped cut shall be made along the specimen at a point approximately midway between the specimen ends, and the thermocouple shall be placed beneath the resulting triangular tab. The thermocouple shall be positioned such that it is in firm contact with any single specimen conductor in the innermost specimen conductor layer. (For specimens with only a single layer of conductors, any specimen conductor may be used.) The thermocouple shall be used with its associated temperature indicating device to measure specimen conductor temperature during heat aging.

(2) The resistance meter shall be connected to measure the resistance between the ends of any single specimen conductor in the innermost conductor layer. (For specimens with only a single layer of conductors, any specimen conductor may be used.) The meter leads shall be positioned on the conductor extremities such that all conductor interconnections are excluded from resistance measurements. The conductor temperature shall then be determined by using the following formula:

$$\text{Conductor temperature, in } °C = (234.5 + t)(R/r) - 234.5$$

Where: R = Measured conductor resistance, in ohms during heat aging.

t = Measured room temperature, in °C, to which the specimen is exposed prior to heat aging. Temperature t shall be measured in the immediate vicinity of the specimen; measurement accuracy shall be within plus or minus 0.5°C.

r = Measured conductor resistance, in ohms, when the specimen is exposed to room temperature t. t shall be held constant, within plus or minus 0.5°C for a period of not less than 2 hours prior to this measurement.

The specimen shall next be placed into the oven such that a length of not less than the sum of 12 inches plus 20 times the maximum specified overall cable or cord diameter (see 3.1) remains straight and horizontal, and such that the remainder of the specimen is formed into the largest practicable horizontal coil (to permit free circulation of convection air currents). The current source shall be connected between the two end conductors of the single series circuit within the specimen, and the specimen shall then be subjected to heat aging: the air temperature

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within the oven shall be raised to 50 ± 3°C and the current source adjusted to produce a conductor temperature of 125 ± 5°C. These temperatures shall be maintained for a continuous period of not less than 400 hours. The current source shall not be shut off during heat aging, except that momentary shutoff is permissible in order to take resistance measurements when submethod (a)(2) herein is used. Following heat aging, the specimen shall be allowed to cool throughout to room temperature. The conductor temperature shall then be recorded: a recording which differs from room temperature by more than 5°C shall be considered to indicate failure of the conductor temperature measurement means. If any such failure should occur, the heat aging test shall be considered invalid, and shall be repeated using a different specimen removed from the same length of cable or cord. (A possible advance indication of such a failure, and hence of a need to retest, may be the need to make prominent adjustments of conductor current in order to maintain indicated conductor temperature during heat aging.) Following heat aging, the specimen shall be subjected a second time to the insulation resistance and voltage withstand tests as specified in 4.8.5 and 4.8.8, respectively. A sample, of length not less than 12 inches plus 20 times the maximum specified overall cable or cord diameter (see 3.1), shall then be removed from that portion of the specimen which had been kept straight during heat aging. This sample shall be subjected to the bending procedure as specified in 4.7.8.5. A portion of the remainder of the specimen shall be subjected to the cable filler removability test as specified in 4.7.9.

(b) Prior to heat aging, hot air oven method specimen number 1 shall be subjected to the insulation resistance and voltage withstand tests as specified in 4.8.5 and 4.8.8, respectively. The voltage withstand test shall use the voltages specified (see 3.1). Both specimens shall then be placed within the oven; specimen number 1 shall be formed into the largest practicable horizontal coil within the oven, and specimen number 2 shall be positioned such that it is both straight and horizontal. Both specimens shall be simultaneously subjected to heat aging. The air temperature within the oven shall be raised to 125 ± 3°C. This temperature shall be maintained for a continuous period of not less than 400 hours. Following heat aging, both specimens shall be allowed to cool to room temperature. Specimen number 1 shall then be subjected a second time to the insulation resistance and voltage withstand tests as specified in 4.8.5 and 4.8.8, respectively. A portion of specimen number 1 shall next be subjected to the cable filler removability test as specified in 4.7.9. Specimen number 2 shall be subjected to the bending procedure as specified in 4.7.8.5.

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4.7.8.4 Observation. Any of the following shall constitute specimen failure:

(a) The falling away of any material from either end of any specimen during heat aging.
(b) Jacket sagging on any unarmored specimen.
(c) Jacket exudation through the armor of any armored specimen. (Jacket bulging, without actual flow or cutting, is not considered to be exudation.)
(d) Specimen (specimen number 1 if the hot air oven method is used) fails either of the insulation resistance tests (see 4.8.5.4).
(e) Specimen (specimen number 1 if the hot air oven method is used) fails either of the voltage withstand tests (see 4.8.8.4).
(f) Specimen (specimen number 1 if the hot air oven method is used) fails the cable filler removability test (see 4.7.9.3).

4.7.8.5 Further procedure. The mandrel shall be secured to a fixed surface, and one end of the specimen shall be secured such that it is both in contact with the curved surface of the mandrel, and also perpendicular to the mandrel center of curvature axis. The opposite end of the specimen shall be bent circumferentially around the curved mandrel surface through an arc of not less than 170 degrees; bending shall be accomplished at a rate of approximately 20 degrees per second. The specimen shall then be secured to maintain its bent shape (as with a piece of twine, stretched taut and tied between each end of the specimen) and shall be removed from the mandrel. Not less than two-thirds of the bent portion of the specimen shall then be dissected, and each of its constituent components visually inspected for deterioration.

4.7.8.6 Further observation. Specimen failure shall be construed if the specimen, upon dissection and inspection, exhibits visible deterioration of such a nature or extent as to impair the performance of the cable or cord in service. Deterioration sufficient to constitute specimen failure shall include, but shall not be limited to: distortion or cracking of any constituent component, hardening of filler material, or any discoloration indicating material incompatibility.

4.7.9 Cable filler removability. This test shall determine whether or not nonfibrous filler materials can be readily removed from the constituent components of cable and cord. This test shall not apply to filler material used to fill voids between either conductor strands or braid strands.

4.7.9.1 Specimen. The specimen shall be a 2 foot ± 2 inch length of completed cable or cord.

4.7.9.2 Procedure. Overall specimen components (such as armors, jackets, overall binders or overall shields) shall be removed from one end of the specimen for a distance of not less than 10 inches, thereby exposing the cable core. Using fingers only, filler material shall then be removed from all exposed insulated conductors, or from all groupings of insulated conductors which have a common covering (such as insulated conductor pairs or triads, which have an overall shield or binder) for a distance of not less than 8 inches. Tools, cloths or solvents shall not be used to assist with filler removal. The common covering and filler material shall then be removed from not less than one

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grouping of insulated conductors, on specimens incorporating such groupings, for a distance of not less than 5 inches. Fingers only shall be used for filler removal. If occasional particles of filler material remain on any constituent component of the cable core, then the removal of these particles shall be attempted by means of light brushing with the fingers or with a dry cloth.

4.7.9.3 Observation. Any of the following shall constitute specimen failure:

(a) Filler material which is nonflexible.
(b) Filler material which adheres to the finger or to any component of the cable core. The adherence of occasional small particles of filler material to cable core components is acceptable unless there are small particles of filler material which cannot be removed from core components by means of light brushing with the fingers or with a dry cloth.
(c) Filler material which leaves a residue on cable core components.

4.7.10 Cold bending cable. This test shall determine the ability of completed cable, which is not intended to be flexed during use, to withstand bending at a reduced temperature, such as might be encountered during shipboard construction or repair.

4.7.10.1 Specimen. The specimen shall consist of a piece of completed cable which shall have a length of not less than the sum of 12 inches plus 1.6 times the specified mandrel diameter (see 3.1).

4.7.10.2 Special apparatus. Apparatus shall include the following:

(a) Refrigeration chamber shall support the requirements as specified in 4.7.10.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.
(b) Unless otherwise specified (see 3.1), the mandrel diameter shall be approximately 12 times the diameter of the cable. Mandrel shall be rigid, smooth cylinder or partial cylinder, with a continuous, curved surface of not less than 180 degrees, of suitable length and construction for the specimen bending of 4.7.10.3. The mandrel shall be provided with a clip for affixing one specimen end (see 4.7.10.3).

4.7.10.3 Procedure. The specimen shall be straightened. If it does not remain straight of itself, it shall be held straight by securing it (as with twine) to a straight wooden bar. The straightened specimen (including attached wooden bar, if applicable) shall be placed within the refrigeration chamber, and shall be subjected to an air temperature of minus 20 ± 2°C (unless otherwise specified, see 3.1) for a continuous period of not less than 6 hours. Prior to any change of chamber temperature, the specimen shall be removed from the chamber (following detachment from the wooden bar, if applicable), and one specimen end shall be inserted into a clip which has been affixed to (or

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adjacent to) the mandrel, such that the specimen is both in contact with the curved surface of the mandrel and also perpendicular to the mandrel center of curvature axis. The opposite end of the specimen shall then be bent circumferentially around the curved mandrel surface through an arc of not less than 170 degrees. This bending shall be accomplished at a rate of not less than 10 degrees per second nor more than 20 degrees per second. The period of time between the removal of the specimen from the refrigeration chamber and the completion of specimen bending shall be not greater than 40 seconds. Care shall be taken to avoid heating the specimen, as by excessive handling, during removal from the refrigeration chamber and during bending. Following bending, the specimen shall be secured to maintain its bent shape (as with a piece of twine, stretched taut and tied between each end of the specimen) and shall be removed from the mandrel. Not less than two-thirds of the bent portion of the specimen shall then be dissected, and each of its constituent components visually inspected for deterioration.

4.7.10.4 Observation. Specimen failure shall be construed if the specimen either during bending or upon dissection and inspection, exhibits distortion or cracking of any specimen component, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service.

4.7.11 Cold bending cord. This test shall determine the ability of completed cord to withstand bending at a reduced temperature, such as might be encountered during shipboard construction or repair.

4.7.11.1 Specimens. The required specimens shall be as follows:

(a) The cord specimen shall consist of a piece of completed cord which shall have a length of not less than the sum of 12 inches plus 60 times the maximum specified overall cord diameter (see 3.1).
(b) Insulated conductor specimens shall consist of not fewer than two different insulated conductors removed from completed cord. Each insulated conductor shall have a length of not less than the sum of 12 inches plus 30 times the maximum specified overall insulated conductor diameter (see 3.1).

4.7.11.2 Special apparatus. Apparatus shall include the following:

(a) Refrigeration chamber for supporting the requirements as specified in 4.7.11.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.
(b) Two mandrels, as specified in (1) and (2), shall be provided and shall be mounted within the test chamber such that the axis of each is horizontal, and such that each supports the requirements as specified in 4.7.11.3. Provision shall be made whereby each mandrel can be rotated about its axis, at the rate specified in 4.7.11.3, by an operator located outside of the chamber (as by a crank, the shaft of which passes through the chamber wall and mandrel axes and to which the mandrels are affixed, or as by a remotely-controlled and appropriately geared motor, which is located inside the chamber) and whereby each mandrel can be locked in a desired rotational position.

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(1) Cord mandrel shall be a rigid, smooth cylinder, and shall have a diameter of not more than three times the specified minimum overall cord diameter (see 3.1), and which shall be of suitable length and construction for the specimen wrapping as specified in 4.7.11.3.
(2) Insulated conductor mandrel shall be a rigid, smooth cylinder, and shall have a diameter of not more than that of the specified nominal insulated conductor diameter (see 3.1), and which shall be of suitable length and construction for the specimen wrapping as specified in 4.7.11.3.

(c) A magnifying glass which shall provide a magnification of not less than three diameters.

4.7.11.3 Procedure. One end of the cord specimen shall be secured to the curved surface of the cord mandrel, and one end of each insulated conductor specimen shall be secured to the curved surface of the insulated conductor mandrel, such that all specimens are vertically suspended without obstruction, and such that free space exists on each mandrel to accommodate the specimen wrapping to follow. A weight shall be attached to the opposite end of each specimen, such that the suspended portion of each specimen shall remain vertical during wrapping. The refrigeration chamber doors shall then be closed, and the specimen shall be subjected to an air temperature of minus 40 ± 2°C for a period of not less than 20 hours. Prior to any change of chamber temperature, the mandrels shall be rotated (from outside the chamber) at a rate of 2.5 ± 0.5 r/min, such that the specimens are wrapped around their mandrels for not fewer than five, closely-spaced turns. The mandrels shall then be locked to maintain the specimens in their wrapped position, the chamber doors shall be opened and the specimens secured to their mandrels to prevent loosening or unwrapping. The mandrels, with specimens attached, shall then be removed from the chamber and the entire wrapped portion of each specimen shall be examined using the magnifying glass.

4.7.11.4 Observation. Specimen failure shall be construed if the specimen, upon examination, exhibits distortion or cracking of any specimen component, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cord in service.

4.7.12 Cold working (minus 20°C). This test shall determine the ability of completed cable, which is not intended to be flexed during use, to withstand a traveling bend at a reduced temperature, such as might be encountered during shipboard construction or repair.

4.7.12.1 Specimen. The specimen shall consist of a 20 ± 1 foot length of completed cable.

4.7.12.2 Special apparatus. Apparatus shall include the following:

(a) A refrigeration chamber for supporting the requirements as specified in 4.7.12.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.
(b) Bending apparatus shall consist of two identical sheaves, as shown on figures 6 and 7, which are attached to a fixed surface such that they are pivoted about their axes and at all times,

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their axes are both horizontal and parallel, the corresponding points on each sheave are located in the same vertical plane, their bending surfaces are separated by a distance of not more than four times the maximum specified cable diameter (see 3.1), and they support the requirements as specified in 4.7.12.3. Each sheave shall have a bending surface width of not more than three times the maximum specified overall cable diameter (see 3.1) and a bending surface diameter of not more than 13 nor less than 11 times the maximum specified overall cable diameter (see 3.1).

(c) Two pieces of rope or twine, each of which shall have a length of not less than 25 feet and which shall support the requirements as specified in 4.7.12.3.

4.7.12.3 Procedure. One end of one piece of rope or twine shall be firmly attached to either end of the specimen, and one end of the second piece of rope or twine shall be firmly attached to the opposite specimen end. Attachment may be made either with a cable grip or by directly tying the rope or twine onto the specimen. The specimen, with rope or twine attached, shall be placed within the refrigeration chamber, and shall be subjected to a temperature of minus 20 ± 2°C for a period of not less than 6 hours. Prior to any change of chamber temperature, the specimen (with rope or twine attached) shall be removed from the chamber, one rope or twine shall be wound through the bending apparatus into an S shape as shown on figure 6 and not more than 20 seconds after removing the specimen from the chamber, the specimen shall be subjected to cold working, as follows. The specimen shall be drawn completely through the bending apparatus, at a rate of 35 ± 5 feet per minute by means of pulling on the rope or twine which initially passes through the apparatus. Immediately thereafter, the specimen shall be drawn completely through the bending apparatus in the opposite direction, at the same rate, by means of pulling on the opposite rope or twine. At all times during cold working, both ropes or twine shall be directed such that the bending surface of each sheave (see figures 5 and 6) is in contact with either the specimen or with rope or twine for an arc of not less than 170 degrees. Following cold working, the specimen shall be dissected for a distance of not less than 3 feet, and each of its constituent components visually inspected for deterioration.

4.7.12.4 Observation. Specimen failure shall be construed if the specimen, either during cold working or upon dissection and inspection, exhibits distortion or cracking of any specimen component, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service.

4.7.13 Cold working (minus 54°C). This test shall determine the ability of completed cable, which is intended for in-sea flexing use external to the hull of a ship, to withstand a traveling bend at a reduced temperature, such as might be encountered during service in polar regions.

4.7.13.1 Specimen. The specimen shall consist of a 20 ± 1 foot length of completed cable.

4.7.13.2 Special apparatus. Apparatus shall include the following:

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(a) A refrigeration chamber for accommodating the bending apparatus with specimen attached and for supporting the requirements as specified in 4.7.13.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.

(b) Bending apparatus shall consist of two identical, rigid and smooth cylindrical mandrels, as shown on figure 8, which shall have a diameter of 12 ± 1/4 inches and which shall be of suitable length and construction for the specimen bending as specified in 4.7.13.3. Both mandrels shall be attached to a fixed surface such that they are pivoted about their axes, their axes are both horizontal and parallel, and the corresponding points on each mandrel are located in the same vertical plane. One mandrel (the idler mandrel shown on figure 8) shall be capable of free rotation at all times; the other mandrel (the drive mandrel as shown on figure 8) shall be attached to a motor (as by a drive chain), such that it can be rotated as specified in 4.7.13.3. The mandrel axes shall be separated by a distance of not more than 24 inches.

(c) Salt water tub (required for qualification only, see 3.2) shall be filled with a uniform solution of salt water comprising 3 ± 1/4 percent salt by weight, for the specimen immersion as specified in 4.7.13.3(b). The salt water solution shall be maintained at a temperature of 22 ± 5°C during specimen immersion.

(d) Hydrostatic (open end) test apparatus (required for qualification only, see 3.2) shall be as specified in 4.7.21.2.

4.7.13.3 Procedure. Specimens not undergoing qualification shall be tested as specified in (a). Specimens undergoing qualification (see 3.2) shall be tested as specified in (b).

(a) The bending apparatus shall be installed in the refrigeration chamber, and the specimen shall be installed in the bending apparatus as shown on figure 8. One end of the specimen shall be secured to the curved surface of the drive mandrel with a clamp, such that the specimen end is perpendicular to the mandrel axis; the free portion of the specimen shall then be held perpendicularly to the mandrel axes, tautly wrapped once about the circumference of the idler mandrel, and then tautly wrapped as many times as the specimen length will allow about the circumference of the drive mandrel. The free end of the specimen shall then be secured to the drive mandrel with a clamp. Wrappings shall be such that the specimen is directed in an opposite sense around each mandrel (that is, the wrapping around the idler mandrel is clockwise, as shown on figure 8, then all wrappings around the drive mandrel shall be counterclockwise). The specimen will form a figure 8 about the mandrels when viewed parallel to the mandrel axes. Specimen wrappings around the drive mandrel shall be as close together as practicable. The refrigeration chamber doors shall then be closed, and the specimen subjected to a temperature of minus 54°C for a period of not less than 6 hours. Prior to any change of chamber temperature, the specimen shall be subjected to cold working. Cold working shall consist of operating the motor of the bending apparatus such that as much of the specimen length

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as practicable is drawn over the idler pulley at the rate of 50 ± 5 feet per second, the direction of motor rotation shall then be reversed and as much of the specimen length as practicable drawn over the idler pulley in the opposite direction at the same rate. This process shall then be repeated three more times (the specimen shall be drawn over the idler pulley a total of four times in each direction). Following cold working, the specimen shall be removed from the bending apparatus, and the approximate middle of the specimen shall be dissected for a distance of not less than 3 feet. Each of its constituent components shall then be visually inspected for deterioration.

(b) The entire specimen, except for its extreme ends (which shall remain exposed to the air) shall be submerged in the solution of the salt water tub for a period of not less than 21 days. The specimen shall then be removed from the salt water tub and immediately subjected to the procedure of (a), above, except for the following:

(1) The specimen shall be drawn over the idler pulley a total of 10 times in each direction instead of a total of four times in each direction.

(2) Following cold working, the specimen shall be allowed to return to room temperature, and a 60 ± 2-inch piece shall be removed from the approximate middle of the specimen. This piece shall be subjected to the hydrostatic (open end) test as specified in 4.7.21. Specimen dissection shall be performed on a cable section which has not been subjected to this hydrostatic (open end) test.

4.7.13.4 Observation. Either of the following shall constitute specimen failure:

(a) The specimen, upon dissection and inspection, exhibits visible distortion or cracking of any specimen component, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service.

(b) (For specimens undergoing qualification only.) The specimen fails the hydrostatic (open end) test (see 4.7.21.4).

4.7.14 Crack resistance. This test shall determine the ability of glass braid coverings to withstand bending, and to determine the propensity of glass braids to puncture overlying coverings.

4.7.14.1 Specimen. Each specimen shall consist of a single insulated conductor (including insulation covering) of a length not less than that specified in table XIX. Specimens shall be removed from completed cable or cord as follows:

(a) Cables or cords containing four or fewer conductors: one specimen shall be removed from each insulated conductor.

(b) Cables or cords containing more than four but fewer than 21 conductors: one specimen shall be removed from each of four different insulated conductors (a total of four specimens).

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(c) Cables or cords containing 21 or more conductors: the number of specimens shall be equal to the square root (rounded, if necessary, to the nearest whole number) of the total number of conductors. Each specimen shall be removed from a different insulated conductor.

An additional specimen, which shall consist of a piece of insulated conductor (including insulation covering) of not less than 10 feet in length, may be required (see 4.7.14.5).

TABLE XIX. Specimen and mandrel dimensions for the crack resistance test.

Conductor size (AWG)	Minimum specimen length, (inches)	Mandrel diameter (inches)
7	24	1-1/2 ± 1/16
9	18	1 ± 1/16
10	12	1/2 ± 1/32
14	12	1/2 ± 1/32
16	8	1/3 ± 1/64
18	8	1/3 ± 1/64
Greater than 18	6	1/4 ± 1/64

4.7.14.2 Special apparatus. Apparatus shall include the following:

(a) Mandrel shall be a rigid, smooth cylinder, of the diameter specified in table XIX, and of suitable length and construction for the specimen wrapping as specified in 4.7.14.3.
(b) An aqueous solution of gentian violet, of a quantity and concentration for the staining as specified in 4.7.14.3 or 4.7.14.5, as applicable.

4.7.14.3 Procedure. Each specimen shall be tightly hand-wrapped, for not fewer than three circumferential and contiguous, 360-degree turns around the mandrel. Following this, each specimen shall be removed from the mandrel and wrapped around the same mandrel for not fewer than three circumferential and contiguous, 360-degree turns in the opposite lateral direction. Each specimen shall next be removed from the mandrel and straightened, to the extent that the specimen is nowhere bent along a radius of less than 12 times the maximum measured specimen diameter. Each straightened specimen shall then be submerged, except for its extreme ends (which shall remain exposed to the air), in the gentian violet for a period of not less than 30 minutes, and shall then be removed from the gentian violet and visually inspected for defects.

4.7.14.4 Observation. Specimen failure shall be construed if the insulation covering of any specimen exhibits visible cracking or peeling which allows penetration of the gentian violet. If the insulation covering of any specimen exhibits an average of more than one pinhole, which allows penetration of the gentian violet for each 12 inches of length, then the further procedure as specified in 4.7.14.5 shall be performed.

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4.7.14.5 Further procedure (may not be required, see 4.7.14.4). An additional specimen, which shall consist of a piece of insulated conductor (including insulation covering) of not less than 10 feet in length, shall be prepared. This entire specimen, except for its extreme ends (which shall remain exposed to the air) shall be submerged in gentian violet for a period of not less than 30 minutes. To facilitate submersion, the specimen may be coiled to the extent that it is nowhere bent along a radius of less than 12 times the maximum measured specimen diameter. The specimen shall then be removed from the gentian violet and visually inspected for defects.

4.7.14.6 Further observation. Specimen failure shall be construed if the insulation covering of the specimen exhibits an average of more than one pinhole, which allows penetration of the gentian violet for each 12 inches of length.

4.7.15 Drip. This test shall determine whether or not nonfibrous fillers will exude and drip from cable ends when cables containing these fillers are subjected to high temperatures. Such filler dripping may produce deleterious effects, such as the coating or bridging of underlying electrical contacts.

4.7.15.1 Specimen. The specimen shall consist of an 18 ± 1/2 inch length of completed cable.

4.7.15.2 Special apparatus. Apparatus shall include a heating chamber for supporting the requirements as specified in 4.7.15.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.

4.7.15.3 Procedure. The specimen shall be straightened, and shall be suspended from one end within the heating chamber, such that the specimen hangs freely downward. The temperature within the chamber shall then be raised to the specified value (see 3.1) within 1°C. This chamber temperature shall be maintained for a continuous period of not less than 18 hours. Provision shall be made to catch any material which may fall away from the lower specimen end during this test.

4.7.15.4 Observation. Specimen failure shall be construed if any material falls away from the lower specimen end during the test.

4.7.16 Durometer hardness. This test shall determine whether or not semirigid polyvinylchloride insulation material, polyvinylchloride jacketing material and silicone rubber jacketing material exhibit sufficient hardness (see 4.7.16.4) for use in cable construction.

4.7.16.1 Specimen. The specimen shall consist of a block of material, of the dimensions specified in ASTM D 2240, which shall have been cured in effectively the same manner as when used in cable manufacture.

4.7.16.2 Special apparatus. Apparatus shall include a type A or type D durometer, as specified (see 3.1), which shall be in accordance with ASTM D 2240.

4.7.16.3 Procedure. The specimen shall be tested in accordance with ASTM D 2240.

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4.7.16.4 Observation. Specimen failure shall be construed if the specimen exhibits a durometer hardness other than that specified (see 3.1).

4.7.17 Flame propagation. This test shall determine the relative ability of grouped cable to resist the propagation of fire along its length.

4.7.17.1 Specimen. Each specimen shall consist of a piece of completed cable of not less than 8 feet in length. The total number of specimens required shall be as specified in the vertical tray flame test as specified in IEEE 383.

4.7.17.2 Special apparatus. Apparatus shall include all equipment specified in the vertical tray flame test in accordance with IEEE 383.

4.7.17.3 Procedure. The specimens shall be tested as specified in the vertical tray flame test as specified in IEEE 383.

4.7.17.4 Observation. Specimen failure shall be construed when any portion of any specimen burns to the top of the tray, either during or after operation of the ribbon burner. The following information (which does not constitute failure criteria) shall additionally be reported to help expand a performance data base: flame temperature, period of time between burner shutoff and cessation of flame on the specimen, overall distance of specimen jacket damage above the burner, and a brief description of any dripping or flaming material which falls from the specimen to the floor during the test.

4.7.18 Flammability. This test shall determine the ability of a single length of completed cable or cord to resist ignition from the spatter of welding operations, and to self-extinguish should ignition occur.

4.7.18.1 Specimen. The specimen shall consist of an 18 ± 2-inch length of completed cable or cord. (Two additional specimens (a total of three specimens) may be required, see 4.7.18.4 and 4.7.18.5.)

4.7.18.2 Special apparatus. Apparatus shall include the following:

(a). Flammability apparatus shall support the specimen vertically, and apply controlled sources of heat and ignition to the specimen as shown on figure 9 and as further described below.

(1) A cup shall be placed over the lower specimen end to direct any gases which might emanate from this end upwards into the heating coil. The sides of this cup shall extend not less than 1 inch up the outside of the specimen from the specimen end, but shall not cover that portion of the specimen which lies beneath the heating coil. The cup shall be shaped from thin metal foil, or from any other fire-resistant material which will not absorb a significant amount of heat from the heating coil, such that it forms an approximately conformal fit around the specimen. This fit shall be sufficiently loose that gases can easily pass out the specimen end, up between the cup and the specimen and out towards the heating coil.

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(2) A specimen holder (such as a lathe chuck) shall support the specimen, by its lower end, in a vertical position. The specimen holder shall not deform the cup to an extent that the cup fails to perform its intended function.

(3) A heating coil shall furnish a controlled source of heat to the specimen when used with the current source (see 4.7.18.2 (b)). The coil shall consist of a 7-turn helix of 10 AWG resistance wire, which is an 80 percent nickel and 20 percent chromium alloy in accordance with ASTM B 344. The coil shall exhibit a uniform axial spacing of 1/4 ± 1/32 inch between the centers of each adjacent turn, and shall have an id which shall be larger than the maximum measured specimen diameter by 0.55 ± 0.05 inch. The heating coil shall be placed over the specimen coaxially, such that the lower end of the coil is located 1-1/2 ± 1/8 inches above the top of the specimen holder.

(4) Two spark plugs shall furnish a source of ignition to the specimen when used with the voltage source (see 4.7.18.2(c)). These plugs shall be mounted such that they are on diametrically opposite sides of the specimen, their axes are horizontal and their electrode gaps are spaced both 1/8 ± 1/32 inch from the surface of the specimen and 1/2 ± 1/16 inch above the top of the heating coil. Means (such as handles) shall be provided whereby both spark plugs may be withdrawn from the specimen following ignition (see 4.7.18.3) to allow unimpeded flame travel and to prevent the collection of soot on spark plug electrodes.

(5) A scale shall be used to measure the height of flame travel (see 4.7.18.3) along the specimen above the heating coil. This scale shall be graduated at intervals of not more than 1/8 inch, and shall be affixed near the specimen in a position which will not significantly impede the progress of the flame.

(b) Current source shall be an adjustable source of 60 Hz single-phase ac, which shall maintain the current specified in table XX through the heating coil, as shown on figure 9.

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TABLE XX. Current in heating coil.

Measured overall cable or cord diameter, inches	Current, amps rms + 1/4 A
Less than 0.099	45
0.100 to 0.199	46
0.200 to 0.299	47
0.300 to 0.399	48
0.400 to 0.499	49
0.500 to 0.599	50
0.600 to 0.699	51
0.700 to 0.799	52
0.800 to 0.899	53
0.900 to 0.999	54
1.000 to 1.099	55
1.100 to 1.399	56
1.400 to 1.799	57
1.800 to 2.299	58
2.300 or greater	59

(c) A source of voltage for producing continuous inter-electrode sparking on both spark plugs, as shown on figure 9.

(d) Enclosure shall be a chamber for enclosing the flammability apparatus with the specimen mounted. This enclosure shall incorporate an exhaust fan at its top, and shall incorporate apertures for admitting air along its sides, adjacent to its base. The exhaust fan shall be operated to produce no more suction than is necessary to carry off smoke and gases, and the enclosure shall be designed to minimize all interior air drafts to the greatest practicable extent. Shatterproof windows, which shall provide a clear and unobstructed view of the flammability apparatus and mounted specimen shall be provided in the walls of the enclosure. The means for withdrawing the spark plugs from the specimen shall be located outside of the enclosure (for example, withdrawal may be accomplished by means of handles passing through the enclosure walls).

(e) A stop watch or equal, which shall resolve 1-second increments, and which shall support the requirements as specified in 4.7.18.3.

4.7.18.3 Procedure. The flammability apparatus shall be placed within the enclosure, and the specimen shall be mounted in the flammability apparatus. The enclosure door shall then be shut, the exhaust fan turned on, and the voltage source applied to the spark plugs. The current source shall next be applied to the heating coil and the stop watch started simultaneously. When specimen ignition occurs, the ignition time (that is, elapsed since starting the stop watch) shall be recorded, the voltage source shall be removed from the spark plugs and the spark plugs shall be withdrawn from the specimen. (Ignition shall be considered to have occurred when flames transfer from volatizing gases to the specimen surface and remain there. Flashes which may occur prior to sustained flame on the specimen surface shall be disregarded.)

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The current source shall be removed from the heating coil in accordance with the following schedule:

(a) The current source of the armored specimen shall be removed from the heating coil 30 ± 3 seconds after ignition occurs.
(b) The current source of the unarmored specimen shall be removed from the heating coil either when the stop watch shows 60 ± 3 seconds, or else when ignition occurs, whichever occurs later.

The burning time (that is, the number of seconds elapsing between the removal of the current source from the heating coil and the complete extinction of all flame) shall be recorded. The flame travel distance (that is, the maximum height above the top of the heating coil that the base of the flame reaches on the specimen surface) shall be noted on the scale, prior to flame extinction, and shall also be recorded. If the specimen exhibits an ignition time of 250 seconds or more, the test shall be discontinued.

4.7.18.4 Observation. Any of the following shall constitute the need for further testing, as specified in 4.7.18.5:

(a) Any specimen: a flame travel distance in excess of that specified (see 3.1).
(b) Armored specimen: an ignition time of less than 60 seconds, or a burning time of more than 60 seconds.
(c) Unarmored specimen (except type MRI): an ignition time of less than 30 seconds or a burning time of more than 120 seconds.
(d) Type MRI specimen only: an ignition time of less than 25 seconds or a burning time of more than 60 seconds.

Any specimen not requiring further testing, including any specimen exhibiting an ignition time of 250 seconds or more, shall be construed as having passed the flammability test.

4.7.18.5 Further procedure (may not be required, see 4.7.18.4). Two additional specimens shall be prepared from the same length of cable or cord as was the original specimen. Each of these additional specimens shall be tested in accordance with 4.7.18.3. An average ignition time, average burning time and average flame travel distance shall be calculated from the results of the three tests (the original specimen test plus the test for each of the two additional specimens).

4.7.18.6 Further observation. Any of the following shall constitute specimen failure:

(a) All specimens: shall have an average flame travel distance in excess of that specified (see 3.1).
(b) Armored specimens: shall have an average ignition time of less than 60 seconds, or an average burning time of more than 60 seconds.
(c) Unarmored specimens (except type MRI): shall have an average ignition time of less than 30 seconds or an average burning time of more than 120 seconds.

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(d) Type MR1 specimens only: shall have an average ignition time of less than 25 seconds or an average burning time of more than 60 seconds.

4.7.19 Gas flame. This test shall determine the ability of completed cable containing silicone-rubber insulation to maintain electrical circuit integrity in the presence of flame.

4.7.19.1 Specimen. The specimen shall consist of a 48 ± 6-inch lengths of completed cable. Any specimen containing more than three conductors shall have its conductors connected together electrically at one or both specimen ends as follows:

(a) Specimens with four conductors shall have physically opposite conductors and be connected together in pairs to form two effective electrical conductors.
(b) Specimens with six conductors shall have physically opposite conductors and shall be connected together in pairs to form three effective electrical conductors.
(c) Specimens with seven or more conductors shall have conductors connected together to form three effective electrical conductors. Connections shall be made such that physically adjacent conductors shall be members of different effective electrical conductors to the maximum practicable extent, and such that the number of conductors constituting each effective electrical conductor does not differ by more than one.

4.7.19.2 Special apparatus. Apparatus shall include the following:

(a) A metered and adjustable source of natural gas or other flammable gas, and an adjustable source of air (see figure 10), support the requirements as specified in 4.7.19.3. The gas and air shall be made available, as a mixture, through a single conduit.
(b) A ribbon type gas burner, American Gas Furnace Co., N.J., type 24X or equal. The ribbon insert utilized by this burner shall be designed to produce the specified flame. This insert shall facilitate the combustion of 27.5 cubic feet per hour of natural gas having a heat content of 1030 ± 2 percent British thermal units (Btu) per cubic feet (ft^3), or another gas having equivalent heat output and temperature values.
(c) A 3-phase, 60 Hz voltage source, as shown on figure 11, shall provide a separate, single-phase test voltage through a normal-blow fuse or overload relay to each specimen conductor. This voltage source shall be in accordance with the following:

(1) Specimens rated for 1 kV which incorporate conductors with cross sections of less than 14,000 circular mils, and all specimens rated for less than 1 kV. The test voltage shall be 120 ± 5 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of 1 A.
(2) Specimens rated for 1 kV which incorporate conductors with cross sections of 14,000 circular mils or more, the test voltage shall be 450 ± 10 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of 1 A.

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(3) Specimens rated for 5 kV: the test voltage shall be 4160 ± 100 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of 1/4 A.

The secondary winding voltage rating of any constituent voltage step-up transformer shall be not less than the required test voltage. Each phase of the voltage source shall have the capacity to supply the specimen with a current which shall be not less than twice the fuse and overload relay rating.

(d) A chamber enclosing the horizontally-mounted burner and the suspended specimen (see 4.7.19.3). This enclosure shall incorporate an exhaust fan at its top, and shall incorporate apertures for admitting air along its sides, adjacent to its base. The exhaust fan shall be operated to produce no more suction than is necessary to carry off smoke and gases, and the enclosure shall be designed to both minimize horizontal air drafts to the greatest practicable extent and to allow free circulation of convection air currents generated during the gas flame test.

4.7.19.3 Procedure. The burner shall be mounted within the chamber, such that the holes in its top are in a common horizontal plane. The specimen shall then be suspended within the enclosure, such that it is parallel to the burner, horizontally centered directly above the burner and spaced 1-7/8 ± 1/8 inches vertically above the top burner surface. Specimen suspension shall be accomplished in a manner (such as by wire loops) which shall not significantly shield the specimen from flame and which shall prevent any portion of the specimen from dropping closer than 3/4 inch to the top surface of the burner during specimen burning. The gas and air supply shall then be attached to the burner, and the gas shall be adjusted to provide a flow rate of not less than that calculated by the following formula:

Gas flow rate, in cubic feet per hour = $1387 (T + 273)/PN$

Where: T = Measured ambient temperature in the vicinity of the burner, in °C; measurement accuracy shall be within plus or minus 0.5°C.

P = Local atmospheric pressure, in lb/in^2 absolute; measurement accuracy shall be within plus or minus 2 percent.

N = Gross heating value, in $Btu's/ft^3$, of the gas at an atmospheric pressure of 14.69 lb/in^2 absolute and a temperature of 23°C, this value shall be accurate within plus or minus 2 percent.

The gas exiting the burner shall then be ignited, and the air flow to the burner adjusted to produce the smallest, blue flame possible. The short circuit monitor test voltages shall then be applied to the specimen, as shown on figure 11. If the specimen is armored, an additional test voltage shall be applied by grounding the armor, as shown on figure 11. (This will effectively apply an additional voltage which is approximately equal to the test voltage divided by the square root of 3 between the armor and the specimen conductors.) The flame and test voltages shall remain continuously applied to the specimen for a period of not less than that specified (see 3.1), unless specimen failure (see 4.7.19.4) occurs prior to the end of this period. The test may then be terminated prematurely.

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4.7.19.4 Observation. Specimen failure shall be construed if any fuse blows or if any overload relay trips prior to completion of the specified test period (see 3.1).

4.7.20 Heat distortion. This test shall determine the ability of cross-linked thermosetting polyethylene insulation to resist physical deformation when subjected to an elevated temperature.

4.7.20.1 Specimen. Each specimen shall be removed from completed cable or cord in accordance with (a) or (b), whichever is indicated. The number of specimens which shall be required shall be in accordance with (c), (d) or (e), as appropriate.

(a) Cables or cords which incorporate conductors with cross sections of more than 190,000 circular mils: each specimen shall consist of a piece of insulation, which has been removed from a conductor and which has been cut to form a rectangular strip. This strip shall be not less than 7/8-inch long nor less than 9/16 ± 1/16-inch wide, and shall be ground and buffed to a thickness of 0.050 ± 0.010 inch with a grinding apparatus (see 4.7.20.2(c)). The thickness of this specimen shall be made as uniform as practicable.

(b) Cables or cords which incorporate conductors with cross sections of 190,000 circular mils or less: Each specimen shall consist of a single insulated conductor of length not less than 7/8 inch.

(c) Cables or cords containing four or fewer conductors: one specimen shall be prepared from each insulated conductor.

(d) Cables or cords containing more than four but fewer than 21 conductors: one specimen shall be prepared from each of four different insulated conductors (a total of four specimens).

(e) Cables or cords containing 21 or more conductors: the number of specimens prepared shall be equal to the square root (rounded, if necessary, to the nearest whole number) of the total number of conductors. Each specimen shall be prepared from a different insulated conductor.

4.7.20.2 Special apparatus. Apparatus shall include the following:

(a) A self-standing thickness gauge, for supporting the requirements as specified in 4.7.20.3, which shall exhibit a measurement error of not more than plus or minus 0.001 inch. This gauge shall incorporate a flat horizontal platen upon which an entire specimen shall be laid flat during measurement, and shall incorporate a foot which rests upon the top of the specimen during measurement. This foot shall be attached to a plunger which moves freely in the vertical direction, and which is attached to a dial or other indicating device from which the specimen thickness may be read. The foot shall present a horizontal disk of 3/8 plus 0, minus 1/64 inch diameter to the specimen, shall bear down upon the specimen with a force of 85 plus 0, minus 4 grams and shall be loaded with weights (see 4.7.20.3) to present additional force to the specimen.

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(b) A micrometer (required for insulated conductor specimens only) with flat, parallel measurement surfaces on both spindle and anvil, for the specimen thickness measurement as specified in 4.7.20.3(b)(1), which shall exhibit a measurement error of not more than plus or minus 0.001 inch.

(c) A motor driven grinding wheel and a motor driven buffing wheel, or their equal, for the specimen preparation as specified in 4.7.20.1(a). Guides shall be provided to assure that the specimen is pulled tangentially to the surface of each wheel during grinding and buffing.

(d) An oven, for accommodating the gauge with specimen inserted and for supporting the requirements as specified in 4.7.20.3. The oven air temperature shall be measured in the immediate vicinity of the specimen.

4.7.20.3 Procedure. The initial insulation thickness of each specimen (T in the following formula) shall be determined (see (a)(1) and (b)) and the oven shall be preheated to 121 ± 1°C. This oven temperature shall be maintained for the remainder of the test. The gauge foot shall then be loaded with weights (see (a)(2) and (b)(2)) and the gauge placed within the oven. Not less than 1 hour later, one specimen shall be selected and also placed within the oven. Not less than 1 hour after placing the specimen within the oven, the specimen shall be placed beneath the gauge foot such that the gauge indicates the specimen thickness. Not less than 1 hour after placing the specimen beneath the gauge foot, the final specimen thickness or diameter (as appropriate) shall be read from the gauge and the final insulation thickness (t in the following formula) determined (see (a)(3) and (b)(3)). In an identical manner, the remaining specimens shall also be tested. The percentage heat distortion of the insulation shall then be calculated by using the following formula:

$$\text{Percentage heat distortion} = \text{Median value of } 100(T-t)/T \text{ for all tested specimens.}$$

Where: T = Initial insulation thickness of the specimen, in thousandths of an inch (see (a) and (b)).
t = Final insulation thickness of the specimen, in thousandths of an inch (see (a) and (b)).

The following additional details apply:

(a) Rectangular strip specimens:
(1) The initial insulation thickness (T) shall be the initial specimen thickness, as measured with the gauge, prior to loading the gauge with weights and prior to inserting either the gauge or the specimen into the oven.
(2) The gauge foot shall be loaded with 2000 ± 100 grams.
(3) The final insulation thickness (t) shall be the final specimen thickness, measured as specified.

(b) Insulated conductor specimen:
(1) The initial insulation thickness (T) shall be defined as 1/2 of the difference between the measured overall specimen

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diameter and the measured specimen conductor diameter. Measurements shall be made using the micrometer. The overall specimen diameter shall be measured in the same radial direction as will be measured when the final insulation thickness (t) (see (b)(3).) is determined.

(2) The gauge foot shall be loaded as follows:

a. Specimens with a conductor cross section of not more than 3999 circular mils nor less than 2700 circular mils: 400 ± 20 grams.

b. Specimens with a conductor cross section of not more than 21,999 circular mils nor less than 4000 circular mils: 500 ± 25 grams.

c. Specimens with a conductor cross section of not more than 94,999 circular mils nor less than 22,000 circular mils: 750 ± 35 grams.

d. Specimens with a conductor cross section of not more than 189,999 circular mils nor less than 95,000 circular mils: 1000 ± 50 grams.

(3) The final insulation thickness (t) shall be defined as 1/2 of the difference between the final specimen diameter, measured as specified, and the previously measured specimen conductor diameter.

4.7.20.4 Observation. The specimen shall be considered to have failed the test if the percentage heat distortion is greater than that specified (see 3.1).

4.7.21 Hydrostatic (open end). This test shall determine the ability of completed cable, which is intended to pass through watertight bulkheads on submersible vehicles, to prevent the longitudinal flow of high-pressure water when properly installed in a bulkhead penetrator.

4.7.21.1 Specimen. The specimen shall consist of a 60 ± 2 inches length of completed cable.

4.7.21.2 Special apparatus. Apparatus shall include a tank which can be filled with water, which can be internally pressurized to the specified value (see 3.1), and which is used for applying this water pressure to one specimen end, as specified in 4.7.21.3. An appropriate stuffing tube, as specified in MIL-S-24235 and MIL-S-24235/1, shall be welded into the tank wall to provide a means for sealing the one specimen end within the tank. An appropriate packing assembly, as specified in MIL-S-24235 and MIL-S-24235/2, shall be available to effect this sealing (see 4.7.21.3). A means shall be provided whereby the pressure within the tank can be measured within an accuracy of plus or minus 5 percent.

4.7.21.3 Procedure. One end of the specimen shall be installed through the tank stuffing tube, and shall be sealed in place by using the packing assembly as specified in section 3, group A of Drawing 803-5001027. The opposite specimen end shall remain freely exposed to air for the remainder of the test. The tank shall then be closed and filled with water, such that the specimen end within the tank remains directly and entirely exposed to water for the remainder of the test. The water shall then be pressurized to a value of

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not less than that specified (see 3.1). This pressure shall be maintained for a continuous period of not less than that specified (see 3.1), unless specimen failure (see 4.7.21.4) occurs prior to the end of this period; the test may then be terminated prematurely. Following the test, the specimen shall be removed from the tank, and the end which had been exposed to water pressure shall be examined.

4.7.21.4 Observation. Either of the following shall constitute specimen failure:

(a) Any water leakage from the free end of the specimen in excess of that specified (see 3.1).
(b) Slippage of more than 1/4 inch, with respect to the specimen jacket, of any specimen component at the end of the specimen which had been exposed to water pressure.

4.7.22 Permanence of printing (conductor insulation). This test shall determine the ability of printed information on insulated conductors to remain legible in the presence of repeated abrasion.

4.7.22.1 Specimen. Each specimen shall consist of a single insulated conductor which shall have sufficient length for use in the test as specified in 4.7.22.3. Specimens shall be removed from completed cable or cord as follows:

(a) Cables or cords containing four or fewer conductors: one specimen shall be removed from each insulated conductor.
(b) Cables or cords containing more than four but fewer than 21 conductors: one specimen shall be removed from each of four different insulated conductors (a total of four specimens).
(c) Cables or cords containing 21 or more conductors: The number of specimens removed shall be equal to the square root (rounded, if necessary, to the nearest whole number) or the total number of conductors. Each specimen shall be removed from a different insulated conductor.

4.7.22.2 Special apparatus. Apparatus shall include an abrading machine, which shall secure a specimen horizontally between two fixed supports, and which shall abrade the specimen printed identification by means of a motor-driven, reciprocating tape consisting of number 50-2/20 unbleached cotton braid. One possible arrangement for the abrading machine is shown on figure 12. The braided cotton tape shall lie flat on the specimen surface, shall remain perpendicular to the specimen axis, and shall abrade an unchanging area of the specimen periphery, over an arc of not less than 135 degrees, at all times during the test. One end of the tape shall be reciprocated at a rate of 28 ± 3 cycles per minute, such that the tape is drawn across the specimen for a distance of 10 plus 1, minus 0 inches in each direction (20 inches minimum total excursion during cycle). The opposite end of the tape shall be maintained under tension by a freely suspended 1/2 plus 1/64, minus 0 pound weight. Each side of the tape shall be used to perform not more than 500 cycles; the tape shall then be discarded and replaced. The abrading machine shall incorporate an automatic counter to total the number of abrasion cycles to which the specimen is subjected during the test.

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4.7.22.3 Procedure. Each specimen shall be wiped with a clean, dry cloth to remove any lubricant or dirt. One specimen shall then be selected and mounted between the two fixed supports of the abrading machine. Mounting shall be accomplished such that the specimen shall be as taut as practicable, and such that the braided cotton tape shall repeatedly abrade the same portion of the specimen printed identification at all times during the test. The automatic counter shall be set initially to zero. The abrading machine motor shall then be turned on, allowing the braided cotton tape to reciprocate and abrade the specimen printed identification either until the printed identification is no longer legible in its abraded region, or until not fewer than the specified number of abrasive cycles (see 3.1) have been completed, whichever occurs first. The number of abrasive cycles completed shall then be noted, and the remaining specimens then tested in an identical manner.

4.7.22.4 Observation. Specimen failure shall be construed if the median value of the number of abrasive cycles completed on each specimen is less than that specified (see 3.1).

4.7.23 Permanence of printing (jacket). This test shall determine the ability of printed information of jacketing material to remain legible in the presence of repeated abrasion.

4.7.23.1 Specimen. The specimen shall consist of a piece of completed cable or cord which shall have sufficient length for use in the test as specified in 4.7.23.3.

4.7.23.2 Special apparatus. Apparatus shall include an abrading machine which shall support and secure the specimen such that it is straight and horizontal, and which shall abrade the specimen printed identification by means of a motor driven, transversely reciprocating steel pin. This steel pin shall have a diameter of 0.025 $\pm$ 0.001 inch where it abrades the specimen, and shall have a surface roughness of not less than 2 microinches, as specified in ANSI B46.1. (A selected sewing needle may satisfy these requirements.) One possible arrangement for the abrading machine is shown on figure 13. The steel pin shall be horizontal and perpendicular to the specimen axis, shall ride along the top of the specimen and shall be weighted to bear down on the specimen with a force of 1 plus 1/16, minus 0 pound at all times during the test. This pin shall be reciprocated at a rate of 60 $\pm$ 2 cycles per minute, such that the pin is drawn along the specimen for a distance of 3/8 plus 1/16, minus 0 inch in each direction (3/4-inch minimum total excursion) during each cycle. The abrading machine shall incorporate an automatic counter to total the number of times that the specimen is abraded by the steel pin during the test.

4.7.23.3 Procedure. The specimen shall be wiped with a clean, dry cloth to remove any lubricant or dirt, and shall be secured in the abrading machine with the specimen printed identification facing upwards, where it is to be abraded by the steel pin. The automatic counter shall be set initially to zero. The abrading machine motor shall then be turned on, allowing the steel pin to reciprocate and abrade the specimen printed identification either until the printed identification is no longer legible in the abraded region, or until not fewer than 250 abrasive cycles have been completed, whichever occurs first. The number of abrasive cycles completed shall then be noted. This test shall next be repeated four more times (five times total), subjecting a fresh portion of the specimen printed identification to abrasion each time.

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4.7.23.4 Observation. Specimen failure shall be construed if the median value of the number of abrasive cycles completed during each of the five tests is less than 250.

4.7.24 Physical tests on insulation and jacket. This test shall determine whether or not various extruded insulating and extruded jacketing materials have been properly processed, by means of tension measurements. This test makes provision for making tension measurements both before and after these materials have been artifically aged.

4.7.24.1 Specimens. Each specimen shall consist of a single piece of insulation or jacketing, which shall have sufficient length for use in the tests as specified in 4.7.24.4, as specified (see 3.1), and which shall have been prepared in accordance with (a) or (b), whichever is indicated. Specimens shall be removed from completed cable or cord. The number of specimens which shall be required each time each test is performed shall be in accordance with (c), (d) or (e), as appropriate; it is recommended, however, that extra specimens be prepared to allow for possible invalid test results (see 4.7.24.4).

(a) For insulation (not jacketing) which has a cross-sectional area of 0.030 in^2 or less, and which can readily be pulled from its associated conductor, the outer diameter of a length of insulated conductor shall be measured with a micrometer (see 4.7.24.2(a)), in two perpendicular directions, at not fewer than three different longitudinal locations (not fewer than six measurements total); an average outer insulation diameter shall then be calculated from these measurements. Next, the maximum conductor diameter shall be measured with the micrometer, and the insulation then removed from the conductor by pulling. The insulation shall not be cut longitudinally or permanently distorted for pulling ease, since these might adversely affect test results. The pulled insulation shall constitute one specimen. The cross sectional area of the specimen shall be determined by the following formula:

Cross sectional area, in square inches = $0.7854 (D + d) (D - d)$

Where: D = Calculated average outer insulation diameter, in inches; accuracy of constituent measurements and calculations shall be within plus or minus 0.001 inch.

d = Measured maximum conductor diameter, in inches; measurement accuracy shall be within plus or minus 0.001 inch.

(b) For all jacketing, and for insulation which cannot readily be pulled from its associated conductor or which has a cross sectional area of greater than 0.030 in^2, longitudinal segment of insulation or jacketing shall be cut from a length of insulated conductor or cable, respectively. This segment shall have a reasonably constant cross sectional shape (such as a circular segment, a circular sector, a rectangle or any

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other cross sectional shape which may be conveniently produced). The segment shall then be ground and buffed with a grinding apparatus (see 4.7.24.2(b)), such that it has two smooth and parallel surfaces which are as flat as practicable. Two straight, parallel and longitudinal cuts shall then be made in the middle of this segment. These cuts shall be perpendicular to the buffed surfaces, such that the cross sectional area between the two cuts shall be rectangular in shape and as large as practicable, except that this cross sectional area shall not exceed 0.030 in^2. Each of these cuts shall be longer than the benchmark spacing specified in 4.7.24.4. Material between each cut and the nearest longitudinal edge of the segment shall then be removed and discarded; material which is longitudinally above or below the region of the cuts may be allowed to remain or may be cut into any configuration appropriate for use with the jaws of the tensile machine (see 4.7.24.2(f)). Care shall be taken at all times to prevent nicking the material, and to avoid cuts which intersect each other at an angle, rather than in a gradual, curving manner. These precautions are necessary to prevent a possible premature specimen failure during tensile strength and elongation testing (see 4.7.24.4(a)). The prepared segment shall constitute one specimen. The maximum width and maximum thickness of the specimen in the region between the two required parallel cuts shall be determined by measurement with a micrometer (see 4.7.24.2(a)), and the cross sectional area of the specimen shall be calculated by the following formula:

Cross sectional area in square inches = TW

Where: T = Measured maximum specimen thickness in region between the two required parallel cuts, in inches; measurement accuracy shall be within plus or minus 0.001 inch.

W = Measured maximum specimen width in region between the two required parallel cuts, in inches; measurement accuracy shall be within plus or minus 0.001 inch.

(c) Cables or cords containing four or fewer conductors: not fewer than four specimens shall be required; not fewer than one of these specimens shall be removed from each insulated conductor.

(d) Cables or cords containing more than four but fewer than 21 conductors: not fewer than four specimens shall be required; each specimen shall be removed from a different insulated conductor.

(e) Cables or cords containing 21 or more conductors: the number of specimens required shall be not less than the square root (rounded, if necessary, to the nearest whole number) of the total number of conductors. Each specimen shall be removed from a different insulated conductor.

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4.7.24.2 Special apparatus. Apparatus shall include the following:

(a) A micrometer with flat, parallel measurement surfaces on both spindle and anvil, which is for the specimen measurement as specified in 4.7.24.1 and which exhibits a measurement error of not more than plus or minus 0.001 inch.
(b) Grinding apparatus (not required for insulation with a cross sectional area of 0.030 in^2 or less, a motor driven grinding wheel and a motor driven buffing wheel, or their equal, for the specimen preparation as specified in 4.7.24.1(b). Guides shall be provided to assure that the specimen is pulled tangentially to the surface of each wheel during grinding and buffing.
(c) A forced-fresh-air circulating oven (if required, see 4.7.24.3), for supporting the requirements as specified in 4.7.24.3(a). The oven air temperature shall be measured in the immediate vicinity of the specimens.
(d) A hermetic chamber (if required, see 4.7.24.3), which shall be filled and internally pressurized with either oxygen or air and then internally heated, as specified in 4.7.24.3(b) or (c), as applicable. (For relative simplicity, it is suggested that a sealable, pressure-proof steel canister be used to contain the specimens, that this canister be pressurized from a regulated source of either oxygen or air, and that this canister be heated by means of immersion in a controlled temperature hot oil bath.) Pressure and temperature shall be automatically regulated, and an appropriate pressure-relief valve shall be provided on the hermetic chamber. These measures will reduce the possibility of spontaneous specimen combustion, and will augment the safety of personnel.
(e) A heated container (if required, see 4.7.24.3), which shall be filled with lubricating oil in accordance with symbol number 2190 TEP of MIL-L-17331, and which shall be for the specimen immersion as specified in 4.7.24.3(d). The oil shall be maintained at a temperature of 121 ± 1°C during specimen immersion.
(f) A motor driven tensile machine which shall be for applying increasing axial tension to the specimen, and which shall automatically indicate within plus or minus 1 percent the maximum axial tension experienced by the specimen prior to rupture. This tensile machine shall incorporate two parallel and opposing jaws, between which the specimen can be secured, as specified in 4.7.24.4(a). The two jaws shall increase their separation at the uniform rate of 20 ± 2 inches per minute by means of the tensile machine motor.

4.7.24.3 Procedure. Specimens shall be maintained at a temperature of 25 ± 3°C for a period of not less than 30 minutes immediately prior to any accelerated aging or testing. Each of the specified physicals (aged) accelerated aging procedures, if any (see 3.1), shall be performed as follows, using the specified number of specimens (see 4.7.24.1) for each procedure. The same specimen shall not be used for more than one physicals (aged) procedure. Each of the specified physicals (unaged) tests, if any (see 3.1), shall be performed as specified in 4.7.24.4.

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(a) Air oven accelerated aging: each specimen shall be freely suspended vertically, secured by one end, within the oven. The air temperature within the oven shall then be raised to the value specified in table XXI, and shall be maintained at this value for a continuous period of not less than that specified in table XXI. Each specimen shall then be removed from the oven and tested as specified in 4.7.24.4.

(b) Air-pressure-heat accelerated aging: each specimen shall be freely suspended vertically, secured by one end, within the heat-pressure chamber. The air pressure within the chamber shall then be raised to 80 ± 2 lb/in^2, and the air temperature within the chamber raised to the value specified in table XXI. This pressure and temperature shall then be simultaneously maintained for a continuous period of not less than that specified in table XXI. Following this, the pressure within the chamber shall be reduced, at a uniform rate, and over a period of not less than 5 minutes, to ambient atmospheric pressure. Each specimen shall then be removed from the chamber and tested as specified in 4.7.24.4.

(c) Oxygen-pressure-heat accelerated aging: each specimen shall be freely suspended vertically, secured by one end, within the heat-pressure chamber. The interior of the chamber shall then be filled with oxygen, the oxygen pressurized to a value of 300 ± 10 lb/in^2, and the oxygen temperature within the chamber raised to the value specified in table XXI. This pressure and temperature shall then be simultaneously maintained for a continuous period of not less than that specified in table XXI. Following this, the pressure within the chamber shall be reduced, at a uniform rate, and over a period of not less than 5 minutes, to ambient atmospheric pressure. Each specimen shall then be removed from the chamber and tested as specified in 4.7.24.4.

(d) Hot-oil immersion accelerated aging: each specimen shall be submerged within the hot oil bath for a continuous period of not less than that specified in table XXI. Each specimen shall then be removed from the hot oil bath, blotted lightly to remove excessive oil and then suspended in air, to ambient room temperature, for 4 ± 1/2 hours. Each specimen shall then be tested as specified in 4.7.24.4.

4.7.24.4 Further procedure. Each of the specified physicals (unaged) tests (see 3.1) shall be performed as follows: the specified number of specimens (see 4.7.24.1) shall be required for each test. In addition, the specimens of each material aged in accordance with 4.7.24.3 shall be tested as specified in (a) below, to the extent specified (see 3.1). Specimens shall be maintained at a temperature of 25 ± 3°C for a period of not less than 30 minutes immediately prior to any testing. Two benchmarks shall be printed on the region of each specimen where the cross sectional area is known (see 4.7.24.1(a) and (b)): these two benchmarks shall be separated by a longitudinal distance of 1 plus 1/64 inch on specimens of polyethylene (thermoplastic) or polypropylene insulation; the two benchmarks shall be separated by a longitudinal distance of 2 ± 1/64 inches on all other specimens. The same specimen shall not be used for more than one test.

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TABLE XXI. Details for accelerated aging procedures.

Specimen material	Procedure	Temperature, (°C)	Minimum period (hours)
Insulation specimens:			
Fluoropolymer (ETFE)	Air oven	180 ± 1	168
Polyethylene (cross-linked)	Air oven	121 ± 2	168
	Air-pressure-heat	127 ± 2	40
Polyethylene (thermoplastic)	Air oven	100 ± 2	96
Polypropylene	Air oven	121 ± 2	96
Polyvinyl chloride (80°C)	Air oven	100 ± 2	96
Polyvinyl chloride (80°C semi-rigid)	Air oven	113 ± 2	168
Polyvinyl chloride (105°C)	Air oven	136 ± 2	96
Rubber (butyl)	Air oven	100 ± 2	168
	Air-pressure-heat	127 ± 2	40
Rubber (ethylene propylene)	Air oven	121 ± 2	168
	Air-pressure-heat	127 ± 2	40
Rubber (synthetic)	Air-pressure-heat	127 ± 2	20
	Oxygen-pressure-heat	80 ± 1	168
Jacketing specimens:			
Polychloroprene	Air-pressure-heat	127 ± 2	20
	Oxygen-pressure-heat	80 ± 1	168
	Hot-oil immersion	121 ± 1	18
Polyether polyurethane	Air oven	80 ± 1	168
Polyethylene (chlorosulphonated)	Air oven	100 ± 2	96
	Air-pressure-heat	127 ± 2	20
	Oxygen-pressure-heat	80 ± 1	168
	Hot-oil immersion	121 ± 1	18
Polyvinyl chloride (special thermoplastic)	Air oven	127 ± 2	168
	Hot-oil immersion	121 ± 1	18
Polyvinyl chloride (standard thermoplastic)	Hot-oil immersion	121 ± 1	18
Silicone rubber	Air oven	260 ± 2	24

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(a) Tensile strength and elongation tests: one specimen shall be selected and secured between the jaws of the tensile machine, such that the region of the specimen which contains both benchmarks shall be suspended between the jaws, and such that the jaws are separated by a distance which shall be not greater than twice the benchmark separation. Care shall be taken during specimen mounting to assure that tension shall be uniformly distributed across the cross sectional area of the specimen during the test, and to assure that the specimen shall be subjected only to axial forces, and not to lateral or torsional forces during the test. The test shall be initiated by starting the tensile machine motor, thereby increasing jaw separation and applying increasing tension to the specimen. The distance between the specimen benchmarks shall be constantly monitored during the test, within an accuracy of plus or minus 1/6 inch. The applied specimen tension and specimen benchmark separation existing immediately prior to specimen rupture shall be recorded if this rupture occurs between benchmarks; if specimen rupture does not occur between benchmarks, then the test results shall be considered invalid, and the test shall be repeated using a different specimen of the same material and aging history. This test shall be successfully repeated, using a different specimen of the same material and aging history each time, until the specified number of specimens (see 4.7.24.1) have been tested. The specimen tensile strength shall then be calculated by using the following formula:

Tensile strength, in lb/in^2 = Median value of F/A for all tested specimens.

Where F = Measured tension in the specimen immediately prior to rupture, in pounds.
A = Measured cross sectional area (see 4.7.24.1) of the specimen, in square inches.

The specimen elongation shall be calculated by using the following formula:

Elongation, in percent = Median value of 100(D/K - 1) for all tested specimens.

Where: D = Measured benchmark separation on the specimen immediately prior to rupture, in inches.
K = Initial benchmark separation (either 1 or 2 inches, as specified in 4.7.24.4).

(b) Set test: one specimen shall be selected and secured between the jaws of the tensile machine, such that the region of the specimen which contains both benchmarks shall be suspended

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between the jaws, and such that the jaws are separated by a distance which shall be not greater than 4 inches. Care shall be taken during specimen mounting to assure that tension shall be uniformly distributed across the cross sectional area of the specimen during the test, and to assure that the specimen shall be subjected only to axial forces, and not to lateral or torsional forces during the test. The tensile machine motor shall then be started, thereby increasing jaw separation and stretching the specimen until the specimen benchmarks are separated by a distance of 6 plus 3/8, minus 1/8 inches. The benchmarks shall be maintained at this separation for a period of 5 ± 2 seconds, after which the specimen shall be immediately released from the lower tensile machine jaw and permitted to contract naturally. Sixty ± 5 seconds after specimen release, the benchmark separation shall be measured, within an accuracy of plus or minus 1/64 inch. This measured benchmark separation shall be recorded, and this test then repeated, using a different specimen of the same material each time, until the specified number of specimens (see 4.7.24.1) have been tested. The specimen set shall then be calculated by using the following formula:

Set, in inches = Median value of (L - 2) for all tested specimens.

Where: L = Measured benchmark separation on the specimen following contraction, in inches.

4.7.24.5 Specimen for (outboard submarine cable) air oven cable jacket test. Specimen shall consist of a length of completed cable approximately 8-1/2 feet long. Ten turns of cable shall be wrapped around a 3-inch diameter mandrel. Upon completion of the conditioning period, specimen shall be allowed to cool to room temperature while still wrapped over the mandrel. Any cracks in the cable jacket which are discernible by visual examination shall be cause for rejection.

4.7.24.6 Specimen for (outboard submarine cable) environmental stress crack test. Specimen shall consist of a length of completed cable approximately 12-feet long. Ten turns of cable shall be wrapped around a 3 inch diameter mandrel and suspended for 24 hours in an aqueous solution containing 10 percent (by volume) of Igepal CO - 630 at 50 ± 2°C. The ends of the specimen shall be kept out of the solution. Upon completion of the conditioning period, specimen shall be allowed to cool to room temperature while still wrapped over the mandrel. Any cracks in the cable jacket which are discernible by visual examination shall be cause for rejection.

4.7.24.7 Observation. Any of the following shall constitute specimen failure:

(a) A calculated specimen tensile strength which is less than that specified (see 3.1).
(b) A calculated specimen elongation which is less than that specified (see 3.1).
(c) A calculated specimen set which is greater than that specified (see 3.1).

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4.7.25 Pressure cycling. This test shall determine the ability of completed cable, which is intended for use external to the hull of submersible vehicles, to maintain constant capacitance and insulation resistance characteristics while repeatedly subjected to high-pressure water.

4.7.25.1 Specimen. The specimen shall consist of a 60 ± 10 foot length of completed cable.

4.7.25.2 Special apparatus. Apparatus shall include the following:

(a) A pressure tank which can be filled with water, and can be internally pressurized to the value specified (see 3.1), and which is for applying this water pressure to the entire specimen, except for its extreme ends, as specified in 4.7.25.3. Either one or two appropriate stuffing tubes shall be welded into the tank wall to provide a means for bringing either one or both specimen ends, respectively, out of the tank while sealing the remainder of the specimen within. Stuffing tubes shall be as specified in MIL-S-24235 and MIL-S-24235/1; a packing assembly, as specified in MIL-S-24235 and MIL-S-24235/2, shall be available for each stuffing tube to allow specimen sealing (see 4.7.25.3). A means shall be provided whereby the pressure within the tank can be measured within an accuracy of plus or minus 5 percent.
(b) Capacitance test apparatus shall be as specified in 4.8.2.2.
(c) Insulation resistance test apparatus shall be as specified in 4.8.5.2.

4.7.25.3 Procedure. One end of the specimen shall be installed through a pressure tank stuffing tube, and shall be sealed in place using the packing assembly as specified in section 3, group A of Drawing 803-5001027. This fitting shall be done such that approximately 1 foot of the specimen end extends outside of the pressure tank, and such that the remainder of the specimen is within the tank. If the tank is provided with a second stuffing tube, the opposite specimen end shall be fitted through this second stuffing tube using the same procedure, such that approximately 1 foot of each specimen end extends outside of the tank. If the tank is not provided with a second stuffing tube, then the opposite specimen end shall be sealed, such that the capacitance and insulation resistance characteristics of the specimen are unaffected, and such that the specimen end will withstand the pressure cycling to follow without incurring water penetration; this specimen end shall then remain within the tank. The tank shall then be closed and filled with water, such that the portion of the specimen within the tank shall remain completely submerged during the remainder of the test. The specimen shall then be subjected to the capacitance and insulation resistance tests as specified in 4.8.2 and 4.8.5, respectively using a specimen end which is external to the tank and which has been prepared. Following this, the specimen shall be subjected to five consecutive pressure cycles as follows:

(a) First and fifth cycle: the water within the tank shall be pressurized to a value of not less than that specified (see 3.1), and shall be maintained at this pressure for a period of not less than 30 minutes. Following this, and while the water pressure remains as specified (see 3.1), the specimen

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shall be subjected to the capacitance and insulation resistance tests as specified in 4.8.2 and 4.8.5, respectively. The water pressure shall then be completely released.

(b) Second, third and fourth cycle: the water within the tank shall be pressurized to a value of not less than that specified (see 3.1). The specimen shall then be immediately subjected to the capacitance and insulation resistance tests as specified in 4.8.2 and 4.8.5, respectively. Following this, the water pressure shall be completely released.

After completion of the fifth cycle, the specimen shall again be subjected to the capacitance and insulation resistance tests as specified in 4.8.2 and 4.8.5, respectively.

4.7.25.4 Observation. Either of the following shall constitute specimen failure:

(a) Specimen fails the capacitance test at any time (see 4.8.2.4).
(b) Specimen fails the insulation resistance test at any time (see 4.8.5.4).

4.7.26 Specific gravity. The specific gravity of extruded silicone rubber insulation shall be determined in accordance with method D of ASTM D 297.

4.7.27 Stress endurance. This test shall determine the ability of completed cable, which is intended for use by divers, to withstand repeated traveling bends, such as might be encountered during in-service cable payout and retrieval.

4.7.27.1 Specimen. The specimen shall consist of a piece of completed cable which shall have sufficient length for use in the test as specified in 4.7.27.3.

4.7.27.2 Special apparatus. Apparatus shall include the following:

(a) Test apparatus: mechanism for drawing the specimen alternately back and forth, for a distance of not less than 18 inches in each direction, around a pulley, while maintaining the specimen under tension. The pulley shall have a diameter of not more than 3 inches, and the specimen shall seat within the pulley periphery over an arc of not less than 90 degrees at all times during the test. Specimen tension shall be maintained by means of freely suspended weights attached to the specimen end opposite that being drawn. A 490 ± 5 pound weight shall be used alternately with a 180 ± 5 pound weight as specified in 4.7.27.3.
(b) Voltage withstand test apparatus shall be as specified in 4.8.8.2.

4.7.27.3 Procedure. The specimen shall be drawn alternately back and forth in the test apparatus for a total of not fewer than 500 cycles (where each cycle shall consist of not less than 18 inches of specimen travel in each direction, a total travel per cycle of not less than 36 inches). The 490 ± 5 pound weight shall be attached to the specimen end during each odd numbered cycle, and the 180 ± 5 pound weight shall be attached to the specimen end

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during each even numbered cycle. The specimen shall be drawn through the pulley at the same constant rate and for the same distance in each direction during each cycle. Following this cycling, the specimen shall be removed from the test apparatus.

4.7.27.4 Observation. Specimen failure shall be construed if the specimen exhibits visible distortion or jacket cracking. If the specimen exhibits no visible distortion or jacket cracking, it shall be further tested as specified in 4.7.27.5.

4.7.27.5 Further procedure (may not be required, see 4.7.27.4). The specimen shall be subjected to the voltage withstand test as specified in 4.8.8. Following this, the region of the specimen which had been drawn back and forth across the pulley of the test apparatus shall be dissected, and each of its components visually inspected for deterioration.

4.7.27.6 Further observation. Any of the following shall constitute specimen failure:

(a) Specimen fails the voltage withstand test (see 4.8.8.4).
(b) Specimen, upon dissection and inspection, exhibits visible distortion or cracking of any specimen component, including strand breakage on the steel strength member or on any conductor, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service.

4.7.28 Tear strength. This test shall determine the ability of elastomeric jacketing material to withstand the propagation of a cut passing through a portion of its length.

4.7.28.1 Specimen. Each specimen shall consist of a piece of jacketing material, which shall have been cut to the dimensions specified in ASTM D 470. Specimens shall be removed from completed cable or cord. Not fewer than five specimens shall be required.

4.7.28.2 Special apparatus. Apparatus shall include the following:

(a) A micrometer with flat, parallel measurement surfaces on both spindle and anvil, which shall be for the specimen measurements as specified in 4.7.28.3 and which shall exhibit a measurement error of not more than plus or minus 0.001 inch.
(b) A motor driven tensile machine which shall be for applying increasing tension to the specimen, and which shall automatically indicate within plus or minus 1 percent the maximum tension experienced by the specimen prior to tearing. This tensile machine shall incorporate two parallel and opposing jaws, between which the specimen can be secured, as specified in 4.7.28.3. The two jaws shall increase their separation at the uniform rate of 20 ± 2 inches per minute by means of the tensile machine motor.

4.7.28.3 Procedure. The specimen shall be tested in accordance with ASTM D 470.

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4.7.28.4 Observation. Specimen failure shall be construed if the median value of the measured tear strengths of all specimens is less than that specified (see 3.1).

4.7.29 Tension. This test shall determine the ability of completed cable, which is intended for in-sea towing use, to withstand axial tension such as might be encountered during service.

4.7.29.1 Specimen. The specimen shall consist of a piece of completed cable, of not less than 6 feet in length.

4.7.29.2 Special apparatus. Apparatus shall include the following:

(a) A tensile machine which shall be for applying and maintaining the specified axial tension (see 3.1) to the specimen, and which shall automatically indicate this tension within plus or minus 1 percent. This tensile machine shall incorporate two opposing jaws, for use with cable grips (see (b) below), and between which the specimen can be secured by using these grips, as specified in 4.7.29.3.
(b) Two woven cable grips, for the use specified in 4.7.29.3.
(c) Voltage withstand test apparatus shall be as specified in 4.8.8.2.

4.7.29.3 Procedure. One cable grip shall be secured to each specimen end, and one jaw of the tensile machine shall then be secured to each cable grip. Care shall be taken during specimen mounting to assure that the specimen will be subjected only to axial forces, and not to lateral or torsional forces during the test. Jaw separation shall next be increased until a tension of not less than that specified (see 3.1) is applied to the specimen. Specimen tension shall be maintained for a continuous period of not less than 1 hour; specimen tension shall then be released and the specimen removed from the tensile machine. The specimen shall then be subjected to the voltage withstand test as specified in 4.8.8.

4.7.29.4 Observation. Either of the following shall constitute specimen failure:

(a) Specimen exhibits visible distortion or cracking, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable in service.
(b) Specimen fails the voltage withstand test (see 4.8.8.4).

4.7.30 Thermoplastic flow. This test shall determine the ability of thermoplastic polyethylene insulation, which serves as the sole separator between a wire conductor and a concentric braid, to resist physical deformation when subjected to an elevated temperature.

4.7.30.1 Specimen. The specimen shall consist of a piece of completed cable or cord which shall have sufficient length for use in the test as specified in 4.7.30.3. Wire conductors (if more than one) shall be connected together electrically at one or both cable ends; similarly, braided conductors (if more than one) shall be connected together electrically at one or both cable ends.

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4.7.30.2 Special apparatus. Apparatus shall include the following:

(a) A mandrel shall be a rigid, smooth cylinder, which shall have a diameter of not more than 12 nor less than 10 times the specified maximum overall cable or cord diameter (see 3.1), and which shall be of length and construction for the specimen wrapping as specified in 4.7.30.3.
(b) A short circuit monitor of not less than 70 V, either dc or rms ac, which shall be for attachment to the specimen through a normal-blow fuse of not greater than 1/4 A rating. The voltage source shall have the capacity to supply the specimen with a current which shall be not less than twice the fuse rating.
(c) A heating chamber for supporting the requirements as specified in 4.7.30.3. The chamber air temperature shall be measured in the immediate vicinity of the specimen.

4.7.30.3 Procedure. The specimen shall be tightly hand-wrapped, for not fewer than two, circumferential and contiguous, 360-degree turns around the mandrel. The mandrel, with specimen attached, shall be placed within the heating chamber, and the short circuit monitor test voltage shall be applied between the wire conductors and the braided conductors of the specimen. The specimen shall then be subjected to the specified air temperatures (see 3.1) for a continuous period of not less than 6 hours.

4.7.30.4 Observation. Specimen failure shall be construed if the short circuit monitor fuse blows prior to completion of the specified test period.

4.7.31 Twisting endurance. This test shall determine the ability of completed cable and cord, which is intended to be flexed during use, to withstand repeated and reversing torsional motion while subjected to a specified temperature.

4.7.31.1 Specimen. The specimen shall consist of a piece of completed cable or cord which shall have a length of not less than the sum of 18 inches plus 10 times the specified maximum overall cable or cord diameter (see 3.1), except that no specimen shall have a length of less than 24 inches. Specimen conductor ends shall be electrically interconnected such that a single, series circuit is formed by the conductors. The two end conductors of this series circuit shall be connected as specified in 4.7.31.3.

4.7.31.2 Special apparatus. Apparatus shall include the following:

(a) A twisting machine for suspending the specimen vertically, and for automatically twisting the upper specimen end alternately clockwise and counterclockwise by 180 plus 10, minus 0 degrees with respect to the lower specimen end as described below.

(1) The specimen shall be suspended and repeatedly twisted from one extremity by means of a clamp, which shall be fastened to a motor driven, rotating and reversing mechanism (such as a pinion connected to a rack, which

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in turn is made to reciprocate by an arm from a motor). The upper specimen end shall be twisted about its axis at a rate of not more than 14, nor less than 12 cycles per minute. Each cycle shall produce 180 plus 10, minus 0 degrees of clockwise axial specimen rotation plus 180 plus 10, minus 0 degrees of counterclockwise axial specimen rotation (360 plus 20, minus 0 degrees total rotation). The upper end of the specimen jacket shall be a distance of not less than 1 inch above the nearest edge of the clamp.

(2) A weight shall be provided for applying tension to the specimen. This weight shall be attached to the lower specimen extremity by means of a second clamp, which shall be constrained from rotating but which shall be free to move in the vertical direction at all times during the test. The weight shall be chosen such that the:

Tension from the second clamp plus the weight in pounds = $10(3.14159)d^2 \pm (3.14159)d^2 \div 2$

Where: d = Specified maximum overall cable or cord diameter, in inches (see 3.1).

This second clamp shall be separated from the rotating clamp by a distance which shall not exceed the sum of 6 inches plus 10 times the specified maximum overall cable or cord diameter (see 3.1). The second clamp and weight shall be constructed such that tension is applied vertically downward along the specimen axis. The lower end of the specimen jacket shall be a distance of not less than 1 inch below the nearest edge of clamp.

(3) Both the rotating clamp and the tensioning weight clamp shall apply radial compression to the specimen that neither the specimen nor any of its internal components will slip in the clamping area.

(4) An automatic counter shall be provided to total the number of twisting cycles performed on the specimen during a test.

(5) The specimen shall not come into contact with any piece of machinery, except for the rotating clamp, the tensioning weight clamp and the circuit continuity monitor (see (b) below), at any time during the test.

(b) Circuit continuity monitor (a means for continuously monitoring the electrical continuity of the series-connected specimen conductors) shall automatically stop the motion of the rotary clamp as by removing motor power when a specimen conductor breaks.

(c) A test chamber for maintaining the specified air temperature (see 3.1) within plus or minus 2°C, for accommodating the twisting machine with specimen attached and for supporting the requirements as specified in 4.7.31.3. The chamber air

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temperature shall be measured in the immediate vicinity of the specimen, approximately midway between the two specimen clamps of the twisting machine. The test chamber shall have a viewing port for observing the specimen under test.

(d) Voltage withstand test apparatus shall be as specified in 4.8.8.2.

4.7.31.3 Procedure. The twisting machine shall be installed in the test chamber, and the specimen shall be installed in the twisting machine. The circuit continuity monitor shall be connected between the two end conductors of the single series circuit within the specimen. The test chamber doors shall then be closed, and the air temperature within the chamber shall be brought to the specified value (see 3.1) within plus or minus 2°C. After the air temperature within the chamber has been maintained at this value for a period of not less than 1 hour, the automatic counter shall be set initially to zero and the twisting endurance test begun by turning on the motor driven, rotating clamp. The specified number of twisting cycles (see 3.1) shall be performed, unless specimen failure (see 4.7.31.4) occurs prior to the completion of this number of cycles. The test may then be terminated prematurely. At all times during the test, the temperature within the chamber shall be maintained as specified (see 3.1) within plus or minus 2°C and the test chamber doors shall remain closed. Following the test, the specimen shall be removed from the twisting machine and the test chamber.

4.7.31.4 Observation. Either of the following shall constitute specimen failure, except that specimen damage within 2 inches of the location where either specimen clamp had been fastened shall not constitute specimen failure, but shall require a complete retest using a different specimen from the same length of cable or cord:

(a) Rotary clamp stoppage caused by the circuit continuity monitor (indicating specimen conductor breakage), prior to completion of the specified number of twisting cycles (see 3.1).

(b) Rupture of the specimen jacket prior to completion of the specified number of bending cycles (see 3.1).

Specimens exhibiting no failure shall be tested further, as specified in 4.7.31.5.

4.7.31.5 Further procedure (may not be required, see 4.7.31.4). The specimen shall be allowed to return to room temperature and shall be subjected to the voltage withstand test as specified in 4.8.8, using the specified voltages (see 3.1). Following this, the portion of the specimen which had been between the specimen clamps of the twisting machine shall be dissected, and each of its constituent components visually inspected for deterioration.

4.7.31.6 Further observation. Either of the following shall constitute specimen failure, except that specimen damage within 2 inches of the location where either specimen clamp had been fastened shall not constitute specimen failure, but shall require a complete retest of twisting endurance using a different specimen from the same length of cable or cord.

(a) Specimen fails the voltage withstand test (see 4.8.8.4).

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(b) Specimen, upon dissection and inspection, exhibits visible distortion or cracking of any specimen component, including strand breakage on any conductor, or exhibits any other visible deterioration of such a nature or extent as to impair the performance of the cable or cord in service.

4.7.32 Visual and dimensional examination. This test shall be to determine whether or not completed cable and cord is assembled in accordance with the requirements of this specification.

4.7.32.1 Specimen. The specimen shall consist of an entire and continuous ordered length of completed cable or cord.

4.7.32.2 Procedure. The specimen shall be examined and measured, as required, to determine conformance with overall jacket dimension, jacket color, surface marking, surface condition, braided armor, armor paint and jacket repair requirements, as applicable (see 3.1, 3.3, 3.4 and 3.7). A portion of the specimen shall then be dissected, and each of its constituent components counted, examined and measured, as required, to determine conformance with all constructional, dimensional, material, identification, color, and insulation repair requirements, as applicable (see 3.1, 3.3, 3.4 and 3.7). Following this, and following installation of end seals on the specimen, these end seals shall be examined to determine conformance with materials and application specifications (see 5.2).

4.7.32.3 Observation. Specimen failure shall be construed if the specimen, its assembly or any of its constituent components is not as specified (see 3.1, 3.3, 3.4 and 3.7).

4.7.33 Watertightness. This test shall determine the ability of completed cable, which is intended to pass through watertight bulkheads, to prevent the longitudinal flow of low-pressure water.

4.7.33.1 Specimen. The specimen shall consist of a 60 $\pm$ 2 inch length of completed cable. The specimen shall be cut to length using a scissors-action cable cutter. (Saws shall not be used.)

4.7.33.2 Special apparatus. Apparatus shall include the following:

(a) A source of pressurized water, which shall be provided at a regulated pressure of 25 $\pm$ 1/2 lb/in^2 and which shall be for use with the terminal fitting (see (b)) as specified in 4.7.33.3.

(b) A metal terminal fitting which applies the source of pressurized water to one end of the specimen, which supports the requirements as specified in 4.7.33.3, and which shall be fabricated as specified in the following. Figure 14 shows one possible arrangement for the terminal fitting. The fitting shall admit the specimen end for the distance specified in 4.7.33.3, and shall have an id, where it fits over the specimen, of not greater than the measured overall specimen diameter plus 1/2 inch. The fitting shall have a means for introducing the source of pressurized water to the specimen end, and a means

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for bleeding off any air which might be trapped between the specimen end and the source of pressurized water. The fitting shall also have an aperture for introducing a hardening sealant (see (c)), to produce a pressure-tight bond between the fitting and the specimen jacket. A plug (such as a thick wrapping of rubber tape around the specimen) shall be provided at the place where the specimen enters the fitting, to prevent sealant loss and to approximately center the specimen within the fitting.

(c) A liquid hardening sealant which shall be for producing a pressure-tight bond between the fitting and the specimen jacket, when applied as specified (see 4.7.33.3) and allowed to harden. The sealant shall be at a temperature of not greater than 100°C when poured into the terminal fitting, and shall not expand in volume while hardening. An appropriate metal alloy is recommended as a sealant since it can be remelted and reused. Any metal alloy used shall have a melting point of not greater than 88°C.

4.7.33.3 Procedure. The specimen shall be secured in a vertical position, the terminal fitting (including the plug) shall be fitted over its upper end, and the hardening sealant shall be poured through the appropriate terminal fitting aperture to bond the fitting to the specimen. These shall be done such that the specimen is approximately centered within the terminal fitting, such that not more than a 6-inch length of specimen jacket is exposed to the sealant, such that the end face of the specimen is located a distance of not more than 1 inch above the sealant surface and itself contains no sealant, and such that the aperture through which the sealant was introduced is itself made pressure-tight. The sealant shall be allowed to harden, following which the terminal fitting plug shall be removed or cut away to the extent that it exerts no radial compression on the specimen. The specimen shall then be positioned such that it is straight, and such that no portion of the specimen is at a greater elevation than the specimen end within the terminal fitting. The fitting shall then be connected to the source of pressurized water, and excess air within the fitting shall be bled off until it can be certain that the entire enclosed specimen end face is exposed to the pressurized water for the remainder of the test. The source of pressurized water shall remain connected to the fitting for a continuous period of 6 plus 1/4, minus 0 hours. During this test period, water (if any) leaking from the specimen end opposite the terminal fitting shall be collected into an initially dry container. Following the test period, the volume of this collected water shall be measured.

4.7.33.4 Observation. Specimen failure shall be construed if the collected water volume exceeds that shown on table XXII.

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TABLE XXII. Limits for water leakage.

Sum of specified cross sections of all specimen conductors (see 3.1) (circular mils)	Collected water volume, (in^3 maximum)
Less than 9,001	4
9,001 to 15,000	5
15,001 to 25,000	6
25,001 to 50,000	8
50,001 to 100,000	9
100,001 to 200,000	10
200,001 to 500,000	11
500,001 to 800,000	12
800,001 or greater	13

4.8 Tests methods (electrical).

4.8.1 Attenuation. This test shall determine the ability of completed cable, which is intended for the transmission of data, voice or control information, to propagate a signal at a specific frequency without causing an unacceptable reduction of signal amplitude.

4.8.1.1 Specimen. The specimen shall consist of a piece of completed cable or cord, which shall have sufficient length to exhibit an electrical attenuation of not less than 3 decibels (dB) at the specified test frequency (see 3.1). Individual transmission lines within this specimen (where each single insulated conductor plus surrounding shield or each insulated conductor pair plus surrounding shield is considered to be a transmission line) shall be selected for testing as follows:

(a) Specimens containing four or fewer transmission lines: every transmission line shall be tested.
(b) Specimens containing more than four but fewer than 21 transmission lines: four different transmission lines shall be tested. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.
(c) Specimens containing 21 or more transmission lines: the number of transmission lines tested shall be equal to the square root (rounded, if necessary, to the nearest whole number) of the total number of transmission lines. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.

An electrical connector, which shall be designed for use at the specified test frequency (see 3.1), and which shall be appropriate for use with the specimen, shall be attached to each end of each transmission line selected for testing.

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4.8.1.2 Special apparatus. Apparatus shall include the following, and shall be used at the specified test frequency (see 3.1), and shall be chosen such that all electrical interconnections (see 4.8.1.3) shall be of the same nominal characteristic impedance throughout (except that connectors and specimens in which a single conductor plus surrounding shield is tested may be of a different impedance than this apparatus, if necessary).

(a) A sinusoidal signal generator, which shall produce an unchanging signal of the specified test frequency (see 3.1) within a frequency accuracy of 0.25 percent. The signal generator may be amplitude modulated in a constant and unchanging manner if the detector (see (b)), requires an amplitude modulated signal for proper operation. The signal generator shall generate power to produce a reasonable indication on the detector when used as specified in 4.8.1.3, however, the signal generator shall not generate so much power that the specimen or any piece of test apparatus will consequently experience a significant temperature rise.

(b) A crystal diode, bolometer or other detector, with its associated indicating device, which shall be used as specified in 4.8.1.3.

(c) A calibrated variable attenuator, which shall produce a maximum attenuation greater than that exhibited by the specimen, and which shall resolve attenuation within an accuracy of plus or minus 0.25 dB.

(d) Two fixed attenuators, each of which shall exhibit an attenuation of not less than 10 dB.

(e) Impedance matching transformers (required only when testing a conductor pair plus surrounding shield): two, two-winding transformers; one winding of each transformer shall have an impedance equal to the nominal impedance of the fixed attenuators; the second winding of each transformer shall be centertapped, and shall have an impedance equal to the specified impedance of the specimen (see 3.1). When testing a single conductor plus surrounding shield, impedance matching transformers may be used, if necessary, to match the nominal unbalanced impedance of each fixed attenuator (see (d)), to the specified unbalanced impedance of the specimen (see 3.1), as shown on figure 15.

(f) An adapter connector (may not be required, see 4.8.1.3) for joining the two fixed attenuators directly together. This adapter connector shall exhibit an attenuation which shall be not greater than 2 percent of the specimen attenuation. If desired, the adapter connector may be fabricated in the same manner and of the same components as the specimen, except that the shortest practicable length of completed cable or cord shall be used to join the two connectors.

4.8.1.3 Procedure. The test apparatus shall be electrically interconnected as shown on figure 15 (when the transmission line to be tested is a single insulated conductor plus surrounding shield) or as shown on figure 16 (when the transmission line to be tested is an insulated conductor pair plus surrounding shield), as applicable; one of the selected transmission lines shall then be connected where shown. Following this, the calibrated variable

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attenuator shall be adjusted to a low value, and the signal generator adjusted to produce a test signal of sufficient amplitude to produce a reasonable indication on the detector. The setting on the calibrated variable attenuator and the indication on the detector shall then be noted. Next, the transmission line shall be disconnected from the test apparatus, and the two fixed attenuators shall be connected together either directly, or, if necessary, by means of an adapter connector. The signal generator shall not be readjusted. The calibrated variable attenuator shall then be readjusted until the detector produces the same indication as was noted when the transmission line was connected to the test apparatus. The new setting on the calibrated variable attenuator shall then be noted, and the transmission line attenuation shall be calculated by using the following formula:

$$\text{Attenuation, in dB per 100 feet} = 100\ (A_2 - A_1)/L$$

Where: A_1 = Initial setting of the calibrated variable attenuator (when the transmission line is connected to the test apparatus), in dB.
A_2 = Final setting of the calibrated variable attenuator (following transmission line removal from the test apparatus), in dB.
L = Measured specimen length, in feet; measurement accuracy shall be within plus or minus 1 percent.

In an identical manner, the remaining selected transmission lines shall also be tested.

4.8.1.4 Observation. Specimen failure shall be construed if any transmission line exhibits an attenuation which is greater than that specified (see 3.1).

4.8.2 Capacitance. This test shall determine the capacitive characteristics of completed cable and cord.

4.8.2.1 Specimen. The specimen shall consist of a 122 ± 1/2 inch length of completed cable or cord, which shall have overall components (such as armors, jackets or overall binders) removed from each end for a distance of 2 plus 1/2, minus 0 inches, thereby exposing the insulated conductors (and shields, if applicable) at each end of the cable core. Each exposed shield (if any) at each cable end shall be unwrapped from the end of its enclosed insulated conductor(s) for a distance of 1 ± 1/8 inch; the unwrapped portion of each shield shall then be tightly twisted to form a pigtail for electrical connection. The insulation shall next be stripped from each end of each insulated conductor for a distance of 1/2 ± 1/8 inch. If the characteristic impedance test is not required (see 3.1), then the insulation may be stripped, for the specified distance from all of the insulated conductors at one specimen end only.

4.8.2.2 Special apparatus. Apparatus shall include an instrument for measuring capacitance (such as a capacitance bridge, Q-meter or other instrument) which shall be for supporting the requirements as specified in 4.8.2.3. This apparatus shall have a two-terminal measurement port, which shall use a sinusoidal wave of the specified frequency (see 3.1) for making measurements, and shall make measurements within an accuracy of plus or minus 1 percent.

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4.8.2.3 Procedure. The appropriate capacitance test method shall be selected and performed from the following:

(a) Specimens incorporating any number of individually shielded, insulated-conductor transmission lines: one transmission line shall be selected for measurement, and each end shall be positioned such that its conductor and shield shall not come into mutual electrical contact, nor into electrical contact with any other conductor or shield which may be within the specimen. One end of this selected transmission line shall then be appropriately connected electrically to the capacitance measuring instrument, and its capacitance shall be measured. In an identical manner, the capacitance of every other transmission line within the specimen shall also be measured. The capacitance per foot of each transmission line shall then be calculated by using the following formula:

Capacitance in picofarads per foot = C/10

Where: C = Measured capacitance of the transmission line, in picofarads.

(b) Specimens incorporating any number of individually shielded, insulated-conductor-pair transmission lines: one transmission line shall be selected for measurement, and each end shall be positioned such that neither its conductors nor its shield come into mutual electrical contact, or into electrical contact with any other conductor or shield which may be within the specimen. Three capacitances shall then be measured at one end of this transmission line as follows, where one conductor of this line has been designated as conductor number 1 and the other conductor of this line has been designated as conductor number 2 as follows:

(1) Conductor number 1 and the shield shall both be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 2 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_a.
(2) Conductor number 2 and the shield shall next be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 1 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated by C_b.
(3) Both conductor number 1 and conductor number 2 shall then be electrically connected to one terminal of the capacitance measuring instrument, and the shield shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_c.

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The mutual capacitance per foot and, if specified (see 3.1), the capacitance unbalance (sometimes called the coefficient of asymmetry) shall then be calculated by using the following approximate formulas:

Mutual capacitance, in picofarads per foot = $(C_a + C_b)/20 - C_c/40$

Capacitance unbalance, in percent = $10|C_a - C_b/C_m$

Where: C_a, C_b and C_c are as defined in (1), (2) and (3) respectively, in picofarads.
C_m = Calculated mutual capacitance, in picofarads per foot.

In an identical manner, the mutual capacitance and, if specified (see 3.1), the capacitance unbalance of every other transmission line shall also be measured and calculated.

(c) Specimens incorporating multiple, unshielded, insulated-conductor-pair transmission lines: the mutual capacitance and, if specified (see 3.1), the capacitance unbalance of every transmission line shall be determined, using the same method as used for specimens incorporating individually-shielded, insulated-conductor-pair transmission lines, with the following change: the overall specimen braid (if any) and all specimen conductors except the conductor pair under test shall be connected together electrically and considered to be the shield of the transmission line under test.

(d) Specimens incorporating any number of individually shielded, insulated-conductor-triad transmission lines: one transmission line shall be selected for measurement, and each end shall be positioned such that none of its conductors, nor its shield, come into mutual electrical contact, nor into electrical contact with any other conductor or shield which may be within the specimen. Four capacitances shall then be measured at one end of this transmission line as follows, where one conductor of this line has been designated as conductor number 1, another conductor of this line has been designated as conductor number 2, and the remaining conductor of this line has been designated as conductor number 3 as follows:

(1) Conductors number 2 and 3 and the shield shall be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 1 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_a.

(2) Conductors number 1 and 3 and the shield shall next be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 2 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_b.

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(3) Conductors number 1 and 2 and the shield shall next be electrically connected to one terminal of the capacitance measuring instrument, and conductor number 3 shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_c.

(4) Conductors number 1, 2 and 3 shall then be electrically connected to one terminal of the capacitance measuring instrument, and the shield shall be electrically connected to the other instrument terminal. The capacitance shall then be measured and designated as C_d.

The mutual capacitance per foot and, if specified (see 3.1), the capacitance unbalance (sometimes called the coefficient of asymmetry) shall then be calculated by using the following approximate formulas:

a. $$C_m = \frac{C_a + C_b + C_c}{40} - \frac{C_d}{120}$$

b. The capacitance unbalance or coefficient of asymmetry (K) of a shielded triad, expressed in percent shall be the greatest of K_1, K_2 or K_3 and shall be determined by the formula:

$$K_1 = \frac{200\ (C_a - C_b)}{(C_a + C_b)}$$

$$K_2 = \frac{200\ (C_a - C_c)}{(C_a + C_c)}$$

$$K_3 = \frac{200\ (C_b - C_c)}{(C_b + C_c)}$$

Where: K_1 = The percent of capacitance unbalance of conductor No. 1 in relation to conductor No. 2.
K_2 = The percent of capacitance unbalance of conductor No. 2 in relation to conductor No. 3.
K_3 = The percent of capacitance unbalance of conductor No. 3 in relation to conductor No. 1.

Note: In the formula, the relative positions of C_a, C_b and C_c may be transposed as necessary to avoid negative values.

In an identical manner, the mutual capacitance and, if specified (see 3.1), the capacitance unbalance of every other transmission line shall also be measured and calculated.

(e) Specimens incorporating multiple unshielded insulated conductor triad lines: the mutual capacitance and, if specified (see 3.1), the capacitance unbalance of every transmission line shall be determined, using the same method as used in (d),

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with the following change: the overall specimen braid (if any) and all specimen conductors except the conductor triad under test shall be connected together electrically and considered to be the shield of the transmission line under test.

4.8.2.4 Observation. Specimen failure shall be construed if the specimen exhibits a mutual capacitance or capacitance unbalance which is greater than that specified (see 3.1).

4.8.2.5 If the characteristic impedance test is specified (see 3.1), then the specimen shall be retained for use in that test.

4.8.3 Characteristic impedance. This test shall determine the characteristic impedance encountered by signals propagating along completed cable and cord. The mutual capacitance (see 4.8.2) shall be calculated for each transmission line within the completed cable or cord prior to performing the characteristic impedance test. This is necessary since values of mutual capacitance are used to calculate values of characteristic impedance.

4.8.3.1 Specimen. The specimen shall be that which was previously used to determine capacitance characteristics (see 4.8.2.1).

4.8.3.2 Special apparatus. Apparatus shall include an instrument for measuring inductance (such as an inductance bridge, Q-meter or other instrument) which shall support the requirements as specified in 4.8.3.3. This apparatus shall have a two-terminal measurement port, which shall use a sinusoidal wave of the specified frequency (see 3.1) for making measurements, and shall make measurements within an accuracy of plus or minus 1 percent.

4.8.3.3 Procedure. Select and perform the appropriate characteristic impedance test method from the following:

(a) Specimens incorporating any number of individually shielded, insulated-conductor transmission lines shall be laid straight. At one specimen end, the exposed conductor of each transmission line shall be electrically connected to the pigtail of its associated shield by tightly twisting both together. At the opposite specimen end, one transmission line shall be selected for measurement, and shall be connected electrically to the inductance measuring instrument. The inductance of this transmission line shall then be measured. In an identical manner, the inductance of every other transmission line within the specimen shall also be measured. The characteristic impedance of each transmission line shall then be calculated by using the following formula:

$$\text{Characteristic impedance, in ohms} = (10^5 L/C_m)^{1/2}$$

Where: C_m = Mutual capacitance, in picofarads per foot, as calculated in 4.8.2.3.
L = Measured inductance, in microhenries/10 feet.

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(b) Specimens incorporating any number of individually shielded or unshielded, insulated-conductor-pair transmission lines shall be laid straight. At one specimen end, the two exposed insulated conductors of each transmission line shall be laid adjacent to each other (to minimize superfluous inductance), and electrically connected by tightly twisting both together. At the opposite specimen end, one transmission line shall be selected for measurement, and its two conductors shall be connected electrically to the inductance measuring instrument. Its shield (if any) shall remain disconnected. The inductance of this transmission line shall then be measured. In an identical manner, the inductance of every other transmission line within the specimen shall also be measured. The characteristic impedance of each insulated conductor pair shall then be calculated by using the formula shown in (a).

4.8.3.4 Observation. Specimen failure shall be construed if the specimen exhibits a characteristic impedance other than that specified (see 3.1).

4.8.4 Conductor resistance. This test shall determine whether or not conductors within completed cable and cord are fabricated from the specified material.

4.8.4.1 Specimen. The specimen shall consist of a piece of completed cable or cord which shall have length to assure that the test as specified in 4.8.4.3 shall be performed with the specified accuracy (see 4.8.4.2). Individual conductors within the specimen shall be selected for testing as follows:

(a) Conductors with cross sections of less than 1400 circular mils shall all be tested.
(b) Conductors with cross sections of 1400 circular mils or more - conductors shall be selected for testing as follows:

(1) Specimens containing four or fewer such conductors: every such conductor shall be tested.
(2) Specimens containing more than four but fewer than 21 such conductors: not fewer than four such conductors shall be tested.
(3) Specimens containing 21 or more such conductors: the number of such conductors tested shall be equal to the square root (rounded, if necessary, to the nearest whole number) of the total number of such conductors.

4.8.4.2 Special apparatus. Apparatus shall include an instrument for measuring resistance (such as a Kelvin-type double bridge, a milliohmmeter or a Wheatstone bridge), which shall support the requirements as specified in 4.8.4.3, and which shall make measurements within an accuracy of plus or minus 1 percent.

4.8.4.3 Procedure. Each selected specimen conductor shall be tested in accordance with ASTM B 193 (except as otherwise specified herein).

4.8.4.4 Observation. Specimen failure shall be construed if any specimen conductor exhibits a conductor resistance in excess of that specified (see 3.1).

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4.8.5 Insulation resistance. This test shall determine the ability of completed cable and cord to transmit dc without exhibiting unacceptable current flow through insulation or jacketing material, either between conductors, or between these conductors and any conducting substance which may be in contact with the surface of the cable or cord during in-service use. The voltage withstand test (see 4.8.8) shall be performed prior to performing the insulation resistance test.

4.8.5.1 The conductor resistance test shall be performed on all conductors which have a nominal circular mil area of less than 1600; for conductors which have a nominal circular mil area of 1600 or greater, the conductor resistance test shall be performed on conductors selected as follows:

No. of insulated conductors in cable or cord	No. of insulated conductors selected for test
1 thru 4	All
5 thru 20	4
21 thru 30	5
31 thru 42	6
43 thru 56	7
57 thru 72	8
73 thru 91	9
92 thru 110	10
111 thru 132	11
133 thru 156	12

4.8.5.2 Specimen. The specimen shall be that which was previously used to perform the voltage withstand test (see 4.8.8.1).

4.8.5.3 Special apparatus. Apparatus shall include the following:

(a) A megohmmeter, (or megohm bridge or other suitable instrument) which shall use a dc test voltage of not less than 200 V nor greater than 500 V, which shall support the requirements as specified in 4.8.5.4, and which shall make measurements within an accuracy of plus or minus 10 percent.
(b) Water tank (may not be required, see 4.8.5.4(d)) which shall be of size to contain the specimen, and which shall be suitable for the use as specified in 4.8.5.4(d).

4.8.5.4 Procedure. The specimen shall be maintained at a constant temperature, within plus or minus 3°C, for a period of not less than 5 hours immediately prior to testing. This temperature shall be noted and shall be used to calculate F (see formula below) when measurements are made in accordance with (a), (b) or (c). The megohmmeter shall then be electrically connected to the specimen as specified (see (a), (b), (c) and (d)). For each connection, the test voltage from the megohmmeter shall be continuously applied to the specimen for a period of 1 minute, plus 25 percent, minus 0 percent, immediately following which the resistance reading of the megohmmeter shall be noted. Care shall be taken during each such measurement to assure that any electrical jumper leads connecting the specimen to the megohmmeter shall be spaced sufficiently far from each other and from any common conducting surface that measurement accuracy shall not be impaired.

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(a) Specimens incorporating one or more individually shielded insulated-conductor transmission lines: one transmission line shall be selected for measurement. One conductor of this transmission line shall be electrically connected to one terminal of the megohmmeter, and the shield and all other conductors (if any) of this transmission line shall be electrically connected to the other megohmmeter terminal. The resistance measurement shall then be performed as specified. In an identical manner, the resistance between each remaining conductor in this transmission line (if any) and the short circuited combination of all other conductors (if any), plus the shield in this transmission line shall be measured. Following this, and in an identical manner, resistance measurements shall be made for all remaining transmission lines within the specimen. (Additional measurements may be required, see (c) and (d).)

(b) Specimens incorporating multiple insulated conductors without individual shields: one insulated conductor shall be selected and electrically connected to one terminal of the megohmmeter. All other specimen conductors, including the overall shield (if any) shall be electrically connected to the other megohmmeter terminal. The resistance measurement shall then be performed as specified. In a similar manner, the resistance between each remaining conductor within the specimen and the short circuited combination of all other conductors plus the overall shield (if any) shall be measured. If the specimen incorporates an overall shield or armor, then an additional resistance measurement shall be made between the shield or armor and the short circuited combination of all remaining conductors. (Additional measurements may be required, see (d).)

(c) Specimens incorporating one or more individually shielded insulated-conductor transmission lines (these measurements shall be required only when shield-to-shield measurements are specified, see 3.1): one shield shall be selected and electrically connected to one terminal of the megohmmeter. Other specimen shields shall be electrically connected to the other megohmmeter terminal, and the resistance measurement shall then be performed as specified. In a similar manner, the resistance between each remaining shield within the specimen and the short circuited combination of all other shields shall be measured.

(d) Specimens incorporating double overall shields: one shield shall be electrically connected to one terminal of the megohmmeter and the other shield shall be connected to the other terminal. The resistance measurement shall be performed as specified. (Additional measurements required, see (e)).

(e) Specimens incorporating single or multiple insulated conductors without shields or armor and other specimens requiring measurements to water (see 3.1): the specimen shall be submerged for not less than 6 hours, except for its extreme ends (which shall remain exposed to the air), in the water tank. The single conductor, or on multiconductor cables, the conductors which are

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outermost in the specimen core, shall be electrically connected to one terminal of the megohmmeter. For other specimens, the overall outermost shield (if any) or if there is no overall shield, the individual shields that are outermost in the specimen core shall be electrically connected to one terminal of the megohmmeter. The other megohmmeter terminal shall be electrically connected to the water surrounding the specimen (as by means of a bare metal wire which is both connected to the megohmmeter terminal and suspended within the water surrounding the specimen). The resistance measurement shall then be performed as specified. The temperature of the water surrounding the specimen shall also be measured, and shall be used to determine F in the following formula:

The lowest value of resistance measured in accordance with each of (a), (b), (c) and (d), as applicable, shall be modified to include the effects of specimen length and temperature by using the following formula:

Insulation resistance, in megohms-1000 feet = RLF/1000

Where: R = Lowest value of resistance measured in accordance with (a), (b), (c) or (d), as applicable, in megohms.
L = Measured specimen length, in feet; measurement accuracy shall be within plus or minus 2 percent.
F = An ad hoc number, which shall accurately modify the measured resistance R to that value of resistance which would be measured if the specimen were at a uniform temperature of 15.5°C. This number shall be determined by the specimen manufacturer via ad-hoc testing for the specific insulation or jacketing material separating the conductors between which the resistance measurement was made, and shall be normalized to unity for a material temperature of 15.5°C.

4.8.5.5 Observation. Specimen failure shall be construed if any calculated value of insulation resistance is less than that specified (see 3.1).

4.8.6 Mutual inductance. This test shall determine the ability of shielded, insulated conductor pairs within completed cable or cord incorporating both shielded, insulated conductor pairs and unshielded insulated conductors to minimize crosstalk caused by the inductive effects of signal currents flowing in adjacent insulated conductors.

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4.8.6.1 Specimen. The specimen shall consist of a 120 ± 1/2 inch length of completed cable or cord, which shall have all overall components (such as armors, jackets or overall binders) and all shields removed from each end for a distance of 1-1/2 ± 1/8 inch, except that shields may be pushed back for this distance rather than removed. The insulation shall then be stripped from each end of each insulated conductor for a distance of 1/2 ± 1/8 inch, and both insulated conductors within each shield shall be electrically connected by twisting each conductor pair tightly together at one specimen end.

4.8.6.2 Special apparatus. Apparatus shall include the following:

(a) A sinusoidal signal generator, which shall produce an unchanging signal at a constant frequency of 1000 ± 200 Hz, which shall produce a signal current appropriate for use in the test as specified in 4.8.6.3 (a signal current of approximately 1 A is typically found necessary), and the signal frequency and signal current shall be known within an accuracy of plus or minus 2 percent.

(b) A rms indicating microvoltmeter, intended for use at the signal generator frequency, exhibiting a signal input impedance of not less than 10,000 ohms, shall be used for the specimen measurements as specified in 4.8.6.3, and shall exhibit a measurement error of not more than plus or minus 6 percent.

4.8.6.3 Procedure. One shielded, insulated conductor pair within the specimen shall be selected, and the microvoltmeter shall be electrically connected across this conductor pair at its unshorted end. The shield shall remain disconnected. The signal generator shall then be electrically connected across any two unshielded, insulated conductors at the specimen end opposite that of the microvoltmeter connection. These same two unshielded, insulated conductors shall be connected together electrically at the opposite specimen end (the end with the attached microvoltmeter) by twisting their two conductors tightly together. The signal current from the signal generator shall next be adjusted until a reasonable indication is noted on the microvoltmeter, and both the microvoltmeter reading and the signal current magnitude shall then be recorded. In a similar manner, additional measurements shall be taken such that microvoltmeter measurements are taken for each shielded, insulated conductor pair as the signal generator is sequentially reconnected to circulate signal current in each possible pairing of unshielded, insulated conductors. (Examples: cable type MCOS-5 shall require three measurements, since it incorporates one shielded, insulated conductor pair plus three unshielded insulated conductors; cable type MCOS-6 shall require two measurements, since it incorporates two shielded, insulated conductor pairs plus two unshielded, insulated conductors.) The mutual inductance represented by each measurement shall then be calculated by using the following formula:

$$\text{Mutual inductance, in microhenries} = V/2\ 3.14159\ fI.$$

Where: V = Measured potential between two shielded, insulated conductors, in microvolts.
f = Signal frequency, in Hz.
I = Measured signal current, in A.

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4.8.6.4 Observation. Specimen failure shall be construed if any calculated value of mutual inductance is greater than that specified (see 3.1).

4.8.7 Pulse response time. This test shall determine the ability of completed cable or cord which incorporates one or more shielded, insulated conductor pairs for pulse data transmission to propagate a square-wave signal without causing unacceptable signal distortion.

4.8.7.1 Specimen. The specimen shall consist of a piece of completed cable or cord, which shall have a length of not less than 200 feet. Individual transmission lines within this specimen (where each insulated conductor pair plus surrounding shield is considered to be a transmission line) shall be selected for testing as follows:

(a) Specimens containing four or fewer transmission lines: every transmission line shall be tested.
(b) Specimens containing more than four but fewer than 21 transmission lines shall be tested. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.
(c) Specimens containing 21 or more transmission lines: the number of transmission lines tested shall be equal to the square root (rounded, if necessary, to the nearest whole number) of the total number of transmission lines. Not fewer than one of these transmission lines shall be selected from each concentric cabling layer within the specimen.

4.8.7.2 Special apparatus. Apparatus shall include the following:

(a) Pulse generator: a signal generator which shall produce uniform square-wave voltage pulses at a constant rate of repetition, and which shall be used for the specimen measurements as specified in 4.8.7.3. The pulses produced by this generator shall each exhibit a 10 to 90-percent voltage rise time of not more than 50 nanoseconds, a 90 to 10-percent voltage decay time of not more than 50 nanoseconds, and a pulsewidth of not less than 2 microseconds.
(b) Oscilloscope: a dual-beam oscilloscope shall have a band width of not less than 10 megahertz, and a time base capable of producing a horizontal beam sweep rate of not slower than 0.1 milliseconds (ms) per centimeter. This oscilloscope shall have provision for a high-impedance, balanced or differential input on both signal channels, and shall be used for the specimen measurements as specified in 4.8.7.3.
(c) Pulse transformers: two, two-winding transformers, each of which shall provide essentially distortion-free transmission of the voltage pulses from the pulse generator, and shall be used for the specimen measurements as specified in 4.8.7.3. One of the pulse transformers (the "input" pulse transformer, see figure 17) shall have a primary winding impedance equal to the nominal signal-terminal impedance of the pulse generator; the secondary winding of this transformer shall be centertapped, and shall have an impedance equal to the specified impedance of the

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specimen (see 3.1). The other pulse transformer (the "output" pulse transformer, see figure 17) shall have a primary winding which shall be centertapped, and shall have an impedance equal to the specified impedance of the specimen (see 3.1); the secondary winding impedance of this transformer shall equal the nominal resistance of the matching resistor (see (d)).

(d) Matching resistor: an electrical resistor, shall be essentially noninductive in construction (such as a carbon-type resistor), which shall have a power rating appropriate for its intended use (see 4.8.7.3), and shall exhibit an electrical resistance appropriate for use with the output pulse transformer (see (c)).

4.8.7.3 Procedure. The test apparatus shall be electrically interconnected as shown on figure 17, and one of the selected transmission lines shall be connected as shown. The connections between this transmission line and the pulse transformers shall be as short as practicable. In addition, both connecting cables between the transmission line and the oscilloscope shall be of identical length and construction, and shall be not longer than 5-feet each, to minimize pulse distortion. The pulse generator shall then be turned on, and both signal channels on the oscilloscope shall be adjusted such that the pulse waveform displayed by each on the oscilloscope screen shall be of the same height. The horizontal beam sweep rate of the oscilloscope shall then be increased, if necessary and the time required for the leading edge of each signal channel pulse to rise from 10 to 90 percent of its final value shall be measured as shown on figure 18. The rise time of the transmission line shall then be calculated by using the following formula:

$$\text{Rise time, in ns per 500 feet} = 500(Y_r^2 - X_r^2)^{1/2}/L$$

Where: X_r = Measured rise time from signal channel number one, in microseconds.
Y_r = Measured rise time from signal channel number two, in microseconds.
L = Measured specimen length, in feet; measurement accuracy shall be within plus or minus 1 percent.

Following this, the time required for the trailing edge of each signal channel pulse to decay from 90 to 10 percent of its initial value shall be measured as shown on figure 19. The decay time of the transmission line shall then be calculated by using the following formula:

$$\text{Decay time, in ns per 500 feet} = 500(Y_d^2 - X_d^2)^{1/2}/L$$

Where: X_d = Measured decay time from signal channel number one, in microseconds.
Y_d = Measured decay time from signal channel number two, in microseconds.
L = Measured specimen length, in feet; measurement accuracy shall be within plus or minus 1 percent.

In an identical manner, the remaining selected transmission lines shall also be tested.

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4.8.7.4 Observation. Specimen failure shall be construed if any transmission line exhibits either a rise time or a decay time which is greater than that specified (see 3.1).

4.8.8 Voltage withstand. This test shall determine the ability of completed cable and cord to withstand overvoltages without exhibiting electrical discharge through insulation or jacketing material, either between conductors, or between these conductors and any conducting material which may be in contact with the surface of the cable or cord during in-service use.

4.8.8.1 Specimen. The specimen shall consist of an entire and continuous ordered length of completed cable or cord.

4.8.8.2 Special apparatus. Apparatus shall include the following:

(a) An adjustable voltage source, which shall be in accordance with either of the following, as appropriate (see 3.1).

(1) An ac voltage source which shall produce a single-phase, approximately sinusoidal voltage under all test conditions, which shall have a frequency of not more than 100 Hz, and which can be adjusted to the specified rms magnitude (see 3.1). This source shall have the capacity to provide an effective power of not less than 5 kVA, and shall incorporate an overload relay which shall reduce the source voltage to zero whenever an applied load (that is, the specimen, see 4.8.8.3) attempts to draw electrical current to exceed this effective power capacity.

(2) A source of dc voltage which can be adjusted to the specified magnitude (see 3.1), and which shall have the capacity to provide an electrical current of not less than 1/4 A. This source shall incorporate an overload relay which shall reduce the source voltage to zero whenever an applied load (that is, the specimen, see 4.8.8.3) attempts to draw electrical current to exceed this current capacity.

(b) When required, a water tank, filled with water, of sufficient size to contain the specimen, shall be used as specified in 4.8.8.3(d) (may not be required, see 4.8.8.3(d)).

4.8.8.3 Procedure. The voltage source shall be electrically connected to the specimen as specified (see (a), (b), (c) and (d)). For each connection, the test voltage from the voltage source shall be increased, in a uniform manner, from 0 V to the specified voltage (see 3.1) during a period of not more than 60 seconds nor less than 10 seconds, except that specified voltages of less than 600 V may be applied instantaneously. (This gradual voltage application is generally necessary to prevent a current on rush, and the consequent tripping of the voltage source overload relay caused by specimen capacitance.) This specified voltage shall then be maintained for a period of 1 plus 1/2, minus 0 minute (unless the voltage source overload relay trips, see 4.8.8.4), following which the test voltage shall be reduced to zero at approximately the same rate as which it was applied.

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(a) Specimens incorporating one or more individually shielded, insulated-conductor transmission lines: one transmission line shall be selected for measurement. One conductor of this transmission line shall be electrically connected to one terminal of the voltage source, and the shield and all other conductor(s) (if any) of this transmission line shall be electrically connected to the other voltage source terminal. The voltage withstand test shall then be performed as specified. In a similar manner, the voltage withstand test shall be performed between each remaining conductor in this transmission line (if any) and the short circuited combination of all other conductors (if any) plus the shield in this transmission line. Following this, and in an identical manner, the voltage withstand test shall be performed on all remaining transmission lines with the specimen. (Additional measurements may be required, see (c) and (d).)

(b) Specimens incorporating multiple insulated conductors without individual shields: one insulated conductor shall be selected and electrically connected to one terminal of the voltage source. Other specimen conductors, including the overall shield (if any) shall be electrically connected to the other voltage source terminal. The voltage withstand test shall then be performed as specified. In a similar manner, the voltage withstand test shall be performed between each remaining conductor within the specimen and the short-circuited combination of all other conductors plus the overall shield (if any). If the specimen incorporates an overall shield, then the voltage withstand test shall additionally be performed between the shield and the short circuited combination of all remaining conductors. (Additional measurements may be required, see (d).)

(c) Specimens incorporating one or more individually shielded insulated-conductor transmission lines (these measurements shall be required only when shield-to-shield measurements are specified, see 3.1): one shield shall be selected and electrically connected to one terminal of the voltage source. Other specimen shields shall be electrically connected to the other voltage source terminal, and the voltage withstand test shall then be performed as specified. In a similar manner, the voltage withstand test shall be performed between each remaining shield within the specimen and the short circuited combination of all other shields.

(d) All specimens (these measurements shall be required only when measurements to water or to ground are specified, see 3.1): the specimen shall be submerged, except for its extreme ends (which shall remain exposed to the air), in the water tank for a period of not less than 1 hour. Following this, and while the specimen is still submerged, the specimen shall be tested as follows. The overall specimen shield (if any), or if there is no overall shield, all individual shields which are outermost in the specimen core plus all unshielded conductors which are outermost in

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the specimen core, shall be electrically connected to one terminal of the voltage source. The other voltage source terminal shall be electrically connected to the water surrounding the specimen (as by means of a bare metal wire which is both connected to the voltage source terminal and suspended within the water surrounding the specimen). The voltage withstand test shall then be performed as specified.

4.8.8.4 Observation. Specimen failure shall be construed if the voltage source overload relay trips prior to removal of the test voltage from any specimen connection. If the insulation resistance test is specified (see 3.1), then the specimen shall be retained for use in that test.

4.9 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Shipment of cables. Cables covered by this specification shall be shipped on reels, unless otherwise specified on the specification sheet.

5.1.1 Reels. Reels shall be of a design, strongly constructed, with flanges of sufficient diameter to protect the cable from injury during handling and storage. Where the gross weight of cable and reel exceed 1000 pounds, the reels shall have metal hubs. The diameter of the drum shall be not less than 14 times the overall diameter of the cable.

5.1.1.1 Placement of cable on the reel. The cable on each reel shall be one continuous length and shall have both ends readily available for testing without re-reeling. The inner end shall extend through a hole in the flange near the drum, and the outer end shall be available at the top of the reel. Both ends of the cable shall be secured to reel flange with 1/4-inch rope, applied with two half-hitches, one 6 inches and the other 4 inches (approximately) back of the end seal, then twisting the rope ends together, knotting them, pulling the cable up tight and stapling the rope to the flange with at least two 1-1/2 inch staples driven into the flange. The inner end which extends through the flange hole shall be protected by a metal end cover.

5.1.1.2 Lagging. Lagging shall be used on reels having flange diameters of 42 inches and larger and on all reels which contain 1000 pounds or more of cable. Reels smaller than 42 inches in diameter which contain less than 1000 pounds of cable may be covered in accordance with 5.1.1.3. Lagging boards shall have a thickness equal to the thickness of the reel flange but not greater than 2 inch commercial lumber. Lagging for all reels shall extend to within 1/4 inch of the outside of the flanges. Lagging boards shall be installed so that they are touching adjacent boards. Lagging which rests entirely on the

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rim of the flange shall be nailed with cement-coated nails. Lagging which is notched and recessed may be secured by the use of stapling only. In addition to nailed lagging or for notched and recessed lagging on reels of cable weighing 1000 pounds or more and having a flange diameter less than 78 inches, two 0.162 ± 0.003 inch (8-gauge) high-tensile round steel wires or two 3/4 by 0.035 inch steel straps shall be used. For reels having a flange diameter of 78 inches or more two 1-1/4 by 0.035 inch steel straps shall be used. In addition to nailed lagging or for notched and recessed lagging on reels of cable weighing less than 1000 pounds, two 0.0915 ± 0.003 inch (13-gauge) high-tensile round steel wires or two 5/8 by 0.020-inch steel straps shall be used. Straps shall be stapled at approximately 15-inch intervals. A minimum clearance of 2 inches shall be provided between inner face of lagging and outer layer of cable.

5.1.1.3 Alternative to lagging. As an alternative to lagging for reels having a flange diameter smaller than 42 inches and containing less than 1000 pounds of cable, a covering may be used which consists of the following:

(a) A cushioning inner layer of asphalt-impregnated, corrugated, fiberboard, or the equivalent, wrapped to give 100-percent coverage.
(b) An outer covering consisting of two layers of weather-resistant veneer paper in accordance with PPP-V-205 or two layers of three-ply weather-resistant toxic-treating plywood wrapped to give 100-percent coverage.

This covering shall be secured in place between the reel flanges by at least two 3/8 by 0.020 inch steel straps.

5.1.1.4 Year marking. In addition to any other required marking, reels shall be marked with a keyed series of colors to indicate the year of manufacture. The cycle of colors shall be repeated every fifth year. The reel marking shall consist of a stripe of one coat of commercial quality outside paint approximately 2-inches wide, colored to designate the particular year of manufacture as follows:

Year of manufacture		Identifying color
1981	1986	Red
1982	1987	Green
1983	1988	Orange
1984	1989	Blue
1985	1990	White

5.1.1.4.1 Location of year marking. The colored stripe on reels shall be applied circumferentially over the lagging or the alternative to lagging and midway between the flanges. The stripe shall consist of one coat of commercial quality outside paint of the appropriate color. In addition to the stripe, both flanges of the reel shall be stenciled with 4 inch high figures to show the year of manufacture.

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5.1.1.5 Standard reel markings. Each reel shall be plainly marked on both flanges with the following information:

(a) Reel number.
(b) Type and size of cable.
(c) Footage.
(d) Contract or order number.
(e) Contractor's name.
(f) Manufacturer's name (if other than contractor).
(g) Gross weight.
(h) One continuous length or the sequence of lengths on the reel indicated in feet, whichever is applicable.

5.1.2 Coils. Coils shall contain one continuous length of cable or cord (standard length, see table XXIV).

TABLE XXIV. Cable or cord lengths.

Type of package	Nominal length (feet)	Standard lengths	Random lengths	Remnant lengths		Scrap lengths
				Price reduction		
		No price reduction (feet)	No price reduction (feet)	5 percent (feet)	10 percent (feet)	Not acceptable (feet)
Coil	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Coil	600	660 to 540	539 to 180	179 to 120	119 to 60	59 to 0
Reel	500	550 to 450	449 to 150	149 to 100	99 to 50	49 to 0
Reel	800	880 to 720	719 to 240	239 to 160	159 to 80	79 to 0
Reel	1000	1100 to 900	899 to 300	299 to 200	199 to 100	99 to 0
Reel	1500	1650 to 1350	1349 to 450	449 to 300	299 to 150	149 to 0
Reel	2000	2200 to 1800	1799 to 600	599 to 400	399 to 200	199 to 0
Reel	2500	2750 to 2250	2249 to 750	749 to 500	499 to 250	249 to 0
Reel	3000	3300 to 2700	2699 to 900	899 to 600	599 to 300	299 to 0

5.1.2.1 Year marking. In addition to any other required marking, coils shall be marked with a keyed series of colors to indicate the year of manufacture. This marking shall consist of a stripe approximately 2-inches wide and colored for the particular year of manufacture. The cycle of colors shall be the same as those used for year marking of reels (see 5.1.1.4).

5.1.2.2 Identification. Two shipping tags shall be securely attached to each coil, both inside and outside the wrapping, and marked with the following information:

(a) Type and size of cable or cord.
(b) Footage.
(c) Contract or order number.
(d) Contractor's name.
(e) Manufacturer's name (if other than contractor).
(f) Gross weight.

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5.2 Packaging and packing. Cable and cord as specified (see 6.2.1) shall be packaged and packed level A, B, or C in accordance with MIL-C-12000, as specified (see 6.2.1). Cables and cords shall have both ends of each shipping length sealed to prevent the entrance of moisture. Materials used and method of application shall be that specified for tape wrapped end seals in accordance with MIL-C-12000 or as approved by NAVSEA. Marking shall be in accordance with MIL-C-12000 and with the requirements specified herein or as specified in the acquisition document (see 6.2.1). All markings shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use. Cables and cords specified herein are intended for use in various applications involving naval ships and shore stations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and size of cable or cord required with reference to applicable specification sheet (see 3.1).
(c) Classification of cables and cords (see 1.2).
(d) Quantity required (see 6.2.3).
(e) Level of packaging and packing required (see 5.2).
(f) Special marking required (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1.1	Inspection system program plan	DI-R-4803	——
4.4.7 and 4.5.7	Reports, test	DI-T-2072	10.1.b

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.3 Quantity. The quantity of each type and size on a contract or order should be specified as an integral multiple of the unit ordering length shown by the specification sheet.

6.2.4 Lengths. The range of standard, random, remnant and scrap lengths for each nominal length is shown in table XXIV. In order to compensate for handling and probable loss on issue, remnant lengths are subject to the price reductions specified in table XXIV. Scrap lengths are not acceptable. It is not intended that items being acquired in exact lengths for a specific job or ship shall necessarily be shipped in lengths which are integral multiples of the unit ordering length or that price reductions shall apply in the event that exact footages or fractional lengths are required and so indicated in acquisition of cable or cord other than stock purposes.

6.3 Special clause. Except when small quantities are to be purchased, invitations for bid and contracts or orders should specify the following:

"In order to compensate for handling and probable loss on issue, remnant lengths shall be subject to the price reductions specified in table XXIV. In order that the number of lengths be kept to a minimum consistent with good manufacturing practices, for each type and size of cable or cord on the contract or order, not less than 70 percent of the total footage to be shipped shall be in standard lengths, and not more than 30 percent may be in any combination of random and remnant lengths. When the total quantity of any one item is six nominal lengths or less, a footage approximately two nominal lengths (in lieu of 30 percent) may be in any combination of random and remnant lengths. When the total quantity of any one item is two nominal lengths or less, the total footage may be in any combination of random lengths."

6.4 Time delay. Comparison inspection will normally require 40 calendar days from the date the sample is sent to the Government designated laboratory to the date the inspection results are received by the manufacturer. This time delay should be taken into consideration by a contractor when estimating delivery time.

6.5 With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-915 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible

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for the Qualified Products List is the Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5.1).

6.5.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Terms and definitions. For the purpose of the quality conformance inspection, the sampling terms specified in 6.7.1 through 6.7.4 shall apply.

6.7.1 Unit of product. A unit of product is the unit ordering length as specified on the specification sheet, except that when the unit ordering length is 500 feet, two such lengths shall constitute a unit of product. Random and remnant lengths or special order lengths may be added together to equal a unit ordering length. Length tolerances as given for standard lengths shall apply (see 6.2.3).

6.7.2 Sample. A sample is one unit of product selected from an inspection lot.

6.7.3 Specimen. A specimen is an individual length of cable or cord or an individual length of a part of cable or cord which has been taken from a sample.

6.7.4 Defective unit. A defective unit is a unit of product from which a specimen that failed to meet one or more requirements was taken.

6.8 Supersession data. This specification supersedes Specifications MIL-C-915E of 1 August 1972, MIL-C-00915D(SHIPS) of 20 March 1972, MIL-C-915C of 15 March 1968, MIL-C-2194E(SHIPS) of 31 January 1968, MIL-C-23206(SHIPS) of 23 February 1962 and MIL-C-24145A(SHIPS) of 16 October 1967.

6.8.1 New types and obsolete types. Detail specifications covered by the specifications in 6.7 are specified in table XXV both by type designation and by detail specification number. Table XXV also lists: (a) new type designations for cables for which a design change has been made, and a remark on the nature of the design change; and (b) the replacement types, for cable types which are obsolete. Although not indicated in the remarks column, the conductor size for all cables, except those listed in table VIII, has been changed from Navy Standard size to AWG size. In most cases, this has amounted to a decrease in the circular mil area of the conductor.

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TABLE XXV. Type designation cross reference.

Previous type and detail specification number	Present type and applicable MIL-C-915 specification sheet number	Remarks
	1PR-A20E——————/81	
	1PR-16——————/82	New type
	1Q-16——————/86	New type
	1SPR-16——————/89	New type
1S50MU MIL-C-915/42	1S50MA——————/42	Armor added
1S50MU /42	1S50MU——————/42	No change
1S50MU /42	1S50MUS——————/42	Double overall shield
1S75MU /55	1S75MA——————/55	Armor added
1S75MU /55	1S75MU——————/55	No change
1SAU /57	1SA——————/57	Armor added
1SAU /57	1SAU——————/57	No change
1SMU /56	1SMA——————/56	Armor added
1SMU /56	1SMU——————/56	No change
1SMWU /65	1SMWA——————/65	Armor added
1SMWU /65	1SMWU——————/65	No change
1SU /59	1SU——————/59	No change
1SU /59	1SUA——————/59	Armor added
1SWU /44	1SWA——————/44	Armor added
1SWF /47	1SWF——————/47	No change
1SWU /44	1SWU——————/44	No change
	1TR-16——————/87	New type
2AU /41	2A——————/41	Armor added
2AU /41	2AU——————/41	No change
2AU /41	2AUS——————/41	Double overall shield
2CS /80	2CS——————/80	No change
2SU /45	2SA——————/45	Armor added
2SJ /60	2SJ——————/60	No change
2SJ /60	2SJA——————/60	Armor added
	2SPR-16——————/84	New type
2SU /45	2SU——————/45	No change
2SU /45	2SUS——————/45	Double overall shield
2SWAU /46	2SWA——————/46	Armor added
2SWAU /46	2SWAU——————/46	No change
2SWF /47	2SWF——————/48	No change
2SWL-7 /77	2SWL-7——————/77	No change
2SWL-7 /77	2SWLA-7——————/77	Armor added
2SWU /49	2SWU——————/49	No change
2SWU /49	2SWUA——————/49	Armor added
2U /63	2U——————/63	No change
2U /63	2UA——————/63	Armor added
2UW-42 /78	2UW-42——————/78	No change
2UW-42 /78	2UWA-42——————/78	Armor added
2UW-42 /78	2UWS-42——————/78	Double overall shield
2WAU /64	2WA——————/64	Armor added

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TABLE XXV. Type designation cross reference.

Previous type and detail specification number	Present type and applicable MIL-C-915 specification sheet number	Remarks
2WAU MIL-C-915/64	2WAU——————/64	No change
	3PR-16——————/85	New type
3SU /51	3SA——————/51	Armor added
3SF /62	3SF——————/62	No change
3SJ /60	3SJ——————/60	No change
3SJ /60	3SJA——————/60	Armor added
3SU /51	3SU——————/51	No change
3SU /51	3SUS——————/51	Double overall shield
3SWU /52	3SWA——————/52	Armor added
3SWU /52	3SWU——————/52	No change
3SWU /52	3SWUS——————/52	Double overall shield
3U /53	3U——————/53	No change
3U /53	3UA——————/53	Armor added
4NW8 /76	4NW8——————/76	No change
4NW8 /76	4NWA8——————/76	Armor added
4SJ /60	4SJ——————/60	No change
4SJ /60	4SJA——————/60	Armor added
5KVTSGU /36	5KVTSGA——————/36	Armor added
5KVTSGU /36	5KVTSGU——————/36	No change
5SS /74	5SS——————/74	No change
6SGU /33	6SGA——————/33	Armor added
6SGU /33	6SGU——————/33	No change
	7PR-16——————/83	New type
7PS /40	7PS——————/40	Asbestos removed
7SGU /34	7SGA——————/34	Armor added
7SGU /34	7SGU——————/34	No change
	7SPR-16S——————/88	New type
7SS /8	7SS——————/8	No change
8NW6 /75	8NWA6——————/75	Armor added
8NW6 /75	8NWG——————/75	No change
CVSP /1	CVSP——————/1	No change
DCOP /3	DCOP——————/3	No change
DHOP /6	DHOP——————/6	No change
DLT /5	DLT——————/5	No change
DNW /68	DNW——————/68	No change
DNW /68	DNWA——————/68	Armor added
DPS /40	DPS——————/40	Asbestos removed
DRW /73	DRW——————/73	No change
DRW /73	DRWA——————/73	Armor added
DSGU /29	DSGA——————/29	Armor added
DSGU /29	DSGU——————/29	No change
DSS /8	DSS——————/8	No change
DSWS /7	DSWS——————/7	No change
ECM /54	ECM——————/54	No change

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TABLE XXV. Type designation cross reference.

Previous type and detail specification number	Present type and applicable MIL-C-915 specification sheet number	Remarks
EMC MIL-C-915/54	EMCA----------/54	Armor added
FHOF /6	FHOF----------/6	No change
FNW /70	FNW----------/70	No change
FNW /70	FNWA----------/70	Armor added
FPS /40	FPS----------/40	Asbestos removed
FSGU /31	FSGA----------/31	Armor added
FSGU /31	FSGU----------/31	No change
FSS /8	FSS----------/8	No change
JAS /9	JAS----------/9	No change
MU /43	MA----------/43	Armor added
MCOS /11	MCOS----------/11	Asbestos removed
MCSF-4 /10	MCSF-4----------/10	No change
MDU /12	MDU----------/12	No change
MDY /13	MDY----------/13	Reinstated
MHOF /14	MHOF----------/14	No change
MMOP /15	MMOP----------/15	No change
MNW /71	MNW----------/71	No change
MNW /71	MNWA----------/71	Armor added
MRI /16	MRI----------/16	No change
MS /50	MS----------/50	No change
MS /50	MSA----------/50	Armor added
MSCU /32	MSCA----------/32	No change
MSCU /32	MSCS----------/32	No change
MSCU /32	MSCU----------/32	No change
MSP /67	MSP----------/67	No change
MSPW /66	MSPW----------/66	No change
MU /43	MU----------/43	No change
MU /43	MUS----------/43	Double overall shield
MWF /58	MWF----------/58	No change
PBTMU /17	PBTM----------/17	Armor added
PBTMU /17	PBTMU----------/17	No change
PI /39	PI----------/39	Asbestos removed
S2S /61	S2S----------/61	No change
SHOF /6	SHOF----------/6	No change
SRW /73	SRW----------/73	No change
SRW /73	SRWA----------/73	Armor added
SSF /18	SSF----------/18	No change
SSGU /28	SSGA----------/28	Armor added
SSGU /28	SSGU----------/28	No change
TCJU /35	TCJA----------/35	Armor added
TCJU /35	TCJU----------/35	No change
TCJX /38	TCJX----------/38	No change
TCKX /38	TCKX----------/38	No change
TCOP /3	TCOP----------/3	No change

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TABLE XXV. Type designation cross reference.

Previous type and detail specification number	Present type and applicable MIL-C-915 specification sheet number	Remarks
TCTU MIL-C-915/35	TCTA———/35	Armor added
TCTU /35	TCTU———/35	No change
TCTX /38	TCTX———/38	No change
THOF /6	THOF———/6	No change
TNW /69	TNW———/69	No change
TNW /69	TNWA———/69	Armor added
TPNW /72	TPNW———/72	No change
TPNW /72	TPNWA———/72	Armor added
TPS /40	TPS———/40	Asbestos removed
TRF /20	TRF———/20	No change
TPUM-6 /79	TPUM-6———/79	No change
TRW /73	TRW———/73	No change
TRW /73	TRWA———/73	Armor added
TRXF /21	TRXF———/21	No change
TSGU /30	TSGA———/30	Armor added
TSGU /30	TSGU———/30	No change
TSP /22	TSP———/22	No change
TSP /22	TSPA———/22	Armor added
TSS /8	TSS———/8	No change
TTOP /24	TTOP———/24	No change
TTRS /25	TTRS———/25	No change
TTRSA /25	TTRSA———/25	Armor added
TTSU /37	TTSA———/37	Armor added
TTSU /37	TTSU———/37	No change

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes. All changes reflect American Wire Gauge (AWG) sizes.

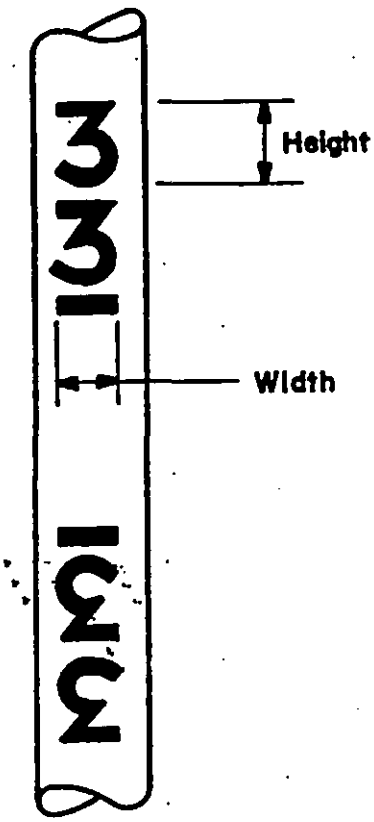
Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Army - CR, ER
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0862)

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SH 13051

FIGURE 1. Conductor of pair number 33.

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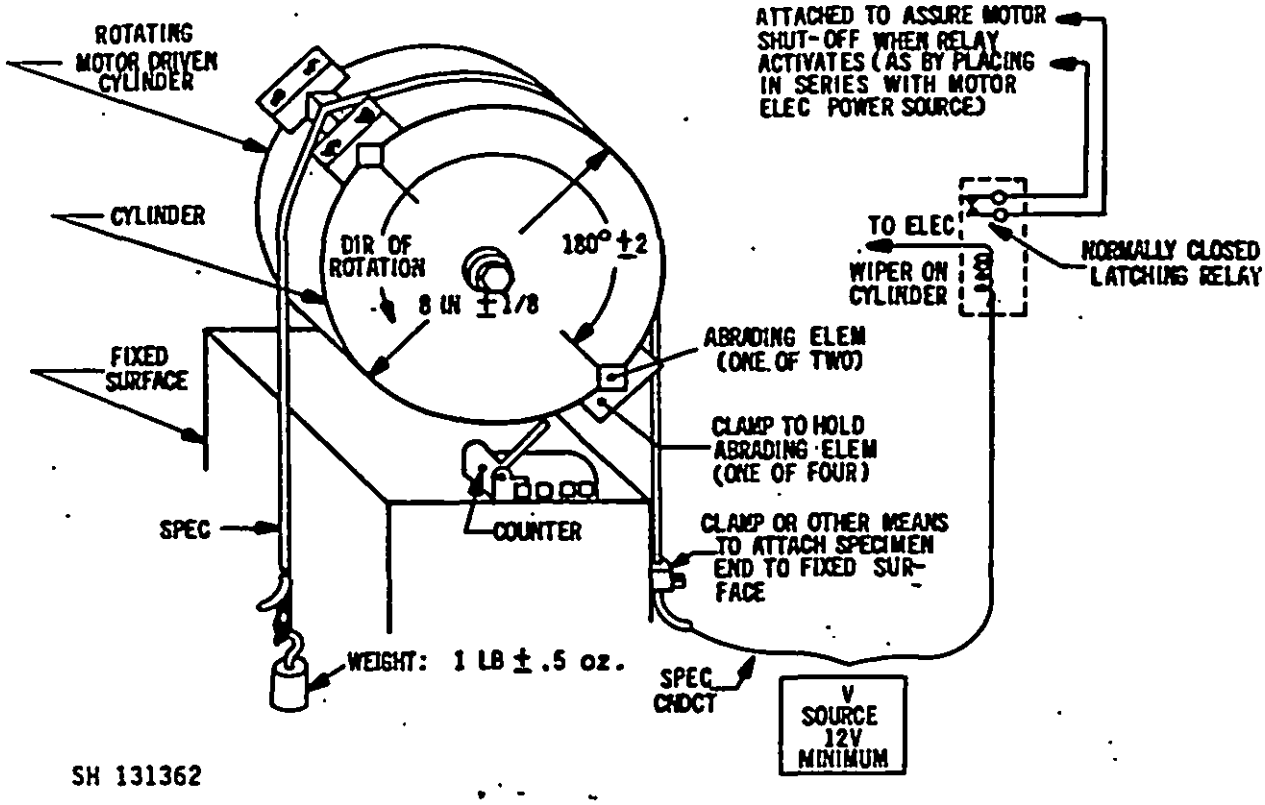


FIGURE 2. Diagram of abrading machine for the abrasion resistance test.

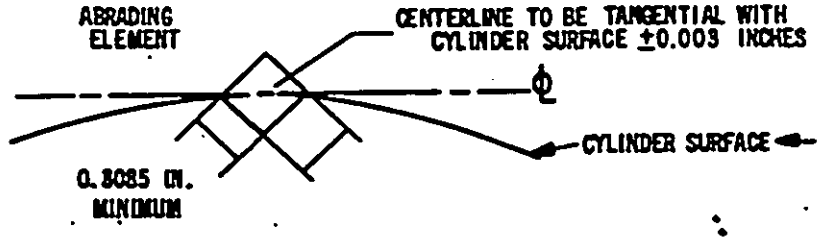
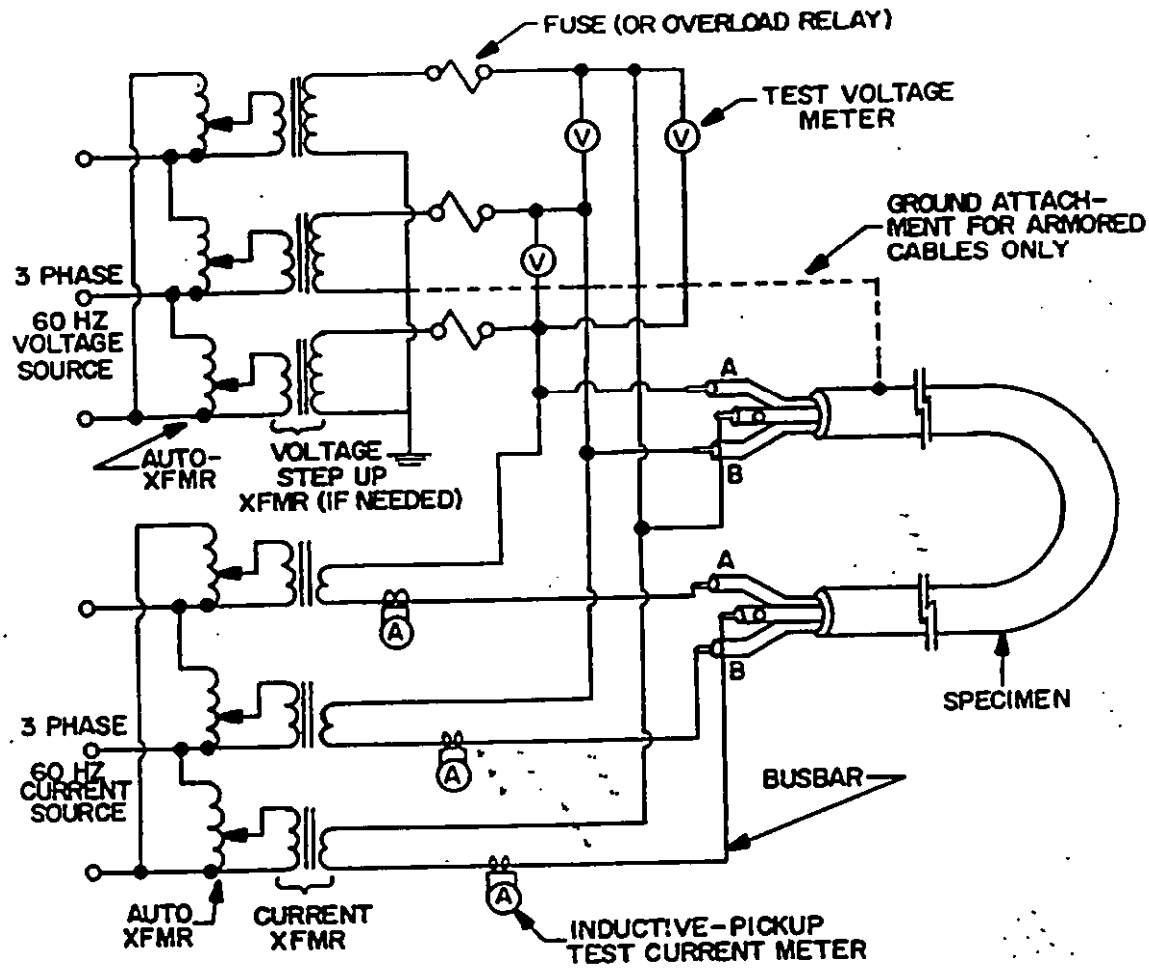


FIGURE 3. Detail diagram of abrading element fit on cylinder surface.

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SH 13054

FIGURE 4. Application of test voltage and test current for accelerated service test.

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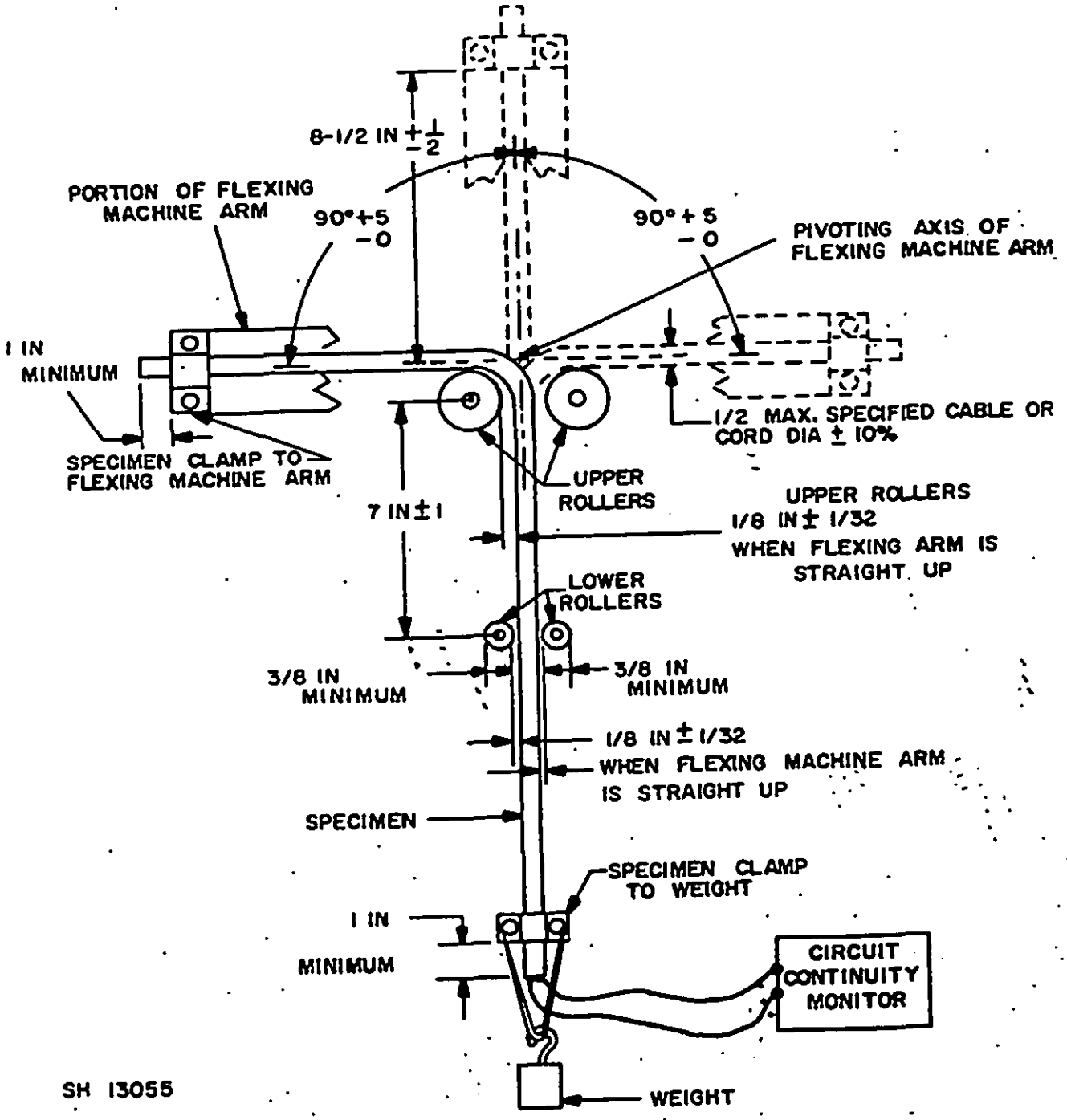
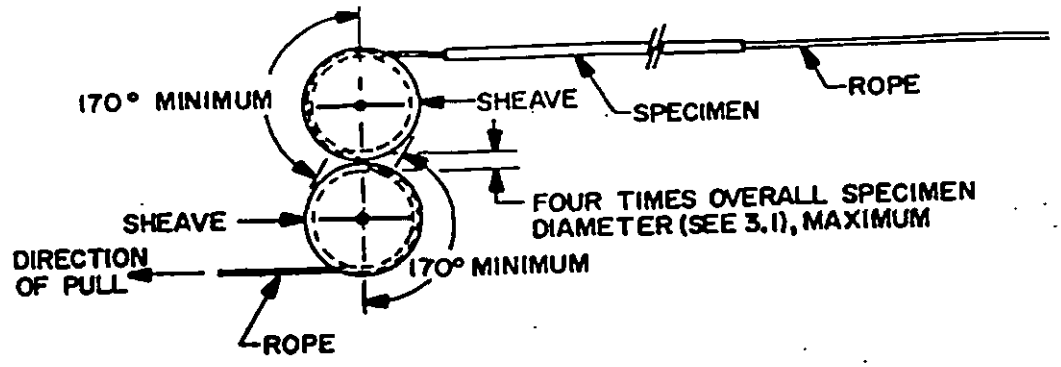


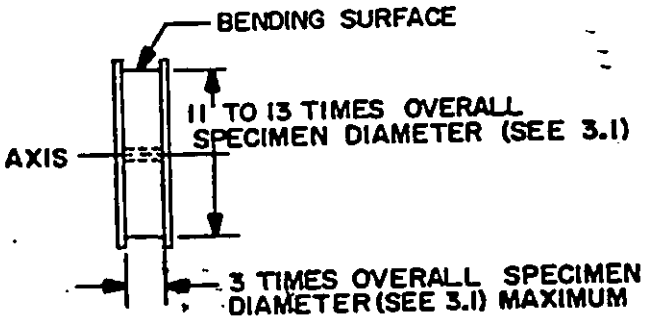
FIGURE 5. Diagram of flexing machine for the bending endurance test.

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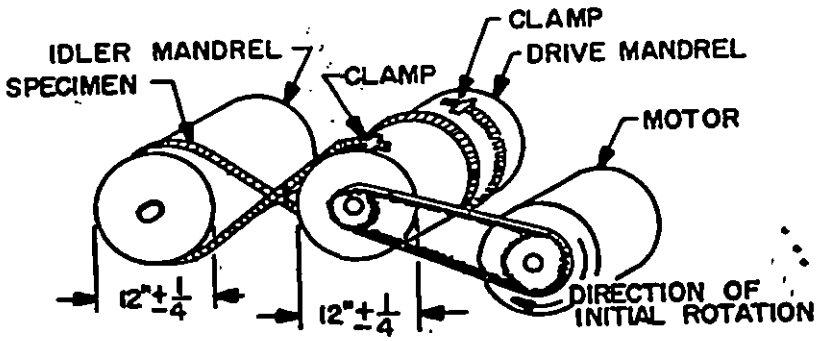
SH 13056

FIGURE 6. Diagram of bending apparatus for the cold working (minus 20°C) test.



SH 13057

FIGURE 7. Edge view of either sheave for the bending apparatus of the cold working (minus 20°C) test.



SH 13058

FIGURE 8. Diagram of bending apparatus for the cold working (minus 54°C) test.

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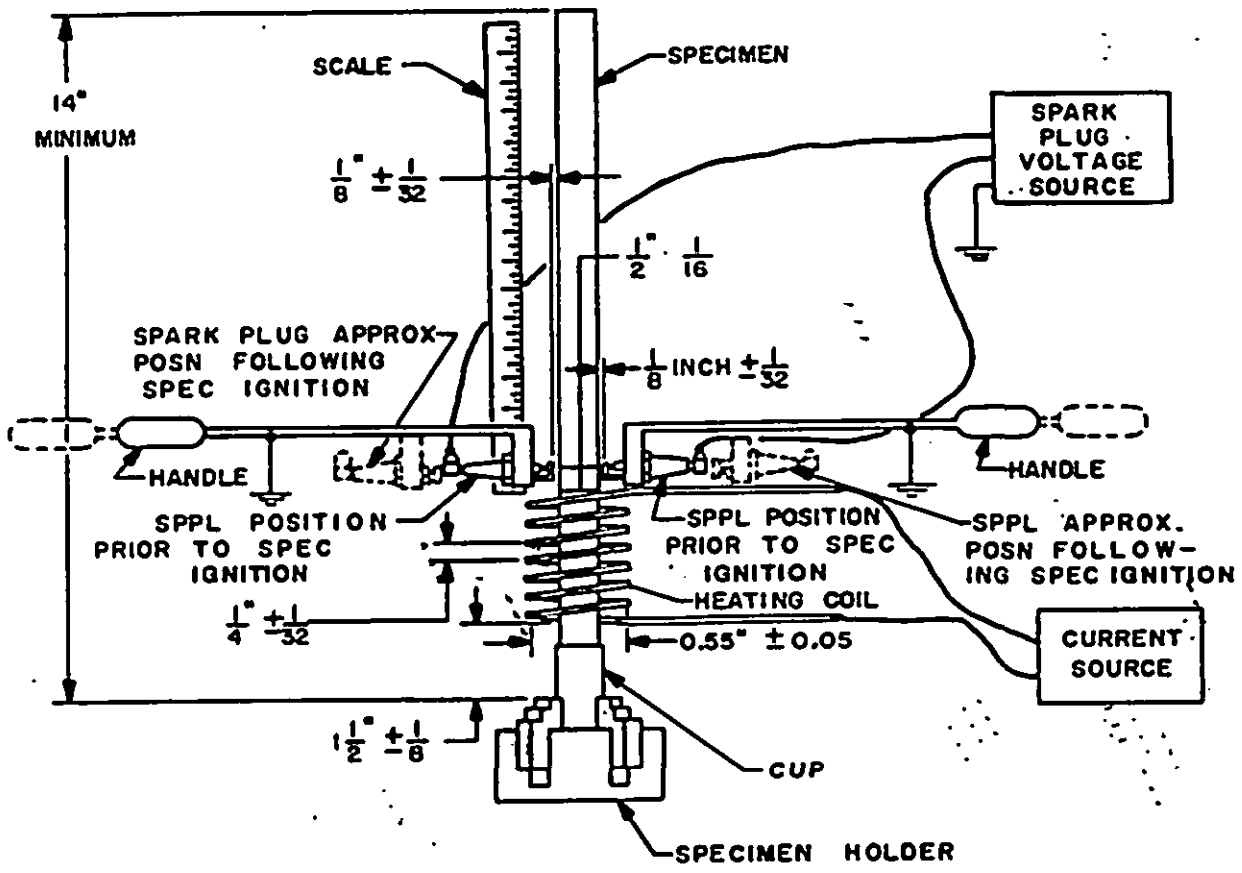
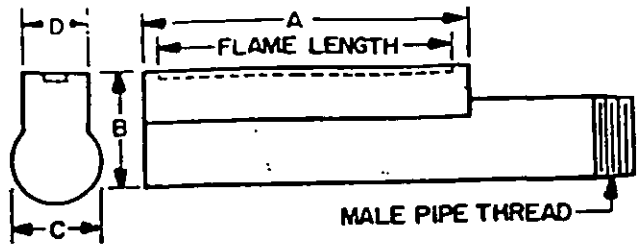


FIGURE 9. Diagram of flammability apparatus for flammability test.

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For Burner Flame Lengths	PHYSICAL DIMENSIONS				
	A	B	C	D	Male Pipe Thd. Conn.
24"	Flame Length plus 3/4"	2"	1-3/16"	1"	1"

FIGURE 10. Style X tubular brass manifold for gas flame test.

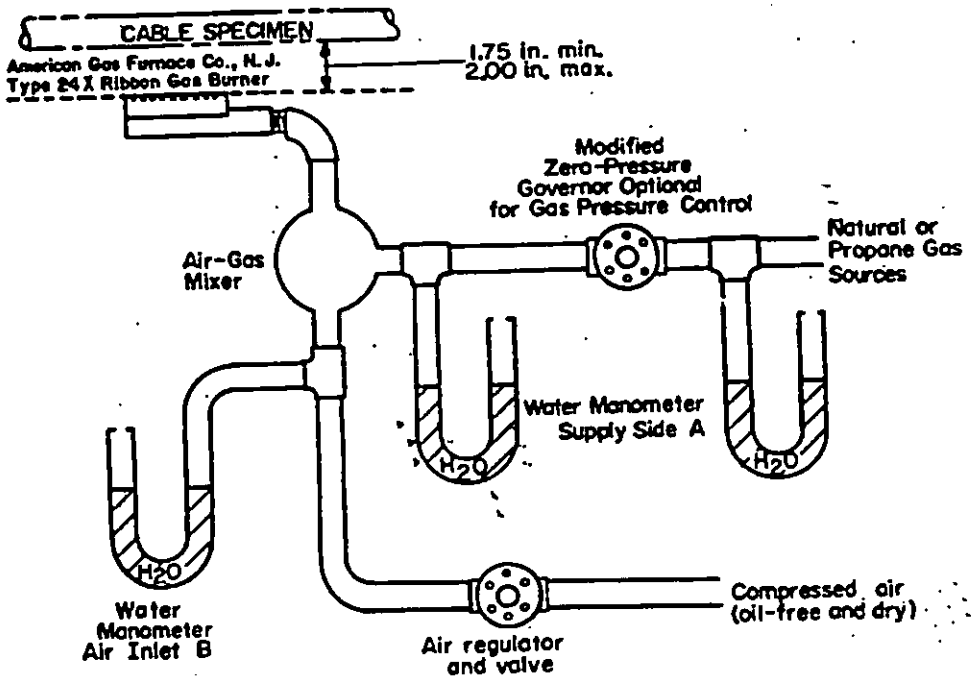
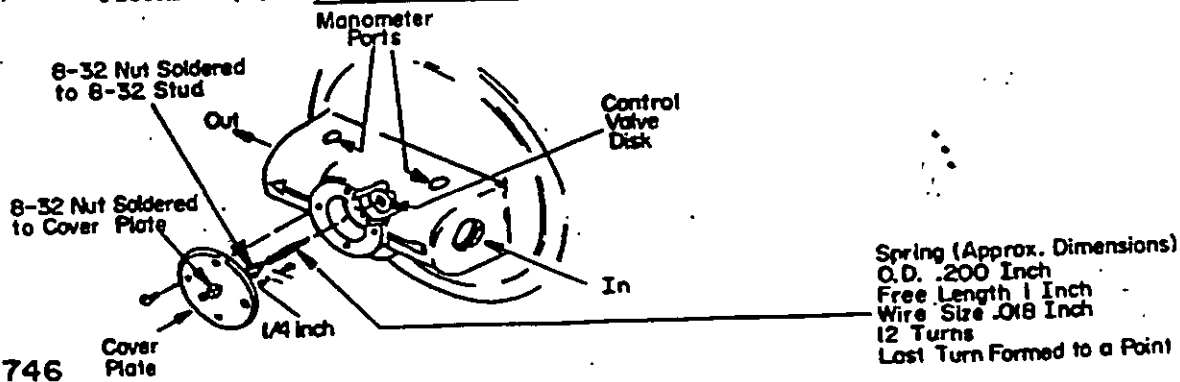


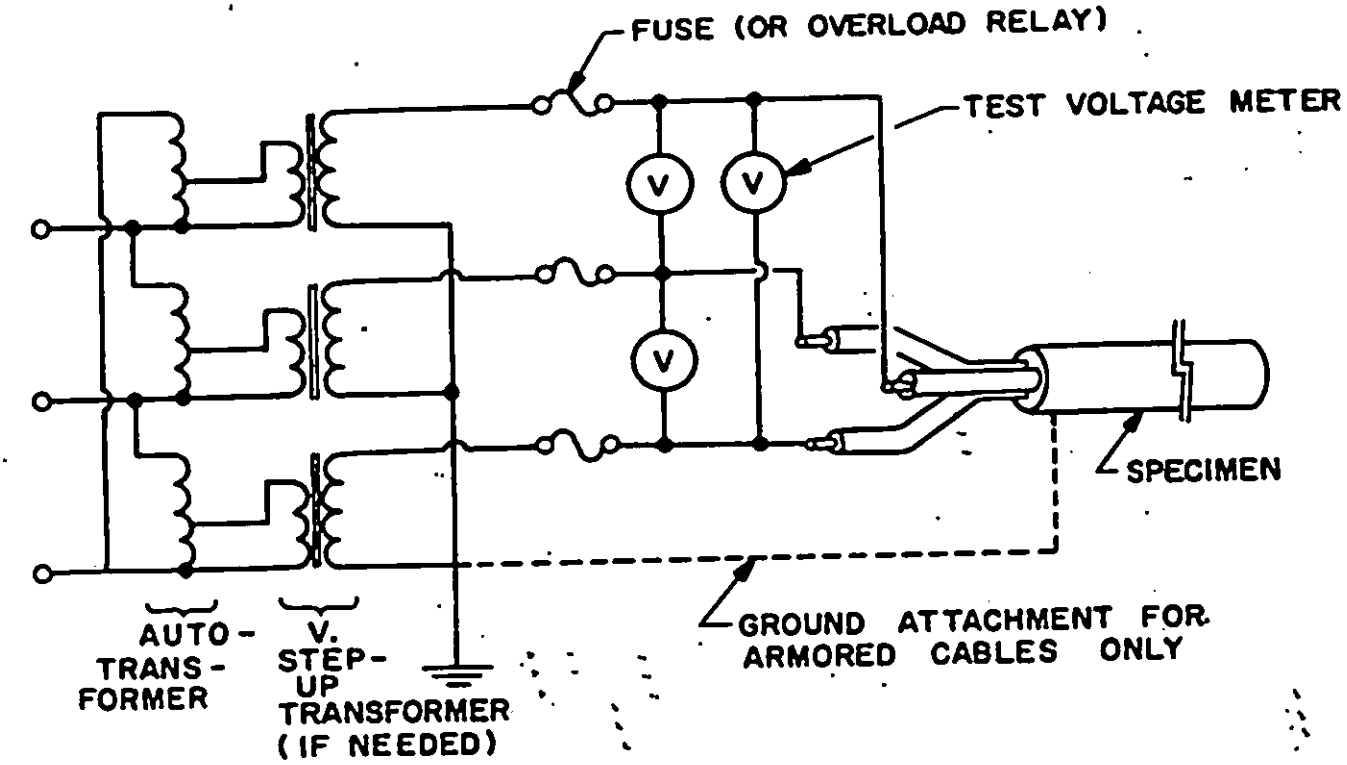
FIGURE 10(A). Schematic drawing of apparatus for gas flame test.



SH 12746

FIGURE 10(B). Detail drawing of zero pressure governor modification.

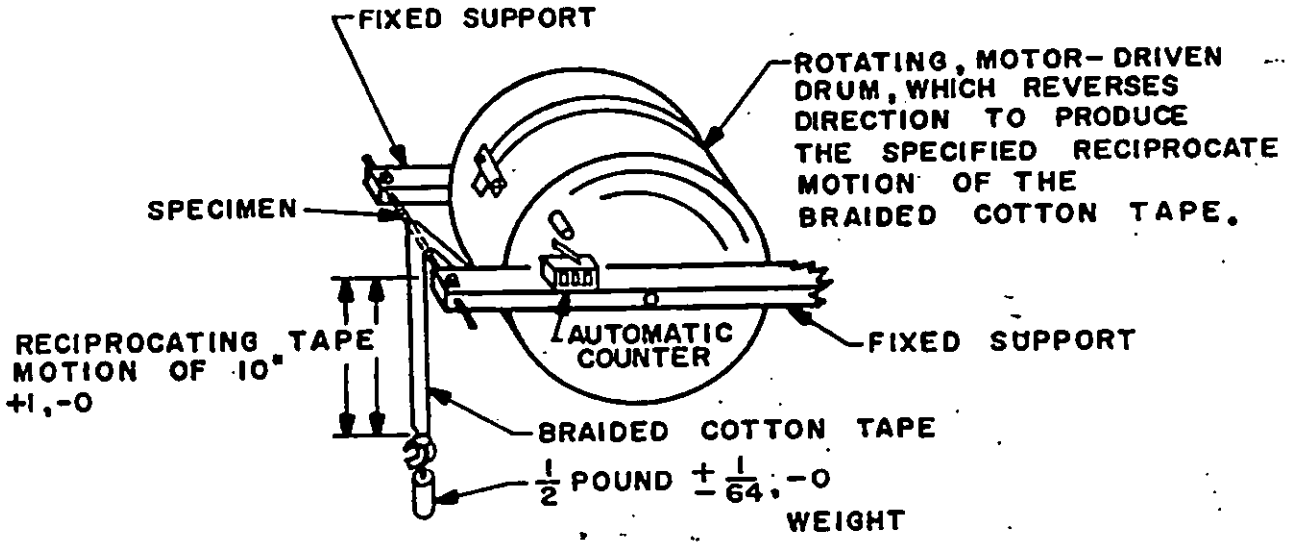
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SH 12747

FIGURE 11. Application of test voltages for the gas flame test.

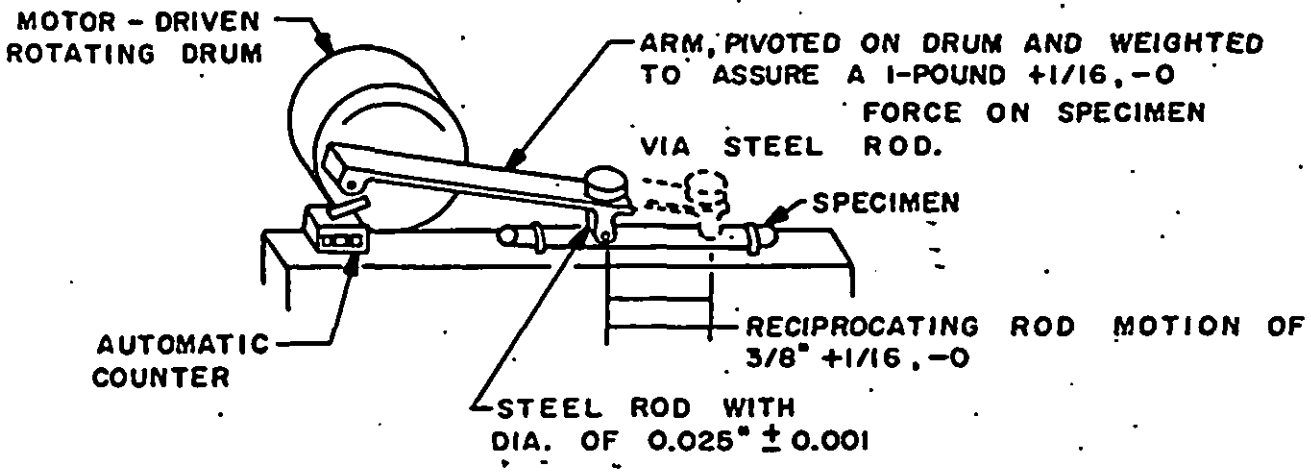
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SH 13062

FIGURE 12. Diagram of typical abrading machine for the permanence of printing (conductor insulation) test.

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SH 13063

FIGURE 13. Diagram of abrading machine for the permanence of printing (jacket) test.

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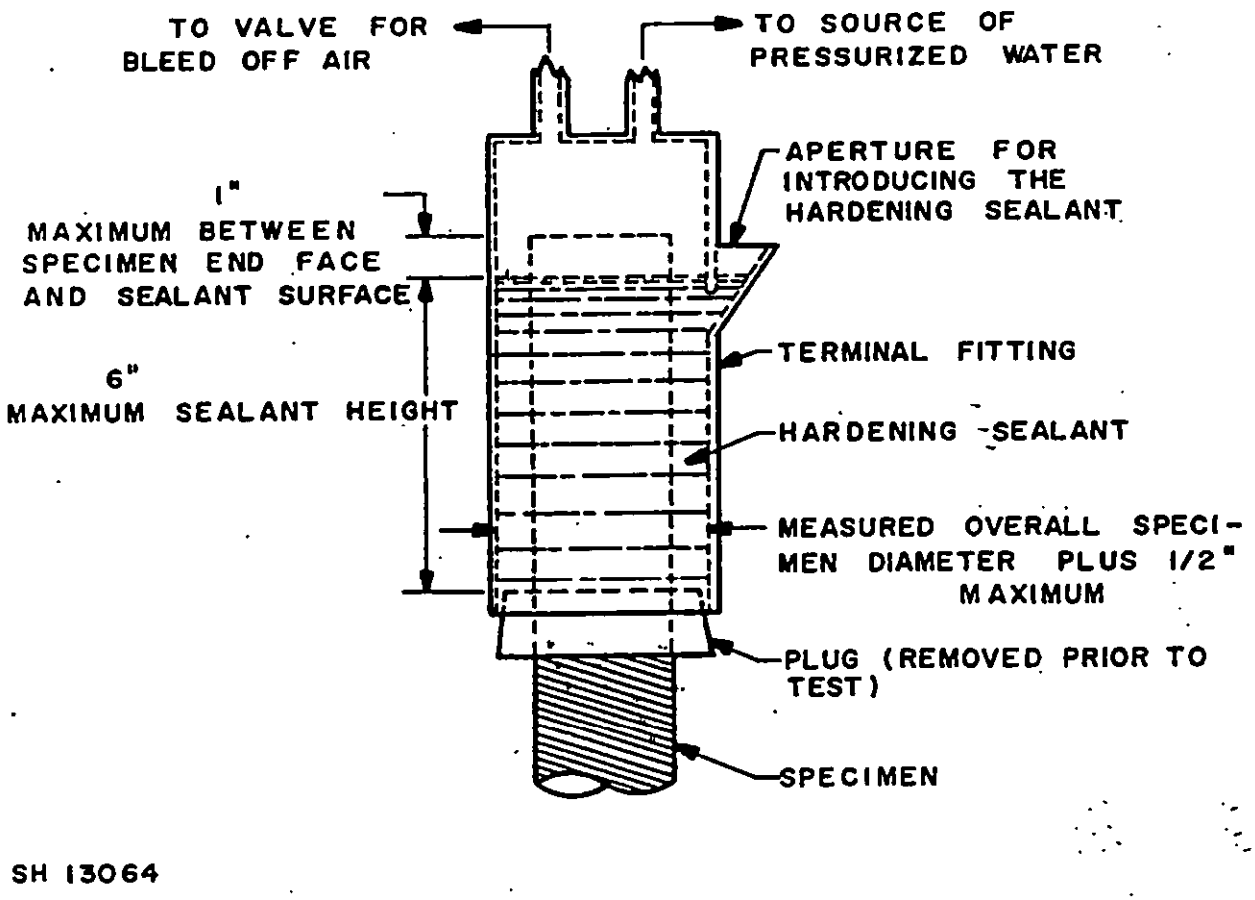


FIGURE 14. Diagram of terminal fitting, showing internal details, for the watertightness test.

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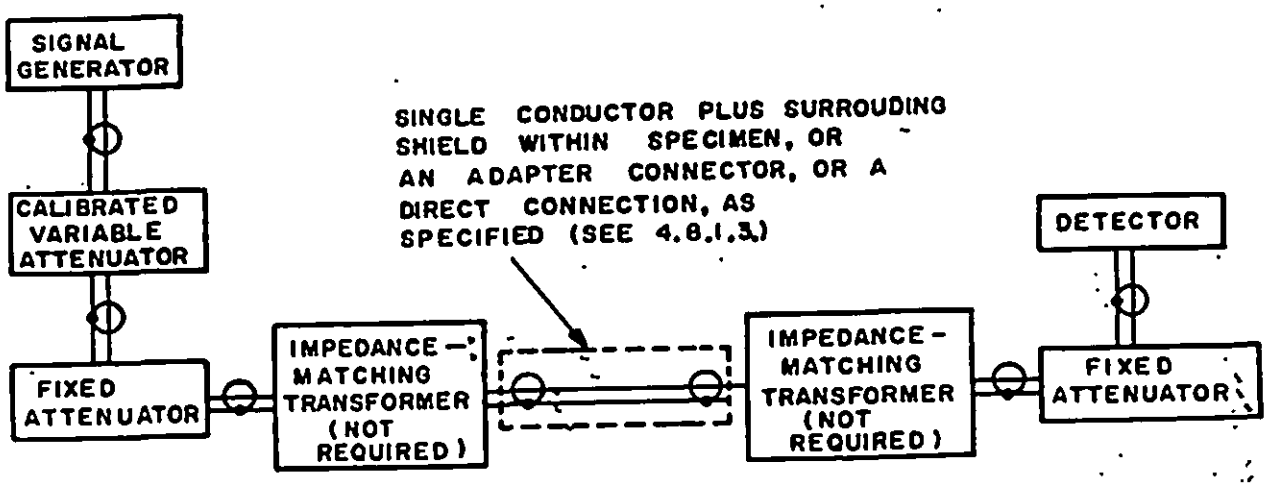


FIGURE 15. Block diagram of electrical connections for the attenuation test when testing a single conductor plus surrounding shield.

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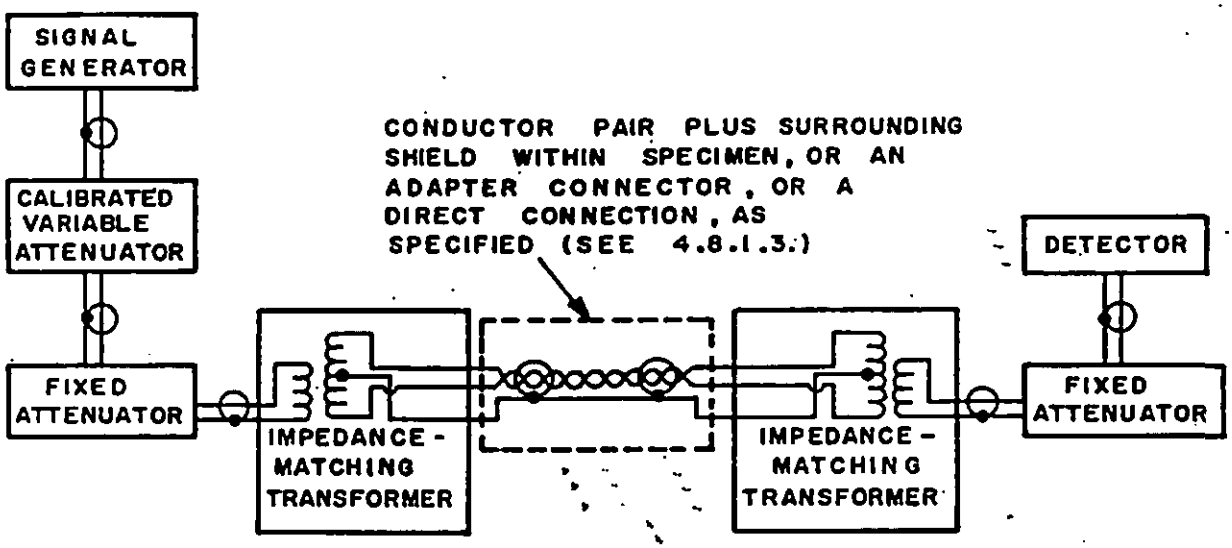


FIGURE 16. Block diagram of electrical connections for the attenuation test when testing a conductor pair plus surrounding shield.

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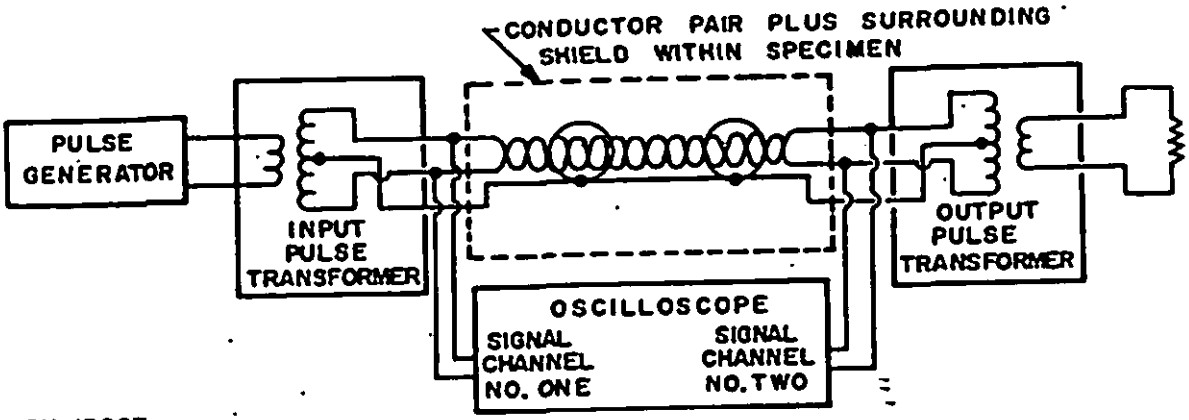


FIGURE 17. Block diagram of electrical connections for the pulse response test.

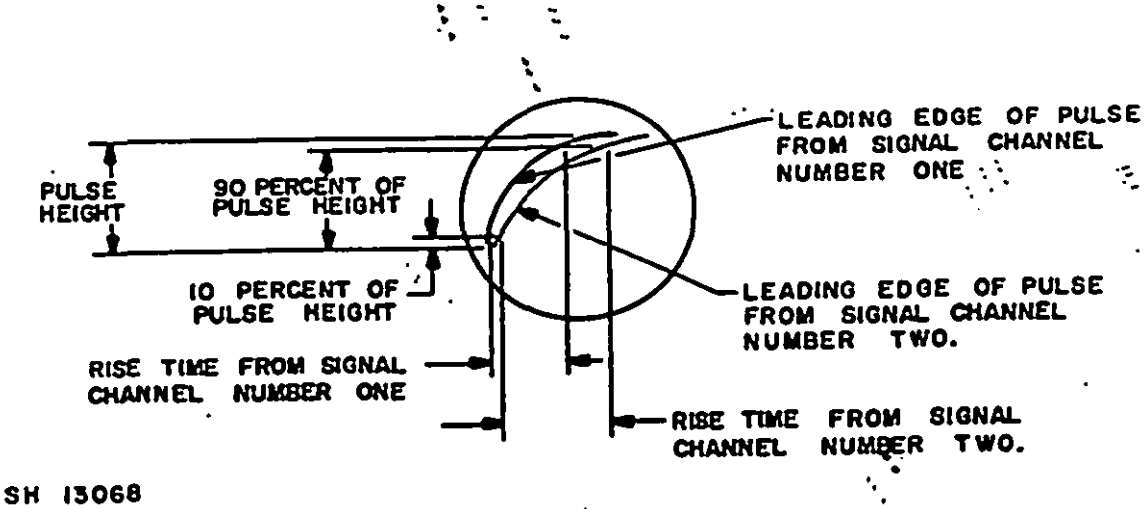
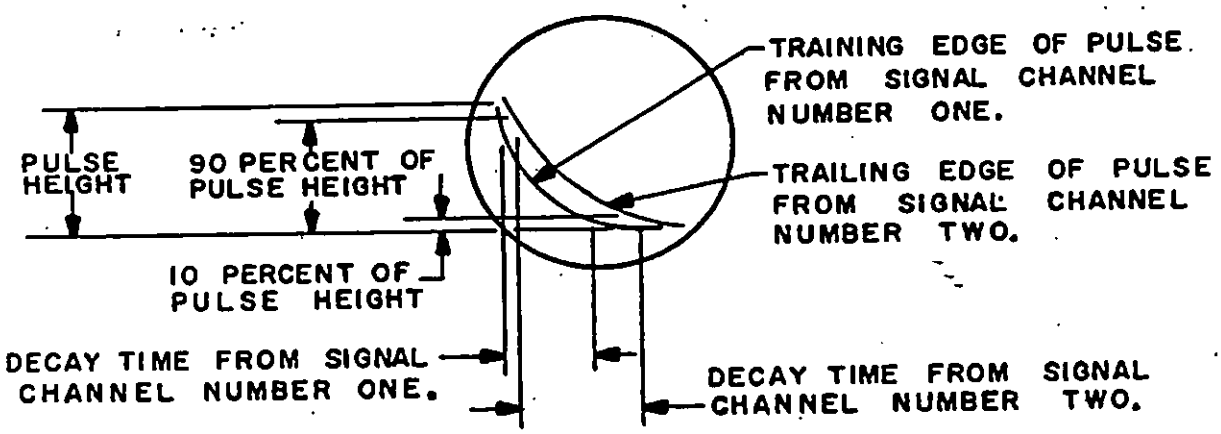


FIGURE 18. Oscilloscope display showing rise time measurements for the pulse response time test.

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SH 13069

FIGURE 19. Oscilloscope display showing decay time measurements for the pulse response time test.

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